

**ANALYSIS ON WIENER SPACE
AND ANTICIPATING STOCHASTIC CALCULUS**

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Introduction

The first two chapters of these notes are devoted to present the basic elements of the stochastic calculus of variations or *Malliavin calculus* on a Gaussian space. That is, we develop an infinite dimensional differential calculus for functionals of an arbitrary Gaussian process. The stochastic calculus of variations on the Wiener space was introduced by Malliavin in [62] in order to provide a probabilistic proof of Hörmander's hypoellipticity theorem. One of the main results of this subject is the equivalence between the L^p norms of the iterations of the derivative and the Ornstein-Uhlenbeck operators, that was established by Meyer in [66].

Chapter 3 deals with the application of the techniques of the stochastic calculus of variations to study regularity properties of probability laws. First we discuss the existence and smoothness of a density for an m -dimensional random vector assuming that its Malliavin matrix is nondegenerate. The basic tool to obtain these results is an integration by parts formula. We deduce also L^p estimates of the density and apply them to the problem of the existence, uniqueness and convergence of approximation schemes for ordinary and partial stochastic differential equations. In Chapter 4 we discuss the properties of the support of the probability distribution of a random element in the underlying Gaussian space. For nondegenerate smooth random vectors, the points where the density is strictly positive are characterized by means of the notion of skeleton. We show this type of characterization and we apply it to deduce Varadhan-type estimates when the variance of the white noise tends to zero.

In chapters 5 and 6 we present some results on the anticipating stochastic calculus. The divergence operator coincides with an extension of the Itô stochastic integral, due to Skorohod, for processes which are not adapted to the Brownian motion. The Skorohod integral turns out to have properties which are similar to the classical Itô integral. For instance, one can deduce a change-of-variable formula which generalizes the Itô formula. On the other hand, one can also define an extended Stratonovich integral by means of suitable approximation by Riemann sums. We discuss the relationship between the Skorohod integral and the extended Stratonovich integral and we establish a change-of-variable formula for the Stratonovich integral which is similar to that in classical analysis. The anticipating stochastic integrals allow us to formulate and solve stochastic differential equations whose coefficients are random, or such that some boundary conditions are in force. We deduce some existence and uniqueness results for these type of equations in Chapter 6.

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Chapter 1

Derivative and divergence operators on a Gaussian space

In this chapter we study the differential calculus on a Gaussian space. That is, we introduce the derivative operator and its adjoint the divergence operator. We show that these operators are local, and we study their behavior when they are applied to a Wiener chaos expansion.

1.1 Derivative operator

We work in the following general context. Suppose that H is a real separable Hilbert space whose norm and inner product are denoted by $\|\cdot\|_H$ and $\langle \cdot, \cdot \rangle_H$, respectively. We associate with H a Gaussian and centered family of random variables $\mathcal{H}_1 = \{W(h), h \in H\}$ such that

$$E(W(h)W(g)) = \langle h, g \rangle_H,$$

for all $h, g \in H$. The family $\mathcal{H}_1$ is defined in some complete probability space $(\Omega, \mathcal{F}, P)$. This means that $\mathcal{H}_1$ is a closed Gaussian subspace of $L^2(\Omega)$ isometric to H . We will assume that the σ -field $\mathcal{F}$ is generated by $\mathcal{H}_1$.

Examples:

1. Let $\{W_t^i, t \geq 0, i = 1, \dots, d\}$ be a d -dimensional Brownian motion. In this case the Gaussian space is obtained by taking $H = L^2(\mathbb{R}_+; \mathbb{R}^d)$, and for any $h \in H$ the variable $W(h)$ is the Wiener stochastic integral $\sum_{i=1}^d \int_0^\infty h_t^i dW_t^i$.
2. Suppose that $(T, \mathcal{B}, \mu)$ is a measure space such that $\mathcal{B}$ is countably generated, and $\{W(A), A \in \mathcal{B}, \mu(A) < \infty\}$ is a family of random variables such that each $W(A)$ has the distribution $N(0, \mu(A))$, $W(A \cup B) = W(A) + W(B)$ if A and B are disjoint, and $\{W(A_1), \dots, W(A_n)\}$ are independent whenever the sets $\{A_1, \dots, A_n\}$ are pairwise-disjoint. In this case the Gaussian space is obtained by taking $H = L^2(T, \mathcal{B}, \mu)$, and $W(h)$ is given by the stochastic integral

$$W(h) = \int_T h(t)W(dt).$$

We will say that W is a *white noise* on $(T, \mathcal{B}, \mu)$. Note that Example 1 is a particular case of example 2 by setting $T = \mathbb{R}_+ \times \{1, \dots, d\}$ and μ equals to the product of the Lebesgue measure times the uniform measure on $\{1, \dots, d\}$.

3. Let ν be a zero mean Gaussian measure on a real separable Banach space $\mathbb{B}$ with full support. Consider the inclusion $j : \mathbb{B}^* \rightarrow L^2(\nu)$ which is linear, continuous and one-to-one. Set $\mathcal{H}_1 = \overline{j(\mathbb{B}^*)}$. Then $\mathcal{H}_1$ is a Gaussian subspace of $L^2(\nu)$.

Remarks:

1. Consider the case of a d -dimensional Brownian motion W defined on the canonical probability space $\Omega = C_0(\mathbb{R}_+, \mathbb{R}^d)$. The subspace H^1 of Ω formed by the functions of the form $\varphi(t) = \int_0^t \dot{\varphi}(s) ds$, $\dot{\varphi} \in H = L^2(\mathbb{R}_+; \mathbb{R}^d)$ is called the Cameron-Martin space. Equipped with the scalar product

$$\langle \varphi_1, \varphi_2 \rangle_{H^1} = \langle \dot{\varphi}_1, \dot{\varphi}_2 \rangle_H$$

H^1 is isometric to H , and the inclusion $i : H^1 \rightarrow \Omega$ is continuous.

2. In the context of Example 3, one can show (cf. Gross [34], Kuo [55]) that the dual map j^* of j is a compact operator from $\mathcal{H}_1$ into $\mathbb{B}$ with a dense image. Then the image $H^1 := j^*(\mathcal{H}_1)$ is a subspace of $\mathbb{B}$, and the triple $(\mathbb{B}, H^1, \nu)$ is called an abstract Wiener space. When $\mathbb{B} = C_0(\mathbb{R}_+, \mathbb{R}^d)$, and ν is the Wiener measure, then H^1 is the Cameron-Martin space. In fact, given a point measure $m = c\delta_{t_0} \in \mathbb{B}^*$, $c \in \mathbb{R}^d$, $t_0 \geq 0$, clearly $j(m) = c \cdot W(t_0)$ and $j^*j(m)(t) = c(t_0 \wedge t)$ because

$$\begin{aligned} j^*j(m)_i(t) &= \langle j^*j(m), e_i \delta_t \rangle_{\mathbb{B}, \mathbb{B}^*} = \langle j(m), j(e_i \delta_t) \rangle_{L^2(\nu)} \\ &= E((c \cdot W(t_0))(e_i \cdot W(t))) = c_i(t_0 \wedge t), \end{aligned}$$

where e_i is the i th vector of the canonical basis of $\mathbb{R}^d$. Notice that the function $\varphi(t) = c(t_0 \wedge t)$ is absolutely continuous and its derivative has an L^2 -norm equals to $\sqrt{t_0} \|c\|$ which is the L^2 -norm of $j(m)$ in $L^2(\nu)$. As a consequence, $j^*(\mathcal{H}_1)$ consists of all absolutely continuous functions $\varphi : \mathbb{R}_+ \rightarrow \mathbb{R}^d$ which vanish at zero and have a square integrable density, that is, $j^*(\mathcal{H}_1)$ is the Cameron-Martin space.

Let us first introduce the derivative operator D . We will follow an approach analogous to the definition of the Sobolev spaces in finite dimensions. We denote by $C_p^\infty(\mathbb{R}^n)$ (resp. $C_b^\infty(\mathbb{R}^n)$) the set of all infinitely differentiable functions $f : \mathbb{R}^n \rightarrow \mathbb{R}$ such that f and all of its partial derivatives have polynomial growth (resp. are bounded).

Let $\mathcal{S}$ (resp. $\mathcal{S}_b$) denote the class of random variables of the form

$$F = f(W(h_1), \dots, W(h_n)), \quad (1.1)$$

where f belongs to $C_p^\infty(\mathbb{R}^n)$ (resp. $C_b^\infty(\mathbb{R}^n)$), $h_1, \dots, h_n$ are in H , and $n \geq 1$. These random variables are called *smooth*. We will denote by $\mathcal{P}$ the subset of $\mathcal{S}$ formed by random variables of the form (1.1) where f is a polynomial. If F has the form (1.1) we define its derivative DF as the H -valued random variable given by

$$DF = \sum_{i=1}^n \frac{\partial f}{\partial x_i}(W(h_1), \dots, W(h_n)) h_i. \quad (1.2)$$

Notice that for any element $h \in H$ the scalar product $\langle DF, h \rangle_H$ coincides with the directional derivative $\frac{d}{d\epsilon} F^{\epsilon h}|_{\epsilon=0}$, where $F^{\epsilon h}$ is the shifted random variable

$$F^{\epsilon h} = f(W(h_1) + \epsilon \langle h, h_1 \rangle_H, \dots, W(h_n) + \epsilon \langle h, h_n \rangle_H).$$

In order to show that the operator D is closable we need the following integration-by-parts formula:

Lemma 1.1.1 *Suppose that F is a smooth random variable and $h \in H$. Then*

$$E(\langle DF, h \rangle_H) = E(FW(h)). \quad (1.3)$$

Proof: We can assume that the norm of h is one. There exist orthonormal elements of H , $e_1, \dots, e_n$, such that $h = e_1$ and F is a smooth random variable of the form

$$F = f(W(e_1), \dots, W(e_n)).$$

Let $\phi(x)$ denote the density of the standard normal distribution on $\mathbb{R}^n$. Then we have

$$E(\langle DF, h \rangle_H) = \int_{\mathbb{R}^n} \frac{\partial f}{\partial x_1}(x) \phi(x) dx = \int_{\mathbb{R}^n} f(x) \phi(x) x_1 dx = E(FW(e_1)),$$

which completes the proof of the lemma. $\square$

Applying the previous result to a product FG , we obtain the following consequence.

Lemma 1.1.2 *Suppose that F and G are smooth functionals, and let $h \in H$. Then we have*

$$E(G\langle DF, h \rangle_H) = E(-F\langle DG, h \rangle_H + FGW(h)). \quad (1.4)$$

As a consequence of the above lemma, D is closable as an operator from $L^p(\Omega)$ to $L^p(\Omega; H)$ for any $p \geq 1$. In fact, let $\{F_N, N \geq 1\}$ be a sequence of smooth random variables such that F_N converges to zero in $L^p(\Omega)$ and the sequence of derivatives DF_N converges to η in $L^p(\Omega; H)$. Then, from Lemma 1.1.2 it follows that η is equal to zero. Indeed, for any $h \in H$ and for any smooth random variable $F \in \mathcal{S}_b$ such that $FW(h)$ is bounded, we have

$$\begin{aligned} E(\langle \eta, h \rangle_H F) &= \lim_N E(\langle DF_N, h \rangle_H F) \\ &= \lim_N E(-F_N \langle DF, h \rangle_H + F_N FW(h)) = 0, \end{aligned}$$

because F_N converges to zero in L^p as N tends to infinity, and the random variables $\langle DF, h \rangle_H$ and $FW(h)$ are bounded. This implies $\eta = 0$.

We will denote the domain of D in $L^p(\Omega)$ by $\mathbb{D}^{1,p}$, meaning that $\mathbb{D}^{1,p}$ is the closure of the class of smooth random variables $\mathcal{S}$ with respect to the norm

$$\|F\|_{1,p} = [E(|F|^p) + E(\|DF\|_H^p)]^{\frac{1}{p}}.$$

For $p = 2$, the space $\mathbb{D}^{1,2}$ is a Hilbert space with the scalar product

$$\langle F, G \rangle_{1,2} = E(FG) + E(\langle DF, DG \rangle_H).$$

We can define the iteration of the operator D in such a way that for a smooth random variable F , the derivative $D^k F$ is a random variable with values on $H^{\otimes k}$.

Then for every $p \geq 1$ and any natural number k we introduce a seminorm on $\mathcal{S}$ defined by

$$\|F\|_{k,p}^p = E(|F|^p) + \sum_{j=1}^k E(\|D^j F\|_{H^{\otimes j}}^p). \quad (1.5)$$

As in the case $k = 1$ one can show that the operator D^k is closable from $\mathcal{S} \subset L^p(\Omega)$ into $L^p(\Omega; H^{\otimes k})$, $p \geq 1$. For any real $p \geq 1$ and any natural number $k \geq 0$, we will denote by $\mathbb{D}^{k,p}$ the completion of the family of smooth random variables $\mathcal{S}$ with respect to the norm $\|\cdot\|_{k,p}$. Note that $\mathbb{D}^{j,p} \subset \mathbb{D}^{k,q}$ if $j \geq k$ and $p \geq q$.

Let V be a real separable Hilbert space. We can also introduce the corresponding Sobolev spaces of V -valued random variables. More precisely, $\mathcal{S}_V$ will denote the family of V -valued smooth random variables of the form

$$F = \sum_{j=1}^n F_j v_j, \quad v_j \in V, \quad F_j \in \mathcal{S}.$$

We define $D^k F = \sum_{j=1}^n D^k F_j \otimes v_j$, $k \geq 1$. Then D^k is a closable operator from $\mathcal{S}_V \subset L^p(\Omega; V)$ into $L^p(\Omega; H^{\otimes k} \otimes V)$ for any $p \geq 1$. For any integer $k \geq 1$ and any real number $p \geq 1$ we can define the seminorm on $\mathcal{S}_V$ by

$$\|F\|_{k,p,V}^p = E(\|F\|_V^p) + \sum_{j=1}^k E(\|D^j F\|_{H^{\otimes j} \otimes V}^p).$$

We denote by $\mathbb{D}^{k,p}(V)$ the completion of $\mathcal{S}_V$ with respect to the norm $\|\cdot\|_{k,p,V}$. For $k = 0$ we put $\|F\|_{0,p,V} = [E(\|F\|_V^p)]^{\frac{1}{p}}$, and $\mathbb{D}^{0,p}(V) = L^p(\Omega; V)$.

Consider the intersection

$$\mathbb{D}^\infty(V) = \bigcap_{p \geq 1} \bigcap_{k \geq 1} \mathbb{D}^{k,p}(V).$$

Then $\mathbb{D}^\infty(V)$ is a complete, countably normed, metric space. We will write $\mathbb{D}^\infty(\mathbb{R}) = \mathbb{D}^\infty$. For every integer $k \geq 1$ and any real number $p \geq 1$ the operator D is continuous from $\mathbb{D}^{k,p}(V)$ into $\mathbb{D}^{k-1,p}(H \otimes V)$. Consequently, D is a continuous linear operator from $\mathbb{D}^\infty(V)$ into $\mathbb{D}^\infty(H \otimes V)$. Moreover, if F and G are random variables in $\mathbb{D}^\infty$, then the scalar product $\langle DF, DG \rangle_H$ is also in $\mathbb{D}^\infty$.

The following result is the chain rule, which can be easily proved by approximating the random variable F by smooth random variables, and the function φ by $(\varphi * \psi_\epsilon)$, where ψ_ϵ is an approximation of the identity.

Proposition 1.1.1 *Let $\varphi : \mathbb{R}^m \rightarrow \mathbb{R}$ be a continuously differentiable function with bounded partial derivatives, and fix $p \geq 1$. Suppose that $F = (F^1, \dots, F^m)$ is a random vector whose components belong to the space $\mathbb{D}^{1,p}$. Then $\varphi(F) \in \mathbb{D}^{1,p}$, and*

$$D(\varphi(F)) = \sum_{i=1}^m \frac{\partial \varphi}{\partial x_i}(F) DF^i.$$

As a consequence of the chain rule, the space $\mathbb{D}^\infty$ is an algebra. We remark that in the infinite dimensional case, and assuming that Ω is a Banach space, the elements of $\mathbb{D}^\infty$ may not be differentiable or even continuous. For instance, the random variable $F = \int_0^1 W_t^1 dW_t^2$ is not continuous on $C([0, 1]; \mathbb{R}^2)$ and it belongs to $\mathbb{D}^\infty$.

We will write $D_h F = \langle DF, h \rangle_H$.

1.2 Divergence operator

We will denote by δ the adjoint of the operator D as an unbounded operator from $L^2(\Omega)$ into $L^2(\Omega; H)$. That is, the domain of δ , denoted by $\text{Dom } \delta$, is the set of H -valued square integrable random variables u such that

$$|E(\langle DF, u \rangle_H)| \leq c \|F\|_2, \quad (1.6)$$

for all $F \in \mathbb{D}^{1,2}$, where c is some constant depending on u . If u belongs to $\text{Dom } \delta$, then $\delta(u)$ is the element of $L^2(\Omega)$ characterized by

$$E(F\delta(u)) = E(\langle DF, u \rangle_H) \quad (1.7)$$

for any $F \in \mathbb{D}^{1,2}$. The operator δ will be called the *divergence operator* and is closed as the adjoint of an unbounded and densely defined operator.

We denote by S_H the class of smooth elementary elements of the form

$$u = \sum_{j=1}^n F_j h_j, \quad (1.8)$$

where the F_j are smooth random variables, and the h_j are elements of H . From the integration-by-parts formula established in Lemma 1.1.2 we deduce that an element u of this type belongs to the domain of δ and moreover that

$$\delta(u) = \sum_{j=1}^n F_j W(h_j) - \sum_{j=1}^n \langle DF_j, h_j \rangle_H. \quad (1.9)$$

Notice that if $u \in S_H$ then Du is an $H \otimes H$ -valued random variable.

For smooth elements u, v and F we have the three following basic relations between the operators D and δ :

$$D_h(\delta(u)) = \langle u, h \rangle_H + \delta(D_h u), \quad (1.10)$$

$$E(\delta(u)\delta(v)) = E(\langle u, v \rangle_H) + E(\text{Tr}(Du \circ Dv)) \quad (1.11)$$

$$\delta(Fu) = F\delta(u) - \langle DF, u \rangle_H. \quad (1.12)$$

Proof of (1.10): Suppose that u has the form (1.8). From (1.9) we deduce

$$D_h(\delta(u)) = \sum_{j=1}^n F_j \langle h, h_j \rangle_H + \sum_{j=1}^n (D_h F_j W(h_j) - \langle D(D_h F_j), h_j \rangle_H),$$

which implies (1.10).

Proof of (1.11): Let $\{e_i, i \geq 1\}$ be a complete orthonormal system on H . Using the duality relationship (1.7) and property (1.10) we obtain

$$\begin{aligned} E(\delta(u)\delta(v)) &= E(\langle v, D(\delta(u)) \rangle_H) = E\left(\sum_{i=1}^{\infty} \langle v, e_i \rangle_H D_{e_i}(\delta(u))\right) \\ &= E\left(\sum_{i=1}^{\infty} \langle v, e_i \rangle_H (\langle e_i, u \rangle_H + \delta(D_{e_i} u))\right) \\ &= E(\langle u, v \rangle_H) + E\left(\sum_{i,j=1}^{\infty} D_{e_i} \langle u, e_j \rangle_H D_{e_j} \langle v, e_i \rangle_H\right). \end{aligned}$$

Proof of (1.12): For any smooth random variable G we have

$$\begin{aligned} E(G\delta(Fu)) &= E(\langle DG, Fu \rangle_H) = E(\langle u, (D(FG) - GDF) \rangle_H) \\ &= E(G(F\delta(u) - \langle u, DF \rangle_H)), \end{aligned}$$

and the result follows.

Property (1.11) implies the following estimate

$$E[\delta(u)^2] \leq E(\|u\|_H^2) + E(\|Du\|_{H \otimes H}^2) = \|u\|_{1,2,H}^2.$$

As a consequence, the space of weakly differentiable H -valued variables $\mathbb{D}^{1,2}(H)$ is included into $\text{Dom } \delta$, and (1.11) holds for any $u \in \mathbb{D}^{1,2}(H)$. By an approximation argument property (1.10) holds for an element $u \in \mathbb{D}^{2,2}(H)$ (actually, (1.10) holds if $u \in \mathbb{D}^{1,2}(H)$ and $D_h u$ belongs to the domain of δ , as it can be proved using Lemma 1.4.1). By the definition of the operator δ , property (1.12) holds whenever $F \in \mathbb{D}^{1,2}$, u belongs to $\text{Dom } \delta$, $Fu \in L^2(\Omega; H)$, and the right-hand side of (1.12) belongs to $L^2(\Omega)$.

1.3 Local properties

In this section we will show that the operators D and δ are local. The next result shows that the operator D is local in the space $\mathbb{D}^{1,1}$.

Proposition 1.3.1 *Let F be a random variable in the space $\mathbb{D}^{1,1}$ such that $F = 0$ a.s. on some set $A \in \mathcal{F}$. Then $DF = 0$ a.e. on A .*

Proof: We can assume that $F \in \mathbb{D}^{1,1} \cap L^\infty(\Omega)$, replacing F by $\arctan(F)$. We want to show that $\mathbf{1}_{\{F=0\}}DF = 0$ a.s. Suppose that $\phi : \mathbb{R} \rightarrow \mathbb{R}$ is an infinitely differentiable function such that $\phi \geq 0$, $\phi(0) = 1$ and its support is included in the interval $[-1, 1]$. Define the function $\phi_\epsilon(x) = \phi(\frac{x}{\epsilon})$ for all $\epsilon > 0$. Set

$$\psi_\epsilon(x) = \int_{-\infty}^x \phi_\epsilon(y) dy.$$

By the chain rule $\psi_\epsilon(F)$ belongs to $\mathbb{D}^{1,1}$ and $D\psi_\epsilon(F) = \phi_\epsilon(F)DF$. Let u be a smooth H -valued random variable of the form

$$u = \sum_{j=1}^n F_j h_j,$$

where $F_j \in \mathcal{S}_b$ and $h_j \in H$. Observe that the duality relation (1.7) holds for F in $\mathbb{D}^{1,1} \cap L^\infty(\Omega)$ and for an element u of this type. Note that the class of such processes u is total in $L^1(\Omega; H)$ in the sense that if $v \in L^1(\Omega; H)$ satisfies $E(\langle v, u \rangle_H) = 0$ for all u in the class, then $v \equiv 0$. Then we have

$$\begin{aligned} |E(\phi_\epsilon(F)\langle DF, u \rangle_H)| &= |E(\langle D(\psi_\epsilon(F)), u \rangle_H)| \\ &= |E(\psi_\epsilon(F)\delta(u))| \leq \epsilon \|\phi\|_\infty E(|\delta(u)|). \end{aligned}$$

Letting $\epsilon \downarrow 0$, we obtain

$$E(\mathbf{1}_{\{F=0\}}\langle DF, u \rangle_H) = 0,$$

which implies the desired result. $\square$

Let us now state and prove the local property of the divergence.

Proposition 1.3.2 *Let $u \in \mathbb{D}^{1,2}(H)$ and $A \in \mathcal{F}$, such that $u(\omega) = 0$, P a.e. on A . Then $\delta(u) = 0$ a.e. on A .*

Proof: Let F be a smooth random variable of the form

$$F = f(W(h_1), \dots, W(h_n)),$$

with $f \in C_0^\infty(\mathbb{R}^n)$ (f is an infinitely differentiable function with compact support). We want to show that

$$\delta(u) \mathbf{1}_{\{\|u\|_H=0\}} = 0, \quad \text{a.s.}$$

Consider a function $\phi : \mathbb{R} \rightarrow \mathbb{R}$ such as that in the proof of Proposition 1.3.1. It is easy to show that the product $F\phi_\epsilon(\|u\|_H^2)$ belongs to $\mathbb{D}^{1,2}$. Then by the duality relation (1.7) we obtain

$$\begin{aligned} E\left(\delta(u)\phi_\epsilon(\|u\|_H^2)F\right) &= E\left(\langle u, D[F\phi_\epsilon(\|u\|_H^2)] \rangle_H\right) \\ &= E\left(\phi_\epsilon(\|u\|_H^2)\langle u, DF \rangle_H\right) + 2E\left(F\phi'_\epsilon(\|u\|_H^2)\langle u, Du \rangle_H\right). \end{aligned}$$

We claim that the above expression converges to zero as ϵ tends to zero. In fact, first observe that the random variables

$$V_\epsilon = \phi_\epsilon(\|u\|_H^2)\langle u, DF \rangle_H + 2F\phi'_\epsilon(\|u\|_H^2)\langle u, Du \rangle_H$$

converge a.s. to zero as $\epsilon \downarrow 0$, since $\|u\|_H = 0$ implies $V_\epsilon = 0$. Second, we can apply the Lebesgue dominated convergence theorem because we have

$$\begin{aligned} \left|\phi_\epsilon(\|u\|_H^2)\langle u, DF \rangle_H\right| &\leq \|\phi\|_\infty \|u\|_H \|DF\|_H, \\ \left|\phi'_\epsilon(\|u\|_H^2)\langle u, Du \rangle_H\right| &\leq \sup_x |x\phi'_\epsilon(x)| \|Du\|_{H \otimes H} \leq \|\phi'\|_\infty \|Du\|_{H \otimes H}. \end{aligned}$$

The proof is now complete. $\square$

Note that D is also local when acting on Hilbert-valued random variables. We can localize the domains of the operators D and δ as follows. We will denote by $\mathbb{D}_{\text{loc}}^{1,p}(V)$, $p \geq 1$, the set of V -valued random variables F such that there exists a sequence $\{(\Omega_n, F_n), n \geq 1\} \subset \mathcal{F} \times \mathbb{D}^{1,p}(V)$ with the following properties:

- (i) $\Omega_n \uparrow \Omega$, a.s.
- (ii) $F = F_n$ a.s. on Ω_n .

We then say that (Ω_n, F_n) localizes F in $\mathbb{D}^{1,p}(V)$, and DF is defined without ambiguity by $DF = DF_n$ on Ω_n , $n \geq 1$. The spaces $\mathbb{D}_{\text{loc}}^{k,p}(V)$ can be introduced analogously.

Then, if $u \in \mathbb{D}_{\text{loc}}^{1,2}(H)$, the divergence $\delta(u)$ is defined as a random variable determined by the conditions

$$\delta(u)|_{\Omega_n} = \delta(u_n)|_{\Omega_n} \quad \text{for all } n \geq 1,$$

where (Ω_n, u_n) is a localizing sequence for u .

1.4 Wiener chaos expansions

We will denote by $\{H_n, n \in \mathbb{N}\}$ the sequence of the Hermite polynomials defined from the series expansion

$$\exp(tx - \frac{t^2}{2}) = \sum_{n=0}^{\infty} t^n H_n(x). \quad (1.13)$$

Then $\{\sqrt{n!} H_n, n \in \mathbb{N}\}$ is a complete orthonormal system in $L^2(\mathbb{R}, \mu)$ where μ is the normal distribution $N(0, 1)$. We will denote by Λ the set of all sequences $a = (a_1, a_2, \dots)$, $a_i \in \mathbb{N}$, such that $|a| = a_1 + a_2 + \dots < \infty$. Let $\{e_i, i \geq 1\}$ be a complete orthonormal system in H . For any $a \in \Lambda$ we set

$$\Phi_a = \sqrt{a!} \prod_{i=1}^{\infty} H_{a_i}(W(e_i)),$$

where $a! = \prod_{i=1}^{\infty} a_i!$. Then the family $\{\Phi_a, a \in \Lambda\}$ constitutes an orthonormal basis of $L^2(\Omega, \mathcal{F}, P)$ (we recall that we assume that the σ -field $\mathcal{F}$ is generated by W). For any $n \geq 0$ we will denote by $\mathcal{H}_n$ the closed subspace of $L^2(\Omega)$ spanned by $\{\Phi_a, a \in \Lambda, |a| = n\}$. Then $\mathcal{H}_n$ is called the Wiener chaos of order n and we have the orthogonal decomposition

$$L^2(\Omega) = \oplus_{n=0}^{\infty} \mathcal{H}_n.$$

We will denote by J_n the orthogonal projection on the n th Wiener chaos $\mathcal{H}_n$. The Wiener chaos expansion can be extended to the space $L^2(\Omega; V)$ of Hilbert valued random variables: $L^2(\Omega; V) = \oplus_{n=0}^{\infty} \mathcal{H}_n(V)$, where $\mathcal{H}_n(V) = \mathcal{H}_n \otimes V$.

The following result characterizes the domain of the operator D in L^2 in terms of the Wiener chaos expansion.

Proposition 1.4.1 *We have*

$$E(\|DF\|_H^2) = \sum_{n=1}^{\infty} n \|J_n(F)\|_2^2, \quad (1.14)$$

in the sense that this sum is finite if and only if $F \in \mathbb{D}^{1,2}$. Moreover, for all $n \geq 1$ we have $D(J_n(F)) = J_{n-1}(DF)$.

Proof: It suffices to compute the derivative of a random variable of the form Φ_a , using the relationship $H'_n = H_{n-1}$, which follows immediately from (1.13):

$$D(\Phi_a) = \sqrt{a!} \sum_{j=1}^{\infty} \prod_{i=1, i \neq j}^{\infty} H_{a_i}(W(e_i)) H_{a_{j-1}}(W(e_j)) e_j.$$

Then $D(\Phi_a) \in \mathcal{H}_{n-1}(H)$ if $|a| = n$, and

$$E(\|D(\Phi_a)\|_H^2) = \sum_{j=1}^{\infty} \frac{a!}{\prod_{i=1, i \neq j}^{\infty} a_i! (a_j - 1)!} = |a|.$$

Now the proposition follows easily. □

By iteration we obtain

$$E(\|D^k F\|_{H^{\otimes k}}^2) = \sum_{n=k}^{\infty} n(n-1) \cdots (n-k+1) \|J_n(F)\|_2^2, \quad (1.15)$$

and $F \in \mathbb{D}^{k,2}$ if and only if $\sum_{n=1}^{\infty} n^k \|J_n(F)\|_2^2 < \infty$. Moreover, for all $n \geq k$ we have $D^k(J_n(F)) = J_{n-k}(D^k F)$.

The next lemma is a consequence of the expression of the operator D in terms of the Wiener chaos.

Lemma 1.4.1 *Let $G \in L^2(\Omega)$ and $\eta \in L^2(\Omega; H)$ be such that*

$$E(G\delta(v)) = E(\langle \eta, v \rangle_H),$$

for all $v \in \mathbb{D}^{1,2}(H)$. Then $G \in \mathbb{D}^{1,2}$ and $DG = \eta$.

Proof: We have

$$E(\langle \eta, v \rangle_H) = E(G\delta(v)) = \sum_{n=1}^{\infty} E(J_n(G)\delta(v)) = \sum_{n=1}^{\infty} E(\langle D(J_n(G)), v \rangle_H),$$

hence, $J_{n-1}\eta = D(J_n(G))$ for each $n \geq 1$ and this implies the result. $\square$

Remarks:

If F is a random variable in the space $\mathbb{D}^{1,1}$ such that $DF = 0$, then $F = E(F)$. This is obvious when $F \in \mathbb{D}^{1,2}$ if we use the Wiener chaos expansion of F and Proposition 1.4.1. In the general case this property can be proved by a duality argument. Indeed, let ψ_N be a function in $C_b^\infty(\mathbb{R})$ such that $\psi_N(x) = x$ if $|x| \leq N$, and $|\psi_N(x)| \leq N + 1$. Then one shows that

$$E(\psi_N(F - E(F))\delta(u)) = 0 \tag{1.16}$$

for any bounded H -valued random variable u in the domain of δ . Approximating an arbitrary random variable $u \in \mathbb{D}^{1,2}(H)$ by the sequence of bounded random variables $\{u\phi_k(\|u\|_H^2), k \geq 1\}$, where $\phi_k(x) = 1$ if $|x| \leq k$, $\phi_k(x) = 0$ if $|x| \geq 2k$, and $|\phi'_k(x)| \leq 2/k$, we obtain that (1.16) holds for any $u \in \mathbb{D}^{1,2}(H)$. Finally, by Lemma 1.4.1 this implies that $\psi_N(F - E(F)) = E(\psi_N(F - E(F)))$, and letting N tend to infinity we deduce the result.

As an application of the chain rule and the above property we can show the following result (see Sekiguchi and Shiota [96]).

Lemma 1.4.2 *Let $A \in \mathcal{F}$. Then the indicator function of A belongs to $\mathbb{D}^{1,1}$ if and only if $P(A)$ is equal to zero or one.*

Proof: Applying the chain rule to a function $\varphi \in C_0^\infty(\mathbb{R})$ which is equal to x^2 on $[0, 1]$ yields

$$D\mathbf{1}_A = D(\mathbf{1}_A)^2 = 2\mathbf{1}_A D\mathbf{1}_A.$$

Consequently, $D\mathbf{1}_A = 0$ because from the above equality we get that this derivative is zero on A^c and equal to twice its value on A . So, by the previous remark we obtain $\mathbf{1}_A = P(A)$. $\square$

1.5 The white noise case

Consider the particular case of a white noise $\{W(A), A \in \mathcal{B}, \mu(A) < \infty\}$ defined on a measure space $(T, \mathcal{B}, \mu)$. Suppose in addition that the measure μ is σ -finite and without atoms. In this case (cf. Itô [49]) the multiple stochastic integral of order n provides an isometry between the space $L_s^2(T^n, \mathcal{B}^n, \mu^n)$ of square integrable and symmetric functions of n variables and the n th Wiener chaos $\mathcal{H}_n$. Let us briefly describe how the multiple stochastic integral is constructed. Consider the set $\mathcal{E}_n$ of elementary functions of the form

$$f(t_1, \dots, t_n) = \sum_{i_1, \dots, i_n=1}^k a_{i_1 \dots i_n} \mathbf{1}_{A_{i_1} \times \dots \times A_{i_n}}(t_1, \dots, t_n), \quad (1.17)$$

where $A_1, A_2, \dots, A_k$ are pairwise-disjoint sets of finite measure, and the coefficients $a_{i_1 \dots i_n}$ are zero if any two of the indices $i_1, \dots, i_n$ are equal. Then we set

$$I_n(f) = \sum_{i_1, \dots, i_n=1}^k a_{i_1 \dots i_n} W(A_{i_1}) \cdots W(A_{i_n}).$$

The set $\mathcal{E}_n$ is dense in $L^2(T^n)$, and I_n is a linear map from $\mathcal{E}_n$ into $L^2(\Omega)$ which verifies $I_n(f_n) = I_n(\tilde{f}_n)$ and

$$E(I_n(f)I_q(g)) = \begin{cases} 0 & \text{if } n \neq q, \\ n! \langle \tilde{f}, \tilde{g} \rangle_{L^2(T^n)} & \text{if } n = q, \end{cases}$$

where $\tilde{f}$ denotes the symmetrization of f . As a consequence, I_n can be extended as a linear and continuous operator from $L^2(T^n)$ into $L^2(\Omega)$. The image of $L^2(T^n)$ by I_n is the n th Wiener chaos $\mathcal{H}_n$. This is a consequence of the fact that multiple stochastic integrals of different order are orthogonal, and that $I_n(f)$ is a polynomial of degree n in $W(A_1), \dots, W(A_k)$ if f has the form (1.17), and those polynomials are included in the sum of the first n chaos. As a consequence, any square integrable random variable F admits an orthogonal expansion in the form

$$F = E(F) + \sum_{n=1}^{\infty} I_n(f_n), \quad (1.18)$$

where the functions $f_n \in L^2(T^n)$ are symmetric and uniquely determined by F .

The operators D and δ can be represented in terms of the Wiener chaos expansion.

Proposition 1.5.1 *Let $F \in \mathbb{D}^{1,2}$ be a square integrable random variable with an expansion of the form (1.18). Then we have*

$$D_t F = \sum_{n=1}^{\infty} n I_{n-1}(f_n(\cdot, t)). \quad (1.19)$$

Proof: Suppose first that $F = I_n(f_n)$, where f_n is a symmetric and elementary function of the form (1.17). Then

$$D_t F = \sum_{j=1}^n \sum_{i_1, \dots, i_n=1}^k a_{i_1 \dots i_n} W(A_{i_1}) \cdots \mathbf{1}_{A_{i_j}}(t) \cdots W(A_{i_n}) = n I_{n-1}(f_n(\cdot, t)).$$

Then the results follows easily. $\square$

Note that the $L^2(\Omega; H)$ -norm of the right-hand side of (1.19) coincides with expression (1.14). We remark that the derivative $D_t F$ is a random field parametrized by T .

Suppose that F is a random variable in the space $\mathbb{D}^{N,2}$, with a Wiener chaos expansion of the form $F = \sum_{n=0}^{\infty} I_n(f_n)$. Then, applying Proposition 1.5.1 N times we obtain that $D^N F$ is a random field parametrized by T^N given by

$$D_{t_1, \dots, t_N}^N F = \sum_{n=N}^{\infty} n(n-1) \cdots (n-N+1) I_{n-N}(f_n(\cdot, t_1, \dots, t_N)).$$

Note that the L^2 -norm of this expression is given by (1.15). As a consequence, if F belongs to $\mathbb{D}^{\infty,2} = \cap_N \mathbb{D}^{N,2}$ then $f_n = \frac{1}{n!} E(D^n F)$ for every $n \geq 0$ (cf. Stroock [100]).

We will now describe the divergence operator in terms of the Wiener chaos expansion. Any element $u \in L^2(\Omega; H) \cong L^2(T \times \Omega)$ (which can be regarded as a square integrable process parametrized by T) has an orthogonal expansion of the form

$$u(t) = \sum_{n=0}^{\infty} I_n(f_n(\cdot, t)), \quad (1.20)$$

where for each $n \geq 1$, $f_n \in L^2(T^{n+1})$ is a symmetric function in the first n variables.

Proposition 1.5.2 *Let $u \in L^2(T \times \Omega)$ with the expansion (1.20). Then u belongs to $\text{Dom } \delta$ if and only if the series*

$$\delta(u) = \sum_{n=0}^{\infty} I_{n+1}(\tilde{f}_n) \quad (1.21)$$

converges in $L^2(\Omega)$.

Equation (1.21) can also be written without symmetrization, because for each n , $I_{n+1}(f_n) = I_{n+1}(\tilde{f}_n)$. However, the symmetrization is needed in order to compute the L^2 norm of the stochastic integrals.

Proof: Suppose that $G = I_n(g)$ is a multiple stochastic integral of order $n \geq 1$ where g is symmetric. Then we have the following equalities:

$$\begin{aligned} E(\langle u, DG \rangle_H) &= \int_T E(I_{n-1}(f_{n-1}(\cdot, t)) n I_{n-1}(g(\cdot, t))) \mu(dt) \\ &= n(n-1)! \int_T \langle f_{n-1}(\cdot, t), g(\cdot, t) \rangle_{L^2(T^{n-1})} \mu(dt) \\ &= n! \langle f_{n-1}, g \rangle_{L^2(T^n)} = n! \langle \tilde{f}_{n-1}, g \rangle_{L^2(T^n)} \\ &= E(I_n(\tilde{f}_{n-1}) I_n(g)) = E(I_n(\tilde{f}_{n-1}) G). \end{aligned}$$

Suppose first that $u \in \text{Dom } \delta$. Then from the above computations and from formula (1.7) we deduce that

$$E(\delta(u)G) = E(I_n(\tilde{f}_{n-1})G)$$

for every multiple stochastic integral $G = I_n(g)$. This implies that $I_n(\tilde{f}_{n-1})$ coincides with the projection of $\delta(u)$ on the n th Wiener chaos. Consequently, the series in

(1.21) converges in $L^2(\Omega)$ and its sum is equal to $\delta(u)$. The converse can be proved by a similar argument. $\square$

For any set $A \subset T$ we will denote by $\mathcal{F}_A$ the σ -field generated by the random variables $\{W(B), B \subset A, B \in \mathcal{B}_0\}$, where $\mathcal{B}_0 = \{B \in \mathcal{B}, \mu(B) < \infty\}$.

Lemma 1.5.1 *Let $A \in \mathcal{B}$ and let F be a random variable in $L^2(\Omega, \mathcal{F}_{A^c}, P) \cap \mathbb{D}^{1,2}$. Then $D_t F = 0$ for all $(t, \omega) \in A \times \Omega$ a.e. with respect to the measure $\mu \otimes P$.*

Proof: Suppose that $F = f(W(B_1), \dots, W(B_n))$, where $f \in C_p^\infty(\mathbb{R}^n)$, and $B_i \subset A^c$, $B_i \in \mathcal{B}_0$ for all $i = 1, \dots, n$. Then we have

$$D_t F = \frac{\partial f}{\partial x_i}(W(B_1), \dots, W(B_n)) \mathbf{1}_{B_i}(t),$$

which implies that $D_t F = 0$ for all $(t, \omega) \in A \times \Omega$. Finally in the general case it suffices to consider a sequence F_n as above converging to F in L^2 . $\square$

The next lemma allow us to interpret the operator δ as a stochastic integral with respect to the white noise W .

Lemma 1.5.2 *Let $A \in \mathcal{B}_0$ and let F be a random variable in $L^2(\Omega, \mathcal{F}_{A^c}, P)$. Then, the process $u = F \mathbf{1}_A$ belongs to $\text{Dom } \delta$ and*

$$\delta(F \mathbf{1}_A) = FW(A).$$

Proof: Suppose first that $F \in \mathbb{D}^{1,2}$. Then, using (1.9) and Lemma 1.5.2 we obtain

$$\delta(F \mathbf{1}_A) = FW(A) - \int_T D_t F \mathbf{1}_A(t) \mu(dt) = FW(A).$$

The general case follows by a limit argument. $\square$

The following lemma provides additional examples of processes in the domain of δ which are not smooth.

Lemma 1.5.3 *Let $A \in \mathcal{B}_0$. Let $\{u(x), x \in \mathbb{R}^m\}$ be an $\mathcal{F}_{A^c}$ -measurable random field with continuously differentiable paths such that*

$$E \left(\sup_{|x| \leq K} |u'(x)|^4 \right) < \infty \quad \text{and} \quad E(|u(0)|^2) < \infty,$$

for every $K > 0$. Let F be a bounded random variable in the space $\mathbb{D}^{1,4}$. Then $u(F) \mathbf{1}_A$ belongs to the domain of δ and

$$\delta(u(F) \mathbf{1}_A) = u(F)W(A) - \sum_{i=1}^m (\partial_i u)(F) \int_A D_t F^i \mu(dt).$$

Idea of the proof: Approximate $u(F)$ by the convolution of $u(x)$ with an approximation of the identity. $\square$

Consider the following space of elementary processes:

$$\mathcal{R}_0 = \left\{ \sum_{i=1}^n F_i \mathbf{1}_{A_i}, A_i \in \mathcal{B}_0, F_i \in L^2(\Omega, \mathcal{F}_{A_i^c}, P) \right\}.$$

By Lemma 1.5.2 we have that $\mathcal{R}_0 \subset \text{Dom } \delta$ and

$$\delta\left(\sum_{i=1}^n F_i \mathbf{1}_{A_i}\right) = \sum_{i=1}^n F_i W(A_i).$$

As a consequence we deduce the following result:

Proposition 1.5.3 *Let $\mathcal{S}_0 \subset \mathcal{R}_0$ be a subspace such that for all $u \in \mathcal{S}_0$ the isometry property $E[\delta(u)^2] = E(\|u\|_H^2)$ holds. Then $\overline{\mathcal{S}_0}^{\|\cdot\|^2} \subset \text{Dom } \delta$, and the isometry property still holds in the closure of $\mathcal{S}_0$.*

Notice that the isometry property holds for a process u of the form $u = F \mathbf{1}_A \in \mathcal{R}_0$, but, in general, it is not true for a general element u in $\mathcal{R}_0$. Particular examples of subspaces of $\mathcal{R}_0$ where the isometry holds are the following:

Example 1:

Suppose that $W = \{W(t), t \in [0, 1]\}$ is a d -dimensional Wiener process. Set

$$\begin{aligned} \mathcal{S}_0 &= \{u : u(t) = \sum_{i=0}^{n-1} F_i \mathbf{1}_{(t_i, t_{i+1}]}(t), F_i \in L^2(\Omega, \mathcal{F}_{[0, t_i]}, P; \mathbb{R}^d), \\ &\quad 0 = t_0 < \dots < t_n = 1\}. \end{aligned}$$

Then $\overline{\mathcal{S}_0}^{\|\cdot\|^2}$ is the space L_a^2 of d -dimensional square integrable and adapted processes. Therefore, $L_a^2 \subset \text{Dom } \delta$ and $\delta(u)$ coincides with the Itô stochastic integral on this space.

If we take

$$\begin{aligned} \mathcal{S}_1 &= \{u : u(t) = \sum_{i=0}^{n-1} F_i \mathbf{1}_{(t_i, t_{i+1}]}(t), F_i \in L^2(\Omega, \mathcal{F}_{[t_{i+1}, 1]}, P; \mathbb{R}^d), \\ &\quad 0 = t_0 < \dots < t_n = 1\}, \end{aligned}$$

then $\overline{\mathcal{S}_1}^{\|\cdot\|^2}$ is the space L_b^2 of d -dimensional backward adapted processes and $\delta(u)$ coincides with the backward Itô integral on this space.

Example 2:

We will denote by W a white noise on $[0, 1]^2$ with intensity equal to the Lebesgue measure. Then $\{W(s, t) = W([0, s] \times [0, t]), s, t, \in [0, 1]\}$ is a two-parameter Wiener process, that is, a zero mean Gaussian process with covariance given by

$$E[W(s_1, t_1)W(s_2, t_2)] = (s_1 \wedge s_2)(t_1 \wedge t_2).$$

Set

$$\begin{aligned} \mathcal{S}_2 &= \{u : u(s, t) = \sum_{i,j=0}^{n-1} F_{i,j} \mathbf{1}_{(s_i, s_{i+1}] \times (t_j, t_{j+1}]}(s, t), \\ &\quad F_{i,j} \in L^2(\Omega, \mathcal{F}_{[0, s_i] \times [0, 1]}, P), 0 = t_0 < \dots < t_n = 1, 0 = s_0 < \dots < s_n = 1\}. \end{aligned}$$

Then $\overline{\mathcal{S}_2}^{\|\cdot\|^2}$ is the space $L_{1,a}^2$ of square integrable and 1-adapted processes. Therefore, $L_{1,a}^2 \subset \text{Dom } \delta$ and $\delta(u)$ coincides with the Itô stochastic integral on this space.

1.6 Stochastic integral representation of random variables

Suppose that $W = \{W(t), t \in [0, 1]\}$ is a one-dimensional Brownian motion. We know that any square integrable random variable F , measurable with respect to W , can be written as

$$F = E(F) + \int_0^1 \phi(t) dW_t,$$

where the process ϕ belongs to L_a^2 . When the variable F belongs to the space $\mathbb{D}^{1,2}$, it turns out that the process ϕ can be identified as the optional projection of the derivative of F . This is called the Clark-Ocone representation formula (see [24], [83]):

Proposition 1.6.1 *Let $F \in \mathbb{D}^{1,2}$, and suppose that W is a one-dimensional Brownian motion. Then*

$$F = E(F) + \int_0^1 E(D_t F | \mathcal{F}_t) dW_t. \quad (1.22)$$

In order to proof this stochastic integral representation, we will make use of the Wiener chaos expansions, and we need the following technical result:

Lemma 1.6.1 *Let W be a white noise on $(T, \mathcal{B}, \mu)$. Suppose that F is a square integrable random variable with the representation $F = \sum_{n=0}^{\infty} I_n(f_n)$. Let $A \in \mathcal{B}$. Then*

$$E(F | \mathcal{F}_A) = \sum_{n=0}^{\infty} I_n(f_n \mathbf{1}_A^{\otimes n}). \quad (1.23)$$

Proof: It suffices to assume that $F = I_n(f_n)$, where f_n is a square integrable and symmetric kernel. Also, by linearity and density we can assume that the kernel f_n is of the form $\mathbf{1}_{B_1 \times \dots \times B_n}$, where $B_1, \dots, B_n$ are mutually disjoint sets of finite measure. In this case we have

$$\begin{aligned} E(F | \mathcal{F}_A) &= E(W(B_1) \cdots W(B_n) | \mathcal{F}_A) \\ &= E\left(\prod_{i=1}^n (W(B_i \cap A) + W(B_i \cap A^c)) | \mathcal{F}_A\right) \\ &= I_n(\mathbf{1}_{(B_1 \cap A) \times \dots \times (B_n \cap A)}). \end{aligned}$$

□

Proof of Proposition 1.6.1: Suppose that $F = \sum_{n=0}^{\infty} I_n(f_n)$. Using (1.19) and Lemma 1.6.1 we deduce

$$\begin{aligned} E(D_t F | \mathcal{F}_t) &= \sum_{n=1}^{\infty} n E(I_{n-1}(f_n(\cdot, t)) | \mathcal{F}_t) \\ &= \sum_{n=1}^{\infty} n I_{n-1}\left(f_n(t_1, \dots, t_{n-1}, t) \mathbf{1}_{\{t_1 \vee \dots \vee t_{n-1} \leq t\}}\right). \end{aligned}$$

Set $\phi_t = E(D_t F | \mathcal{F}_t)$. We can compute $\delta(\phi)$ using the above expression for ϕ and (1.21), and we obtain

$$\delta(\phi) = \sum_{n=1}^{\infty} I_n(f_n) = F - E(F),$$

which shows the desired result because $\delta(\phi)$ is equal to the Itô stochastic integral of ϕ . $\square$

Bibliographical notes: The integral representation formula (1.22) can be extended to random variables in $\mathbb{D}^{1,1}$ (cf. [50]) and to elements of $\mathbb{D}^{-\infty}$ (cf. [104]). The iterated divergence operator δ^k is an extension of the multiple stochastic integrals introduced by Hajek and Wong [40] (see [80]).

Chapter 2

Ornstein-Uhlenbeck semigroup and equivalence of norms

In this chapter we introduce the operator L , which is the infinitesimal generator of the Ornstein-Uhlenbeck semigroup, and we show the equivalence between the $\|\cdot\|_{k,p}$ norms defined using the derivative operator and the norms $\|F\|_{k,p} = \|(I-L)(F)^{k/2}\|_p$.

Let $W = \{W(h), h \in H\}$ be a centered Gaussian family associated with a real and separable Hilbert space H , defined on a probability space $(\Omega, \mathcal{F}, P)$. Suppose that the σ -field $\mathcal{F}$ is generated by W . Consider the one-parameter semigroup $\{T_t, t \geq 0\}$ of contraction operators on $L^2(\Omega)$ defined by

$$T_t F = \sum_{n=0}^{\infty} e^{-nt} J_n F, \quad (2.1)$$

where J_n denotes the projection on the n th Wiener chaos. This semigroup is called the Ornstein-Uhlenbeck semigroup.

2.1 Mehler's formula

The following result is known as Mehler's formula.

Proposition 2.1.1 *Let $W' = \{W'(h), h \in H\}$ be an independent copy of W . Then for any $t \geq 0$ and $F \in L^2(\Omega)$ we have*

$$T_t F = E'(F(e^{-t}W + \sqrt{1-e^{-2t}}W')), \quad (2.2)$$

where E' denotes the mathematical expectation with respect to W' .

Note that the right-hand side of (2.2) is well defined because

$$\{e^{-t}W(h) + \sqrt{1-e^{-2t}}W'(h), h \in H\}$$

is a centered Gaussian family with the same covariance as W .

Proof: Both T_t and the right-hand side of (2.2) give rise to linear contraction operators on $L^2(\Omega)$. Thus, it suffices to show (2.2) when $F = \exp(\lambda W(h) - \frac{1}{2}\lambda^2)$,

where $h \in H$ is an element of norm one and $\lambda \in \mathbb{R}$, We have

$$\begin{aligned} E' \left(\exp \left(e^{-t} \lambda W(h) + \sqrt{1 - e^{-2t}} \lambda W'(h) - \frac{1}{2} \lambda^2 \right) \right) \\ = \exp \left(e^{-t} \lambda W(h) - \frac{1}{2} e^{-2t} \lambda^2 \right) = \sum_{n=0}^{\infty} e^{-nt} \lambda^n H_n(W(h)) = T_t F, \end{aligned}$$

because $J_n F = \lambda^n H_n(W(h))$. □

Mehler's formula implies that the operator T_t is nonnegative and that it is a contraction on $L^p(\Omega)$ for any $p \geq 1$. Moreover the semigroup $\{T_t, t \geq 0\}$ is continuous in $L^p(\Omega)$ for any $p \geq 1$.

2.2 Hypercontractivity

The operators T_t verify a property (cf. Nelson [73], Neveu [74]) called hypercontractivity, which is stronger than the contractivity in L^p , and which says that

$$\|T_t F\|_{q(t)} \leq \|F\|_p, \quad (2.3)$$

if $q(t) = e^{2t}(p-1) + 1 > p$, $t > 0$ and $F \in L^p(\Omega, \mathcal{F}, P)$.

As a consequence of the hypercontractivity property it can be shown that for any $1 < p < q < \infty$ the norms $\|\cdot\|_p$ and $\|\cdot\|_q$ are equivalent on any Wiener chaos $\mathcal{H}_n$. In fact, let $t > 0$ such that $q = 1 + e^{2t}(p-1)$. Then for every $F \in \mathcal{H}_n$ we have

$$e^{-nt} \|F\|_q = \|T_t F\|_q \leq \|F\|_p. \quad (2.4)$$

In addition, for each $n \geq 1$ the operator J_n is bounded in L^p for any $1 < p < \infty$, and

$$\|J_n F\|_p \leq \begin{cases} (p-1)^{\frac{n}{2}} \|F\|_p & \text{if } p > 2 \\ (p-1)^{-\frac{n}{2}} \|F\|_p & \text{if } p < 2. \end{cases}$$

In fact, suppose first that $p > 2$, and let $t > 0$ be such that $p-1 = e^{2t}$. Using the hypercontractivity property with the exponents p and 2, we obtain

$$\|J_n F\|_p = e^{nt} \|T_t J_n F\|_p \leq e^{nt} \|J_n F\|_2 \leq e^{nt} \|F\|_2 \leq e^{nt} \|F\|_p. \quad (2.5)$$

If $p < 2$, we use a duality argument.

Consider a sequence of real numbers $\{\phi(n), n \geq 0\}$. This sequence determines a linear operator $T_\phi : \mathcal{P} \rightarrow \mathcal{P}$ defined by

$$T_\phi F = \sum_{n=0}^{\infty} \phi(n) J_n F, \quad F \in \mathcal{P}.$$

We remark that the operators T_t are of this type, the corresponding sequence being e^{-nt} . It would be useful to know whether such a multiplication operator is bounded in all L^p , $p > 1$. The following examples provide sufficient conditions for this property to hold.

Examples:

1. Suppose that $\phi(n) = \sum_{k=0}^{\infty} a_k n^{-k}$ for $n \geq N$ and for some $a_k \in \mathbb{R}$ such that $\sum_{k=0}^{\infty} |a_k| N^{-k} < \infty$. Then the operator T_ϕ is bounded in L^p for any $1 < p < \infty$.

Proof: By duality, and taking into account that T_ϕ is selfadjoint, we can assume that $p \geq 2$. Moreover it suffices to show that T_ϕ is bounded in L^p on $\bigoplus_{n=N}^{\infty} \mathcal{H}_n$. Fix $F \in \bigoplus_{n=N}^M \mathcal{H}_n$ with $M \geq N$. We have

$$\|T_\phi F\|_p = \left\| \sum_{n=N}^{\infty} \left(\sum_{k=0}^{\infty} a_k n^{-k} \right) J_n F \right\|_p \leq \sum_{k=0}^{\infty} |a_k| \left\| \sum_{n=N}^{\infty} n^{-k} J_n F \right\|_p.$$

Now, using the equality

$$n^{-k} = \left(\int_0^{\infty} e^{-nt} dt \right)^k = \int_{[0, \infty)^k} e^{-n(t_1 + \dots + t_k)} dt_1 \dots dt_k,$$

we obtain

$$\|T_\phi F\|_p \leq \sum_{k=0}^{\infty} |a_k| \int_{[0, \infty)^k} \|T_{t_1 + \dots + t_k} F\|_p dt_1 \dots dt_k. \quad (2.6)$$

Then it suffices to have an estimate of the L^p -norm of the Ornstein-Uhlenbeck semi-group of the form

$$\|T_t F\|_p \leq K_{N,p} e^{-Nt} \|F\|_p. \quad (2.7)$$

Indeed, from (2.6) and (2.7) we deduce

$$\|T_\phi F\|_p \leq K_{N,p} \sum_{k=0}^{\infty} |a_k| N^{-k} \|F\|_p,$$

which allows us to complete the proof. The estimate (2.7) follows from the hypercontractivity property (2.3). In fact, suppose $p > 2$ (the case $p = 2$ is immediate) and choose t_0 such that $p = e^{2t_0} + 1$. For all $t \geq t_0$ we have

$$\begin{aligned} \|T_t F\|_p^2 &= \|T_{t_0} T_{t-t_0} F\|_p^2 \leq \|T_{t-t_0} F\|_2^2 = \sum_{n=N}^{\infty} e^{-2n(t-t_0)} \|J_n F\|_2^2 \\ &\leq e^{-2N(t-t_0)} \|F\|_2^2 \leq e^{-2N(t-t_0)} \|F\|_p^2, \end{aligned}$$

which implies (2.7) with $K_{N,p} = e^{Nt_0}$. If $t < t_0$ Eq. (2.7) is obvious again with $K_{N,p} = e^{Nt_0}$. $\square$

2. For any $\alpha > 0$ let us consider the sequence $\phi(n) = (1+n)^{-\alpha}$, $n \geq 0$. Then the operator T_ϕ is a contraction in L^p for any $p \geq 1$. This follows immediately from the equation

$$(1+n)^{-\alpha} = \Gamma(\alpha)^{-1} \int_0^{\infty} e^{-(n+1)t} t^{\alpha-1} dt,$$

which implies

$$T_\phi = \Gamma(\alpha)^{-1} \int_0^{\infty} e^{-t} t^{\alpha-1} T_t dt. \quad (2.8)$$

With the notations of the next section we have $T_\phi = (I - L)^{-\alpha}$.

Given a sequence of real numbers $\phi = \{\phi(n), n \geq 0\}$, define $T_{\phi+} = \sum_{n=0}^{\infty} \phi(n+1)J_n$. The following commutativity relationship holds for a multiplication operator T_{ϕ} :

$$D(T_{\phi}F) = T_{\phi+}(DF), \quad (2.9)$$

for any $F \in \mathcal{P}$. In fact, if F belongs to the n th Wiener chaos, we have

$$D(T_{\phi}F) = D(\phi(n)F) = \phi(n)DF = T_{\phi+}(DF).$$

In particular we have $D(T_t F) = e^{-t}T_t(DF)$, and by iteration we obtain

$$D^k(T_t F) = e^{-kt}T_t(D^k F) \quad (2.10)$$

for any $F \in \mathcal{P}$, and $k \geq 1$.

The following two properties hold:

(A) Let $F \in \mathbb{D}^{k,p}$. Then $T_t F$ also belongs to $\mathbb{D}^{k,p}$, and

$$\lim_{t \downarrow 0} \|T_t F - F\|_{k,p} = 0. \quad (2.11)$$

Indeed, from (2.10) it follows that T_t is a contraction operator with respect to any seminorm $\|\cdot\|_{k,p}$. This implies that $T_t F \in \mathbb{D}^{k,p}$. The continuity at the origin with respect to the norm of $\mathbb{D}^{k,p}$ is immediate.

(B) Let $F \in L^p(\Omega)$. Then $T_t F \in \bigcap_{k \geq 1} \mathbb{D}^{k,p}$, and

$$\|D^k(T_t F)\|_p \leq c_{k,p} t^{-k} \|F\|_p, \quad (2.12)$$

for any $k \geq 1$, $t > 0$, and $p > 1$. In particular $T_t F \in \mathbb{D}^{\infty}$ for any random variable F which has moments of all orders.

Proof: It suffices to show (2.12) for any polynomial random variable. Suppose first that $k = 1$. We have from Mehler's formula

$$\begin{aligned} D(T_t F) &= D(E'(F(e^{-t}W + \sqrt{1-e^{-2t}}W'))) \\ &= \frac{e^{-t}}{\sqrt{1-e^{-2t}}} E'(D'(F(e^{-t}W + \sqrt{1-e^{-2t}}W')))). \end{aligned}$$

We recall that for any random variable $G \in \mathbb{D}^{1,2}$ we have

$$\|E(DG)\|_H = \|J_1 G\|_2 = c_p \|J_1 G\|_p \leq c'_p \|G\|_p.$$

Hence,

$$E(\|D(T_t F)\|_H^p) \leq c_p \left(\frac{e^{-t}}{\sqrt{1-e^{-2t}}} \right)^p E(|F|^p),$$

and, by iteration, we deduce (2.12). $\square$

From properties (A) and (B) it follows that the family $\mathcal{P}$ of polynomial random variables is dense in $\mathbb{D}^{k,p}$ for any $k \geq 1$ and $p > 1$. Indeed, we approximate a given random variable F in $\mathbb{D}^{k,p}$ by a polynomial $G \in \mathcal{P}$ in L^p , and, then, we use the inequalities

$$\|F - T_t G\|_{k,p} \leq \|F - T_t F\|_{k,p} + \sum_{j=0}^k c_{j,p} t^{-j} \|F - G\|_p.$$

2.3 Generator of the Ornstein-Uhlenbeck semigroup

The infinitesimal generator of the semigroup T_t in $L^2(\Omega)$ is given by

$$LF = \sum_{n=1}^{\infty} -n J_n F, \quad (2.13)$$

and its domain is

$$\text{Dom } L = \{F \in L^2(\Omega) : \sum_{n=1}^{\infty} n^2 \|J_n F\|_2^2 < \infty\}.$$

From Proposition 1.4.1 it follows that $\text{Dom } L \subset \mathbb{D}^{1,2}$. Actually, $\text{Dom } L = \mathbb{D}^{2,2}$.

The next proposition explains the relationship between the operators D , δ , and L .

Proposition 2.3.1 *Let $F \in L^2(\Omega)$. Then $F \in \text{Dom } L$ if and only if $F \in \mathbb{D}^{1,2}$ and $DF \in \text{Dom } \delta$, and in this case, we have $\delta(DF) = -LF$.*

Proof: For any polynomial random variable G we have, using the duality relationship (1.7) and Proposition 1.4.1,

$$\begin{aligned} E(G\delta(DF)) &= E(\langle DG, DF \rangle_H) = \sum_{n=0}^{\infty} n E(J_n G J_n F) \\ &= E\left(G \sum_{n=0}^{\infty} n J_n F\right) = -E(G LF), \end{aligned}$$

and the result follows easily. $\square$

We are going to show that the operator L behaves as a second-order differential operator.

Proposition 2.3.2 *Suppose that $F = (F^1, \dots, F^m)$ is a random vector whose components belong to $\mathbb{D}^{2,4}$. Let φ be a function in $C^2(\mathbb{R}^m)$ with bounded first and second partial derivatives. Then $\varphi(F) \in \text{Dom } L$, and*

$$L(\varphi(F)) = \sum_{i,j=1}^m \frac{\partial^2 \varphi}{\partial x_i \partial x_j}(F) \langle DF^i, DF^j \rangle_H + \sum_{i=1}^m \frac{\partial \varphi}{\partial x_i}(F) LF^i. \quad (2.14)$$

Proof: Suppose first that F is a smooth random variable of the form

$$F = f(W(h_1), \dots, W(h_n)),$$

with $f \in C_p^\infty(\mathbb{R}^n)$. In this case using Proposition 2.3.1 we obtain

$$\begin{aligned} LF &= -\delta(DF) = -\delta\left(\sum_{i=1}^n \frac{\partial f}{\partial x_i}(W(h_1), \dots, W(h_n)) h_i\right) = \\ &= \sum_{i,j=1}^n \frac{\partial^2 f}{\partial x_i \partial x_j}(W(h_1), \dots, W(h_n)) \langle h_i, h_j \rangle_H \\ &\quad - \sum_{i=1}^n \frac{\partial f}{\partial x_i}(W(h_1), \dots, W(h_n)) W(h_i). \end{aligned}$$

In the general case we approximate F by smooth random variables in the norm $\|\cdot\|_{2,4}$ and use the continuity of the operator L in the norm $\|\cdot\|_{2,2}$. $\square$

For any natural $k \geq 0$ we can define the norm

$$\|F\|_{k,2}^2 = E(|(I - L)^{k/2}(F)|^2) = \sum_{n=0}^{\infty} (n+1)^k E(|J_n F|^2).$$

Notice that this seminorm is equivalent to the norm $\|F\|_{k,2}$, taking into account (1.15). In the next section we generalize this equivalence of norms to the case of $p \neq 2$. Set $C = -\sqrt{-L}$, then $\mathbb{D}^{1,2}$ coincides with the domain of C .

2.4 Meyer's inequalities

We are going to establish Meyer inequalities following Pisier's approach. In order to do this we need some preliminaries. Consider the function $\varphi : [-\frac{\pi}{2}, 0) \cup (0, \frac{\pi}{2}] \rightarrow \mathbb{R}_+$ defined by

$$\varphi(\theta) = \frac{1}{\sqrt{2\pi|\log \cos^2 \theta|}} \text{sign } \theta. \quad (2.15)$$

Notice that when θ is close to zero this function tends to infinity as $\frac{1}{\sqrt{2\pi|\theta|}}$. Moreover, we have, for all $n \geq 0$ (making the changes of variable $\cos \theta = y$ and $y = e^{-\frac{x^2}{2}}$)

$$\int_0^{\frac{\pi}{2}} \frac{\sin \theta \cos^n \theta}{\sqrt{\pi|\log \cos^2 \theta|}} d\theta = \frac{1}{\sqrt{2(n+1)}}. \quad (2.16)$$

Suppose that $\{W'(h), h \in H\}$ is an independent copy of the Gaussian family $\{W(h), h \in H\}$. For any $\theta \in \mathbb{R}$, $F \in L^0(\Omega, \mathcal{F}, P)$ we set

$$R_\theta F = F(W \cos \theta + W' \sin \theta).$$

With these notations we can write the following expression for the operator $D(-C)^{-1}$.

Lemma 2.4.1 *For every $F \in \mathcal{P}$ such that $E(F) = 0$ we have*

$$D(-C)^{-1} F = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} E'(D'(R_\theta F)) \varphi(\theta) d\theta. \quad (2.17)$$

Proof: Suppose that $F = p(W(h_1), \dots, W(h_n))$, where $h_1, \dots, h_n \in H$ and p is a polynomial in n variables. For any $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$ we have

$$R_\theta F = p(W(h_1) \cos \theta + W'(h_1) \sin \theta, \dots, W(h_n) \cos \theta + W'(h_n) \sin \theta),$$

and therefore

$$\begin{aligned} D'(R_\theta F) &= \sum_{i=1}^n \frac{\partial p}{\partial x_i} (W(h_1) \cos \theta + W'(h_1) \sin \theta, \\ &\quad \dots, W(h_n) \cos \theta + W'(h_n) \sin \theta) \sin \theta h_i = \sin \theta R_\theta (DF). \end{aligned}$$

Consequently, using (2.2) we obtain

$$E'(D'(R_\theta F)) = \sin \theta E'(R_\theta(DF)) = \sin \theta T_t(DF),$$

where $t \geq 0$ is such that $\cos \theta = e^{-t}$. This implies

$$E'(D'(R_\theta F)) = \sum_{n=0}^{\infty} \sin \theta (\cos \theta)^n J_n(DF).$$

Note that since F is a polynomial random variable the above series is actually the sum of a finite number of terms. By (2.16) the right-hand side of (2.17) can be written as

$$\sum_{n=0}^{\infty} \left(\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin \theta (\cos \theta)^n \varphi(\theta) d\theta \right) J_n(DF) = \sum_{n=0}^{\infty} \frac{1}{\sqrt{n+1}} J_n(DF).$$

Finally, applying the commutativity relationship (2.9) to the multiplication operator defined by the sequence $\phi(n) = \frac{1}{\sqrt{n}}$, $n \geq 1$, $\phi(0) = 0$, we get

$$T_{\phi+} DF = DT_{\phi} F = D(-C)^{-1} F,$$

and the proof of the lemma is complete. $\square$

Now with the help of the preceding equation we can show that the operator DC^{-1} is bounded from $L^p(\Omega)$ into $L^p(\Omega; H)$ for any $p > 1$. This property will be proved using the boundedness in L^p of the Hilbert transform. We recall that the Hilbert transform of a function $f \in C_0^\infty(\mathbb{R})$ is defined by

$$Hf(x) = \int_{\mathbb{R}} \frac{f(x+t) - f(x-t)}{t} dt.$$

The transformation H is bounded in $L^p(\mathbb{R})$ for any $p > 1$ (see Dunford and Schwarz [27], Theorem XI.7.8). Henceforth c_p and C_p denote generic constants depending only on p , which can be different from one formula to another.

Proposition 2.4.1 *Let $p > 1$. There exists a finite constant $c_p > 0$ such that for any $F \in \mathcal{P}$ with $E(F) = 0$ we have*

$$\|DC^{-1}F\|_p \leq c_p \|F\|_p.$$

Proof: Using (2.17) we can write

$$\begin{aligned} E(\|DC^{-1}F\|_H^p) &= E\left(\left\|\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} E'(D'(R_\theta F))\varphi(\theta)d\theta\right\|_H^p\right) \\ &= \alpha_p^{-1} EE'\left(\left|W'\left(\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} E'(D'(R_\theta F))\varphi(\theta)d\theta\right)\right|^p\right), \end{aligned}$$

where $\alpha_p = E(|\xi|^p)$ with ξ an $N(0,1)$ random variable. We recall that for any $G \in L^2(\Omega', \mathcal{F}', P')$ which belongs to the domain of D' the Gaussian random variable $W'(E'(D'G))$ is equal to the projection $J_1'G$ of G on the first Wiener chaos. Therefore, we obtain that

$$E(\|DC^{-1}F\|_H^p) = \alpha_p^{-1} EE'\left(\left|\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} J_1' R_\theta F \varphi(\theta) d\theta\right|^p\right).$$

If $g : [-\frac{\pi}{2}, \frac{\pi}{2}] \rightarrow \mathbb{B}$ is a Lipschitz function with values in some separable Banach space $\mathbb{B}$ the product $\varphi(\theta)g(\theta)$ does not belong to $L^1([-\frac{\pi}{2}, \frac{\pi}{2}]; \mathbb{B})$ unless $g(0) = 0$. Nevertheless we can define the integral of the product φg in the following way

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \varphi(\theta)g(\theta)d\theta = \int_0^{\frac{\pi}{2}} \varphi(\theta)[g(\theta) - g(-\theta)]d\theta.$$

Notice that $J'_1 R_\theta F$ vanishes at $\theta = 0$ but this is no longer true for $R_\theta F$. Taking into account this remark we can write

$$\begin{aligned} E \left(\|DC^{-1}F\|_H^p \right) &= \alpha_p^{-1} EE' \left(\left| J'_1 \left(\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} R_\theta F \varphi(\theta) d\theta \right) \right|^p \right) \\ &\leq c_p EE' \left(\left| \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} R_\theta F \varphi(\theta) d\theta \right|^p \right), \end{aligned}$$

for some constant $c_p > 0$. For any $\xi \in \mathbb{R}$ we define the process

$$\overline{W}_\xi(h) = (W(h) \cos \xi + W'(h) \sin \xi, -W(h) \sin \xi + W'(h) \cos \xi).$$

The law of this process is the same as that of $\{(W(h), W'(h)), h \in H\}$. On the other hand, $\overline{R}_\xi R_\theta F = R_{\xi+\theta} F$, where we set

$$\overline{R}_\xi G((W(h_1), W'(h_1)), \dots, (W(h_n), W'(h_n))) = G(\overline{W}_\xi(h_1), \dots, \overline{W}_\xi(h_n)).$$

Therefore, we get

$$\begin{aligned} \left\| \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} R_\theta F \varphi(\theta) d\theta \right\|_p &= \left\| \overline{R}_\xi \left[\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} R_\theta F \varphi(\theta) d\theta \right] \right\|_p \\ &= \left\| \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} R_{\xi+\theta} F \varphi(\theta) d\theta \right\|_p, \end{aligned}$$

where $\|\cdot\|_p$ denotes the L^p norm with respect to $P \times P'$. Integration with respect to ξ yields

$$E \left(\|DC^{-1}F\|_H^p \right) \leq c_p EE' \left(\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left| \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} R_{\xi+\theta} F \varphi(\theta) d\theta \right|^p d\xi \right). \quad (2.18)$$

Furthermore, there exists a bounded continuous function $\tilde{\varphi}$ and a constant $c > 0$ such that

$$\varphi(\theta) = \tilde{\varphi}(\theta) + \frac{c}{\theta},$$

on $[-\frac{\pi}{2}, \frac{\pi}{2}]$. Consequently, using the L^p boundedness of the Hilbert transform, we see that the right-hand side of (2.18) is dominated up to a constant by

$$EE' \left(\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} |R_\theta F|^p d\theta \right) = \pi \|F\|_p^p.$$

In fact, the term $\tilde{\varphi}(\theta)$ is easy to treat. On the other hand, to handle the term $\frac{1}{\theta}$ we write

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left| \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{R_{\xi+\theta} F}{\theta} d\theta \right|^p d\xi = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left| \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\tilde{R}_{\xi+\theta} F}{\theta} d\theta \right|^p d\xi$$

$$\begin{aligned}
&\leq c_p \left(\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left| \int_{\mathbb{R}} \frac{\tilde{R}_{\xi+\theta} F}{\theta} d\theta \right|^p d\xi + \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left| \int_{[-\frac{\pi}{2}, \frac{\pi}{2}]^c} \frac{\tilde{R}_{\xi+\theta} F}{\theta} d\theta \right|^p d\xi \right) \\
&\leq c'_p \left(\int_{\mathbb{R}} |H(\tilde{R} \cdot F)(\xi)|^p d\xi + \left(\frac{2}{\pi} \right)^p \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left| \int_{\mathbb{R}} \tilde{R}_{\xi+\theta} F d\theta \right|^p d\xi \right) \\
&\leq c''_p \left(\int_{\mathbb{R}} |\tilde{R}_{\xi} F|^p d\xi \right) \leq c'''_p \left(\int_{-\frac{3\pi}{2}}^{\frac{3\pi}{2}} |R_{\xi} F|^p d\xi \right),
\end{aligned}$$

where $\tilde{R}_{\theta} F = \psi(\theta) R_{\theta} F$, and ψ is a smooth function with support included in $[-\frac{3\pi}{2}, \frac{3\pi}{2}]$ such that $0 \leq \psi \leq 1$ and $\psi(\theta) = 1$ if $\theta \in [-\pi, \pi]$. Hence,

$$EE' \left(\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left| \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{R_{\xi+\theta} F}{\theta} d\theta \right|^p d\xi \right) \leq c_p E(|F|^p).$$

□

Proposition 2.4.2 *Let $p > 1$. Then there exist positive and finite constants c_p and C_p such that for any $F \in \mathcal{P}$ we have*

$$c_p \|DF\|_{L^p(\Omega; H)} \leq \|CF\|_p \leq C_p \|DF\|_{L^p(\Omega; H)}. \quad (2.19)$$

Proof: Set $G = CF$. Applying Proposition 2.4.1 to the random variable $G = CF$ we have

$$\|DF\|_{L^p(\Omega; H)} = \|DC^{-1}G\|_{L^p(\Omega; H)} \leq c_p \|G\|_p = c_p \|CF\|_p,$$

which shows the left inequality. The right inequality is proved by means of a duality argument. Let $F, G \in \mathcal{P}$. Set $\tilde{G} = C^{-1}(I - J_0)(G)$. Let q be such that $\frac{1}{p} + \frac{1}{q} = 1$. Then we have

$$\begin{aligned}
|E(GCF)| &= |E((I - J_0)(G)CF)| = |E(CFC\tilde{G})| = |E(\langle DF, D\tilde{G} \rangle_H)| \\
&\leq \|DF\|_{L^p(\Omega; H)} \|D\tilde{G}\|_{L^q(\Omega; H)} \leq c_q \|DF\|_{L^p(\Omega; H)} \|C\tilde{G}\|_q \\
&= c_q \|DF\|_{L^p(\Omega; H)} \|(I - J_0)(G)\|_q \leq c'_q \|DF\|_{L^p(\Omega; H)} \|G\|_q.
\end{aligned}$$

Taking the supremum with respect to $G \in \mathcal{P}$ with $\|G\|_q \leq 1$, we obtain

$$\|CF\|_p \leq c'_q \|DF\|_{L^p(\Omega; H)}.$$

□

Now we can state Meyer's inequalities in the general case.

Theorem 2.4.1 *For any $p > 1$ and any integer $k \geq 1$ there exist positive and finite constants $c_{p,k}$ and $C_{p,k}$ such that for any $F \in \mathcal{P}$,*

$$c_{p,k} E \left(\|D^k F\|_{H^{\otimes k}}^p \right) \leq E \left(|C^k F|^p \right) \leq C_{p,k} \left[E \left(\|D^k F\|_{H^{\otimes k}}^p \right) + E(|F|^p) \right]. \quad (2.20)$$

Proof: The proof can be done by induction on k . The case $k = 1$ corresponds to Proposition 2.4.2. In order to illustrate the method let us describe the proof of the left-hand side of (2.20) for $k = 2$.

Notice that if $\{\xi_n, n \geq 1\}$ is a family of independent random variables defined on the probability space $([0, 1], \mathcal{B}([0, 1]), \lambda)$ (λ is the Lebesgue measure), with distribution $N(0, 1)$, and $\{a_n, 1 \leq n \leq N\}$ is a sequence of real numbers, we have, for each $p > 1$,

$$\int_{[0,1]} \left| \sum_{n=1}^N a_n \xi_n(t) \right|^p dt = A_p \left(\sum_{n=1}^N a_n^2 \right)^{\frac{p}{2}}, \quad (2.21)$$

where $A_p = \int_{\mathbb{R}} |x|^p \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx$.

Suppose that $F = p(W(h_1), \dots, W(h_n))$, where the h_i 's are orthonormal elements of H . We fix a complete orthonormal system $\{e_i, i \geq 1\}$ in H which contains the h_i 's.

With these notations, using Eq. (2.21) and Proposition 2.4.2, we can write

$$\begin{aligned} E(\|D^2 F\|_{H \otimes H}^p) &= E \left(\left\| \sum_{i,j=1}^{\infty} \langle D^2 F, e_i \otimes e_j \rangle_H^2 \right\|_H^{\frac{p}{2}} \right) \\ &\leq c_p \int_0^1 \int_0^1 E \left(\left\| \sum_{i,j=1}^{\infty} \langle D^2 F, e_i \otimes e_j \rangle_H \xi_j(t) \xi_i(s) \right\|_H^p \right) dt ds \\ &\leq c'_p \int_0^1 E \left(\left\| \sum_{i=1}^{\infty} \left\langle D \left(\sum_{j=1}^{\infty} \langle DF, e_j \rangle_H \xi_j(t) \right), e_i \right\rangle_H^2 \right\|_H^{\frac{p}{2}} \right) dt \\ &= c'_p \int_0^1 E \left(\left\| D \left(\sum_{j=1}^{\infty} \langle DF, e_j \rangle_H \xi_j(t) \right) \right\|_H^p \right) dt \\ &\leq c''_p \int_0^1 E \left(\left\| C \left(\sum_{j=1}^{\infty} \langle DF, e_j \rangle_H \xi_j(t) \right) \right\|_H^p \right) dt \\ &\leq c'''_p E \left(\left\| \sum_{j=1}^{\infty} \langle CDF, e_j \rangle_H^2 \right\|_H^{\frac{p}{2}} \right) = c'''_p E(\|CDF\|_H^p). \end{aligned}$$

Consider the operator $R = \sum_{n=1}^{\infty} \sqrt{1 - \frac{1}{n}} J_n$. This operator verifies $CDF = DCRF$ and it is bounded in L^p for all $p > 1$ (see example 1). Hence, we obtain

$$E(\|D^2 F\|_{H \otimes H}^p) \leq c_p E(\|DCRF\|_H^p) \leq c'_p E(|C^2 R F|^p) \leq c''_p E(|C^2 F|^p).$$

□

Khintchine's inequalities also allows us to establish the inequalities (2.20) for polynomial random variable taking values in a separable Hilbert space V . Let us now introduce a continuous family of Sobolev spaces.

For any $p > 1$ and $s \in \mathbb{R}$ we will denote by $\|\cdot\|_{s,p}$ the seminorm

$$\|F\|_{s,p} = \|(I - L)^{\frac{s}{2}} F\|_p,$$

where $F \in \mathcal{P}$ is a polynomial random variable. Note that $(I - L)^{\frac{s}{2}} = \sum_{n=0}^{\infty} (1+n)^{\frac{s}{2}} J_n$. These seminorms verify the following properties:

- (i) $\|F\|_{s,p}$ is increasing in both coordinates s and p . The monotonicity in p is clear and in s follows from the fact that the operators $(I - L)^\alpha$ are contractions in L^p , for all $\alpha < 0$, $p > 1$ (see (2.8)).
- (ii) The seminorms $\|\cdot\|_{s,p}$ are compatible, in the sense that for any sequence F_n in $\mathcal{P}$ converging to zero in the norm $\|\cdot\|_{s,p}$, and being a Cauchy sequence in any other norm $\|\cdot\|_{s',p'}$ they also converge to zero in the norm $\|\cdot\|_{s',p'}$.

We define $\mathbb{D}^{s,p}$ as the completion of $\mathcal{P}$ with respect to this norm.

Remarks:

1. $\|F\|_{0,p} = \|F\|_{0,p} = \|F\|_p$, and $\mathbb{D}^{0,p} = L^p(\Omega)$. For $k = 1, 2, \dots$ the seminorms $\|\cdot\|_{k,p}$ and $\|\cdot\|_{k,p}$ are equivalent due to Meyer's inequalities. In fact, we have

$$\|F\|_{k,p} = \|(I - L)^{\frac{k}{2}} F\|_p \leq |E(F)| + \|R(-L)^{\frac{k}{2}} F\|_p,$$

where $R = \sum_{n=1}^{\infty} \left(\frac{n+1}{n}\right)^{\frac{k}{2}} J_n$. Notice that $R = T_\phi$ with

$$\phi(n) = \begin{cases} 0, & n = 0 \\ h(\frac{1}{n}), & n \geq 1 \end{cases}$$

where $h(x) = (1+x)^{\frac{k}{2}}$ is analytic in a neighbourhood of 0. Therefore, the operator R is bounded in L^p for all $p > 1$. Hence,

$$\begin{aligned} \|F\|_{k,p} &\leq c_p(\|F\|_p + \|(-L)^{\frac{k}{2}} F\|_p) \leq c'_p(\|F\|_p + \|D^k F\|_{L^p(\Omega; H^{\otimes k})}) \\ &\leq c''_p \|F\|_{k,p}. \end{aligned}$$

In a similar way one can show the converse inequality.

Thus, the Sobolev spaces $\mathbb{D}^{k,p}$ coincide with those defined by means of the derivative operator.

2. From property (ii) we have $\mathbb{D}^{s,p} \subset \mathbb{D}^{s',p'}$ if $p' \leq p$ and $s' \leq s$.
3. For $s > 0$ the operator $(I - L)^{-\frac{s}{2}}$ is an isometric isomorphism (in the norm $\|\cdot\|_{s,p}$) between L^p and $\mathbb{D}^{s,p}$ and between $\mathbb{D}^{-s,p}$ and L^p for all $p > 1$. As a consequence, the dual of $\mathbb{D}^{s,p}$ is $\mathbb{D}^{-s,q}$ where $\frac{1}{p} + \frac{1}{q} = 1$. If $s < 0$, the elements of $\mathbb{D}^{s,p}$ may not be ordinary random variables and they are interpreted as distributions on the Gaussian space. Set $\mathbb{D}^{-\infty} = \bigcup_{s,p} \mathbb{D}^{s,p}$. The space $\mathbb{D}^{-\infty}$ is the dual of the space $\mathbb{D}^{\infty}$ which is a countably normed space.
4. Suppose that V is a real separable Hilbert space. We can define the Sobolev spaces $\mathbb{D}^{s,p}(V)$ of V -valued functionals as the completion of the class $\mathcal{P}_V$ of V -valued polynomial random variable with respect to the seminorm $\|F\|_{s,p,V}$ defined in the same way as before. The above properties are still true for V -valued functionals.

The main application of Meyer's inequalities is the following continuity theorem.

Proposition 2.4.3 *Let V be a real separable Hilbert space. For every $p > 1$ and $s \in \mathbb{R}$, the operator D is continuous from $\mathbb{D}^{s,p}(V)$ to $\mathbb{D}^{s-1,p}(V \otimes H)$ and the operator δ (defined as the adjoint of D) is continuous from $\mathbb{D}^{s,p}(V \otimes H)$ to $\mathbb{D}^{s-1,p}(V)$.*

Proof: We will only proof the continuity of D for $V = \mathbb{R}$. The continuity of the adjoint operator follows by duality. For any $F \in \mathcal{P}$ we have

$$(I - L)^{\frac{s}{2}} DF = DR(I - L)^{\frac{s}{2}} F,$$

where $R = \sum_{n=1}^{\infty} \left(\frac{n}{n+1}\right)^{\frac{s}{2}} J_n$. Notice that $R = T_\phi$ with

$$\phi(n) = \begin{cases} 0, & n = 0 \\ h(\frac{1}{n}), & n \geq 1 \end{cases}$$

where $h(x) = (\frac{1}{1+x})^{\frac{s}{2}}$ is analytic near 0. Therefore, the operator R is bounded in L^p for all $p > 1$, and we obtain

$$\begin{aligned} \|(I - L)^{\frac{s}{2}} DF\|_p &= \|DR(I - L)^{\frac{s}{2}} F\|_p \leq c_p \|(I - L)^{\frac{1}{2}} R(I - L)^{\frac{s}{2}} F\|_p \\ &= c_p \|R(I - L)^{\frac{s+1}{2}} F\|_p \leq c'_p \|(I - L)^{\frac{s+1}{2}} F\|_p = c'_p \|F\|_{s+1, p}, \end{aligned}$$

which implies the desired continuity. $\square$

The following inequality is also an immediate consequence of Meyer's inequalities.

Proposition 2.4.4 *Let u be an element of $\mathbb{D}^{1,p}(H)$, $p > 1$. Then we have*

$$\|\delta(u)\|_p \leq c_p \left(\|J_0 u\|_H + \|Du\|_{L^p(\Omega; H \otimes H)} \right). \quad (2.22)$$

Proof: From Proposition 2.4.3 we know that the operator δ is continuous from $\mathbb{D}^{1,p}(H)$ into $L^p(\Omega)$. This implies that

$$\|\delta(u)\|_p \leq c_p \left(\|u\|_{L^p(\Omega; H)} + \|Du\|_{L^p(\Omega; H \otimes H)} \right).$$

On the other hand, we have

$$\|u\|_{L^p(\Omega; H)} \leq \|J_0 u\|_H + \|u - J_0 u\|_{L^p(\Omega; H)},$$

and

$$\begin{aligned} \|u - J_0 u\|_{L^p(\Omega; H)} &= \|(I - L)^{-\frac{1}{2}} R C u\|_{L^p(\Omega; H)} \leq c_p \|C u\|_{L^p(\Omega; H)} \\ &\leq c'_p \|D u\|_{L^p(\Omega; H \otimes H)}, \end{aligned}$$

where $R = \sum_{n=1}^{\infty} \left(1 + \frac{1}{n}\right)^{\frac{1}{2}} J_n$. $\square$

The next lemmas are often useful.

Lemma 2.4.2 *Let $\{F_n, n \geq 1\}$ be a sequence of random variables converging to F in L^p , for some $p > 1$. Suppose that $\sup_n \|F_n\|_{s,p} < \infty$ for some $s > 0$. Then $F \in \mathbb{D}^{s,p}$.*

Proof: We know that

$$\sup_n \|(I - L)^{\frac{s}{2}} F_n\|_p < \infty.$$

Therefore, there exists a subsequence $F_{n(i)}$ such that $(I - L)^{\frac{s}{2}} F_{n(i)}$ converges weakly in $\sigma(L^p, L^q)$ (where q is the conjugate of p) to some element G . Then, for any polynomial random variable R we have

$$\begin{aligned} E[F(I - L)^{\frac{s}{2}} R] &= \lim_i E[F_{n(i)}(I - L)^{\frac{s}{2}} R] \\ &= \lim_i E[(I - L)^{\frac{s}{2}} F_{n(i)} R] = E[GR]. \end{aligned}$$

Thus, $F = (I - L)^{-\frac{s}{2}} G$, and this implies that $F \in \mathbb{D}^{s,p}$. $\square$

Lemma 2.4.3 *Let $\eta \in \mathbb{D}^\infty$ be such that $\langle \eta, F \rangle \geq 0$ for any nonnegative smooth random variable $F \in \mathcal{S}$. Fix a complete orthonormal system $\{e_i, i \geq 1\}$ in H , and let $\mathcal{G}$ be the σ -field generated by the random variables $\{W(e_i), i \geq 1\}$. Then there exists a finite measure μ_η on the σ -field $\mathcal{G}$ such that*

$$\int_{\Omega} F d\mu_\eta = \langle \eta, F \rangle, \quad (2.23)$$

for any random variable F of the form $F = f(W(e_1), \dots, W(e_n))$, $n \geq 1$, $f \in C_p^\infty(\mathbb{R}^n)$. In particular $\mu_\eta(\Omega) = \langle \eta, 1 \rangle$.

Proof: The proof will be done in two steps:

Step 1: Consider the increasing sequence of σ -fields $\mathcal{F}_n = \sigma\{W(e_1), \dots, W(e_n)\}$. We claim that there exists a measure ν_n on the σ -field $\mathcal{F}_n$ such that (2.23) holds for any random variable F of the form $F = f(W(e_1), \dots, W(e_n))$, where $f \in C_p^\infty(\mathbb{R}^n)$. Indeed, consider the function $\varphi : \mathbb{R}^n \rightarrow \mathbb{C}$ defined by

$$\varphi(t) = \left\langle \eta, \exp\left(i \sum_{j=1}^n t^j W(e_j)\right) \right\rangle.$$

This mapping is nonnegative definite and continuous, because for all $t_1, \dots, t_N \in \mathbb{R}^n$, $c_1, \dots, c_N \in \mathbb{C}$ we have

$$\sum_{l,k=1}^N \varphi(t_l - t_k) c_l \bar{c}_k = \left\langle \eta, \left| \sum_{k=1}^N c_k \exp\left(i \sum_{j=1}^n t_k^j W(e_j)\right) \right|^2 \right\rangle \geq 0.$$

Therefore, there exists a finite measure $\bar{\nu}_n$ on $\mathbb{R}^n$ which has moments of all orders and such that

$$\int_{\mathbb{R}^n} f(x) \bar{\nu}_n(dx) = \langle \eta, f(W(e_1), \dots, W(e_n)) \rangle,$$

for any function f in $C_p^\infty(\mathbb{R}^n)$. This implies the existence of a finite measure μ_η on $\mathcal{F}_n$ such that (2.23) holds for any F of the form $f(W(e_1), \dots, W(e_n))$ with $f \in C_p^\infty(\mathbb{R}^n)$.

Step 2: It suffices to show that μ_η can be extended to a measure on $\mathcal{G}$. Consider the random variable $G = \sum_{n=1}^\infty n^{-2} W(e_n)^2$. This variable belongs to $\mathbb{D}^\infty$. Let ψ be a smooth function such that $\psi(x) = 0$ if $x \leq 0$, $\psi(x) = 1$ if $x \geq 1$, and $0 \leq \psi \leq 1$. For any $a > 1$, set $\varphi_a(x) = \psi(|x| - (a-1))^+$. For any random variable F of the form $f(W(e_1), \dots, W(e_n))$ with $f \in C_p^\infty(\mathbb{R}^n)$ and such that $0 \leq F \leq 1$, we can write

$$\begin{aligned} \int_{\Omega} F d\mu_\eta &= \langle \eta, F \rangle = \langle \eta, F(1 - \varphi_a(G)) \rangle + \langle \eta, F\varphi_a(G) \rangle \\ &\leq \langle \eta, 1 \rangle \|F(1 - \varphi_a(G))\|_\infty + \langle \eta, \varphi_a(G) \rangle. \end{aligned}$$

Hence, the inequality

$$\int_{\Omega} F d\mu_\eta \leq \langle \eta, 1 \rangle \|F(1 - \varphi_a(G))\|_\infty + \langle \eta, \varphi_a(G) \rangle \quad (2.24)$$

holds for any F in the class $\mathcal{L} := \{f(W(e_1), \dots, W(e_n)), n \geq 1, 0 \leq f \leq 1, f \text{ Borel}\}$.

In order to complete the proof of the lemma and taking into account Daniell's theorem, it suffices to show that given a sequence of random variables $\{F_n, n \geq$

$1\} \subset \mathcal{L}$, uniformly bounded by one, such that $0 \leq F_n(\omega) \downarrow 0$ for all $\omega \in \Omega$, then $\int_{\Omega} F_n d\mu_{\eta} \downarrow 0$. We can write

$$\int_{\Omega} F_n d\mu_{\eta} \leq \langle \eta, 1 \rangle \|F_n(1 - \varphi_a(G))\|_{\infty} + \langle \eta, \varphi_a(G) \rangle \quad (2.25)$$

Suppose that $\eta \in \mathbb{D}^{-k,p}$. The second summand in the right-hand side of (2.25) can be estimated as follows:

$$\langle \eta, \varphi_a(G) \rangle = \langle (I - L)^{-\frac{k}{2}} \eta, (I - L)^{\frac{k}{2}} \varphi_a(G) \rangle \leq \|\eta\|_{-k,p} \|(I - L)^{\frac{k}{2}} \varphi_a(G)\|_q,$$

where $\frac{1}{p} + \frac{1}{q} = 1$, and this tends to zero as a tends to infinity. The first summand converges to zero as n tends to infinity due to Dini's theorem and the fact that the set $\{x_n, n \geq 1\} : \sum_{n=1}^{\infty} n^{-2} x_n^2 < a\}$ is compact in ℓ^2 . $\square$

Bibliographical notes: The fact that the positive distributions on the Wiener space are measures has been proved in the context of an abstract Wiener space by Sugita in [103]. Other contributions to this problem are [79], [110], [92] and [82]. Spaces of Banach-valued Wiener functionals have been studied in [29], [64] and [42].

Chapter 3

Application of Malliavin calculus to study probability laws

In this chapter we will discuss the application of the stochastic calculus of variations on a Gaussian space to the study of different properties of probability distributions of functionals of the underlying Gaussian process. The type of properties that we can analyze with the help of the Malliavin calculus include the absolute continuity, smoothness of the density, estimates (uniform and in L^p) of the density, and application of these estimates to deduce Krylov-type inequalities.

3.1 Computation of probability densities

Suppose that $W = \{W(h), h \in H\}$ is a centered Gaussian family associated with a real and separable Hilbert space, defined on a probability space $(\Omega, \mathcal{F}, P)$. We assume that the σ -field $\mathcal{F}$ is generated by W .

Suppose that $F = (F^1, \dots, F^m)$ is a random vector whose components belong to the space $\mathbb{D}_{\text{loc}}^{1,1}$. We associate with F the following random symmetric nonnegative definite matrix:

$$\gamma_F = (\langle DF^i, DF^j \rangle_H)_{1 \leq i, j \leq m}.$$

This matrix is called the *Malliavin matrix* of the random vector F . The following theorem was established by Bouleau and Hirsch [16] and provides a criterion for the absolute continuity of the law of F .

Theorem 3.1.1 *Let $F = (F^1, \dots, F^m)$ be a random vector whose components belong to the space $\mathbb{D}_{\text{loc}}^{1,p}$, $p > 1$, and suppose that the random matrix $\gamma_F = (\langle DF^i, DF^j \rangle)_{1 \leq i, j \leq m}$ is invertible a.s. Then the law of F is absolutely continuous with respect to the Lebesgue measure on $\mathbb{R}^m$.*

In dimension one the nondegeneracy condition required in the previous theorem reduces to $\|DF\|_H > 0$ a.s. The proof of Theorem 3.1.1 is based on the coarea formula. A simple proof in dimension one can be given as follows (see [81]).

Proof of Theorem 3.1.1 for $m = 1$: We can assume that F belongs to the space $\mathbb{D}^{1,p}$ and that $|F| < k$ for some constant k . We have to show that for any measurable function $g : (-k, k) \rightarrow [0, 1]$ such that $\int_{-k}^k g(y) dy = 0$ we have $E(g(F)) = 0$. We can find a sequence of continuously differentiable functions with bounded derivatives

$g^n : (-k, k) \rightarrow [0, 1]$ such that as n tends to infinity $g^n(y)$ converges to $g(y)$ for almost all y with respect to the measure $P \circ F^{-1} + \lambda$, where λ is the Lebesgue measure. Set

$$\psi^n(y) = \int_{-k}^y g^n(x) dx, \quad y \in (-k, k)$$

and

$$\psi(y) = \int_{-k}^y g(x) dx, \quad y \in (-k, k).$$

By the chain rule, $\psi^n(F)$ belongs to the space $\mathbb{D}^{1,p}$ and we have $D[\psi^n(F)] = g^n(F)DF$. We have that $\psi^n(F)$ converges to $\psi(F)$ a.s. as n tends to infinity, because g^n converges to g a.e. with respect to the Lebesgue measure. This convergence also holds in $L^p(\Omega)$ by dominated convergence. On the other hand, $D[\psi^n(F)]$ converges a.s. to $g(F)DF$ because g^n converges to g a.e. with respect to the law of F . Again by dominated convergence, this convergence holds in $L^p(\Omega; H)$. Observe that $\psi(F) = 0$ a.s. Now we use the property that the operator D is closed to deduce that $g(F)DF = 0$ a.s. Consequently, $g(F) = 0$ a.s., which completes the proof of the theorem. $\square$

We remark that the above proof also works with $p = 1$. The following proposition provides an explicit expression for the density of a one-dimensional random variable.

Proposition 3.1.1 *Let F be a random variable in the space $\mathbb{D}^{1,2}$. Suppose that $\frac{DF}{\|DF\|^2}$ belongs to the domain of the operator δ in L^2 . Then the law of F has a continuous and bounded density given by*

$$p(x) = E \left[\mathbf{1}_{\{F > x\}} \delta \left(\frac{DF}{\|DF\|^2} \right) \right]. \quad (3.1)$$

Proof: Let ψ be a nonnegative smooth function with compact support, and set $\varphi(y) = \int_{-\infty}^y \psi(z) dz$, $y \in \mathbb{R}$. We know that $\varphi(F)$ belongs to $\mathbb{D}^{1,2}$, and making the scalar product of its derivative with DF obtains

$$\langle D(\varphi(F)), DF \rangle_H = \psi(F) \|DF\|_H^2.$$

Using the duality relationship between the operators D and δ (see (1.7)), we obtain

$$E[\psi(F)] = E \left[\left\langle D(\varphi(F)), \frac{DF}{\|DF\|_H^2} \right\rangle_H \right] = E \left[\varphi(F) \delta \left(\frac{DF}{\|DF\|_H^2} \right) \right]. \quad (3.2)$$

By an approximation argument Eq. (3.2) holds for $\psi(y) = \mathbf{1}_{[a,b]}(y)$. As a consequence, we apply Fubini's theorem to get

$$\begin{aligned} P(a \leq F \leq b) &= E \left[\left(\int_{-\infty}^F \mathbf{1}_{[a,b]}(x) dx \right) \delta \left(\frac{DF}{\|DF\|_H^2} \right) \right] \\ &= \int_a^b E \left[\mathbf{1}_{\{F > x\}} \delta \left(\frac{DF}{\|DF\|_H^2} \right) \right] dx, \end{aligned}$$

which implies the desired result. $\square$

Proposition 3.1.1 also holds under the hypotheses $F \in \mathbb{D}^{1,1}$, and $\frac{DF}{\|DF\|_H^2} \in \mathbb{D}^{1,p}(H)$ for some $p, p' > 1$. From expression (3.1) we can deduce estimates for the density. Fix p and q such that $\frac{1}{p} + \frac{1}{q} = 1$. By Hölder's inequality we obtain

$$p(x) \leq (P(F > x))^{1/q} \|\delta \left(\frac{DF}{\|DF\|_H^2} \right)\|_p.$$

In the same way and taking into account the relation $E[\delta(\frac{DF}{\|DF\|_H^2})] = 0$ we can deduce the inequality

$$p(x) \leq (P(F < x))^{1/q} \|\delta \left(\frac{DF}{\|DF\|_H^2} \right)\|_p.$$

As a consequence, we get

$$p(x) \leq (P(|F| > |x|))^{1/q} \|\delta \left(\frac{DF}{\|DF\|_H^2} \right)\|_p, \quad (3.3)$$

for all $x \in \mathbb{R}$. Now using the L^p -estimate of the operator δ established in Proposition 2.4.4 we obtain

$$\|\delta \left(\frac{DF}{\|DF\|_H^2} \right)\|_p \leq c_p \left(\left\| E \left(\frac{DF}{\|DF\|_H^2} \right) \right\|_H + \left\| D \left(\frac{DF}{\|DF\|_H^2} \right) \right\|_{L^p(\Omega; H \otimes H)} \right). \quad (3.4)$$

We have

$$D \left(\frac{DF}{\|DF\|_H^2} \right) = \frac{D^2 F}{\|DF\|_H^2} - 2 \frac{\langle D^2 F, DF \otimes DF \rangle_{H \otimes H}}{\|DF\|_H^4},$$

and, hence,

$$\left\| D \left(\frac{DF}{\|DF\|_H^2} \right) \right\|_{H \otimes H} \leq 3 \frac{\|D^2 F\|_{H \otimes H}}{\|DF\|_H^2}. \quad (3.5)$$

Finally, from the inequalities (3.3), (3.4) and (3.5) we deduce the following result.

Lemma 3.1.1 *Let q, α, β be three positive real numbers such that $\frac{1}{q} + \frac{1}{\alpha} + \frac{1}{\beta} = 1$. Let F be a random variable in the space $\mathbb{D}^{2,\alpha}$, such that $E(\|DF\|_H^{-2\beta}) < \infty$. Then the density $p(x)$ can be estimated as follows:*

$$p(x) \leq c_{q,\alpha,\beta} (P(|F| > |x|))^{1/q} \left(E(\|DF\|_H^{-1}) + \|D^2 F\|_{L^\alpha(\Omega; H \otimes H)} \|\|DF\|_H^{-2}\|_\beta \right). \quad (3.6)$$

Let us apply the preceding lemma to a Brownian martingale.

Example: We will assume that $W = \{W_t, t \in [0, T]\}$ is a Brownian motion, and $H = \{H_t, t \in [0, T]\}$ is an adapted process verifying the following hypotheses:

- (i) $E \int_0^T H_s^2 ds < \infty$, H_t belongs to the space $\mathbb{D}^{2,2}$ for each $t \in [0, T]$, and

$$\lambda := \sup_{s,t \in [0,T]} E(|D_s H_t|^p) + \sup_{r,s \in [0,T]} E\left(\left(\int_0^T |D_{r,s}^2 H_t|^2 dt\right)^{p/2}\right) < \infty,$$

for some $p > 3$.

(ii) $|H_t| \geq \rho > 0$ for some constant ρ .

Consider the martingale $M_t = \int_0^t H_s dW_s$, and denote by $p_t(x)$ the probability density of M_t . Then the following estimate holds

$$p_t(x) \leq \frac{c}{\sqrt{t}} P(|M_t| > |x|)^{\frac{1}{q}}, \quad (3.7)$$

where $q > \frac{p}{p-3}$. The constant c depends on λ , ρ , and p .

Proof of (3.7): We will apply Lemma 3.1.1 to the random variable M_t . We claim that $M_t \in \mathbb{D}^{2,2}$ for each $t \in [0, T]$. In fact, note first that $M_t \in \mathbb{D}^{1,2}$ because the process H belongs to $\mathbb{D}^{2,2}(H)$, due to condition (i), and the operator δ is continuous from $\mathbb{D}^{2,2}(H)$ into $\mathbb{D}^{1,2}$. Furthermore, using (1.10) we get

$$D_s M_t = H_s \mathbf{1}_{\{s \leq t\}} + \delta(D_s H.).$$

For each $s \in [0, T]$, the process $\{D_s H_t, t \in [0, T]\}$ is adapted, square integrable, and it vanishes for $t \leq s$ due to Lemma 1.5.1. Hence it belongs to the domain of δ and $\delta(D_s H.)$ coincides with the Itô stochastic integral $\int_s^t D_s H_r dW_r$. So, we have

$$\begin{aligned} D_s M_t &= H_s + \int_s^t D_s H_r dW_r, \quad s \leq t, \\ D_s M_t &= 0, \quad s > t. \end{aligned} \quad (3.8)$$

Notice that $D_{s_1} H. \in \mathbb{D}^{1,2}(H)$ for all s_1 a.e., the process $D_{s_2} D_{s_1} H.$ belongs to the domain of δ for all s_1, s_2 a.e. (because it is adapted and square integrable), and

$$E \int_0^T |\delta(D_{s_2} D_{s_1} H.)|^2 ds_2 = E \int_0^T \int_0^T |D_{s_2} D_{s_1} H_t|^2 ds_2 dt < \infty.$$

As a consequence, from property (1.10) deduce that for all s_1 a.e. $\delta(D_{s_1} H.)$ belongs to $\mathbb{D}^{1,2}$ and

$$D_{s_2}(\delta(D_{s_1} H.)) = D_{s_1} H_{s_2} + \delta(D_{s_2} D_{s_1} H.).$$

Hence, $M_t \in \mathbb{D}^{2,2}$ and from (3.8) we deduce

$$D_{s_1, s_2}^2 M_t = D_{s_1} H_{s_2} + D_{s_2} H_{s_1} + \int_{s_1 \vee s_2}^t D_{s_1, s_2}^2 H_r dW_r, \quad s_1, s_2 \leq t.$$

We will take $\alpha = p$ in Lemma 3.1.1. Using Hölder's and Burkholder's inequalities we obtain

$$\begin{aligned} E(\|D^2 M_t\|_{H \otimes H}^p) &= E \left(\left(\int_{[0, t]^2} (D_{s_1, s_2}^2 M_t)^2 ds_1 ds_2 \right)^{p/2} \right) \\ &\leq c_p \left\{ E \left(\left(\int_{[0, t]^2} (D_{s_1} H_{s_2})^2 ds_1 ds_2 \right)^{p/2} \right) \right. \\ &\quad \left. + E \left(\left(\int_{[0, t]^2} \left(\int_{s_1 \vee s_2}^t D_{s_1, s_2}^2 H_r dW_r \right)^2 ds_1 ds_2 \right)^{p/2} \right) \right\} \\ &\leq c'_p \lambda t^p. \end{aligned}$$

Set

$$\sigma(t) := \|DM_t\|_H^2 = \int_0^t (H_s + \int_s^t D_s H_r dW_r)^2 ds.$$

We have the following lower estimates for σ for any $h \leq 1$:

$$\begin{aligned} \sigma(t) &\geq \int_{t(1-h)}^t (H_s + \int_s^t D_s H_r dW_r)^2 ds \\ &\geq \int_{t(1-h)}^t \frac{1}{2} H_s^2 ds - \int_{t(1-h)}^t \left(\int_s^t D_s H_r dW_r \right)^2 ds \geq \frac{th\rho^2}{2} - I_h(t), \end{aligned}$$

where

$$I_h(t) = \int_{t(1-h)}^t \left(\int_s^t D_s H_r dW_r \right)^2 ds.$$

Choose h of the form $h = \frac{4}{t\rho^2 y}$, and notice that $h \leq 1$ if $y \geq a := \frac{4}{t\rho^2}$. We have

$$P\left(\sigma(t) \leq \frac{1}{y}\right) \leq P\left(I_h(t) \geq \frac{1}{y}\right) \leq y^{p/2} E(|I_h(t)|^{p/2}). \quad (3.9)$$

Using Burkholder's inequality for square integrable martingales we get the following estimate

$$\begin{aligned} E(|I_h(t)|^{p/2}) &\leq c_p (th)^{\frac{p}{2}-1} \int_{t(1-h)}^t E\left(\left(\int_s^t (D_s H_r)^2 dr\right)^{p/2}\right) ds \\ &\leq c'_p \sup_{s,r \in [0,t]} E(|D_s H_r|^p) (th)^p. \end{aligned} \quad (3.10)$$

Consequently, for $0 < \gamma < \frac{p}{2}$ we obtain, using (3.9) and (3.10),

$$\begin{aligned} E[\sigma(t)^{-\gamma}] &= \int_0^\infty \gamma y^{\gamma-1} P(\sigma(t)^{-1} > y) dy \\ &\leq a^\gamma + \gamma \int_a^\infty y^{\gamma-1} P(\sigma(t) < \frac{1}{y}) dy \\ &\leq \left(\frac{4}{t\rho^2}\right)^\gamma + \gamma \int_{\frac{4}{t\rho^2}}^\infty E(|I_h(t)|^{p/2}) y^{\gamma-1+\frac{p}{2}} dy \\ &\leq c \left(t^{-\gamma} + \int_{\frac{4}{t\rho^2}}^\infty y^{\gamma-1-\frac{p}{2}} dy\right) \leq c' \left(t^{-\gamma} + t^{\frac{p}{2}-\gamma}\right). \end{aligned} \quad (3.11)$$

Substituting (3.11) in Eq. (3.6) with $\alpha = p$, $\beta < \frac{p}{2}$, and with $\gamma = \frac{1}{2}$ and $\gamma = \beta$, we get the desired estimation. $\square$

The inequality (3.7) implies

$$p_t(x) \leq c \frac{|x|^{-\frac{p}{q}}}{\sqrt{t}} \left(E \left(\left| \int_0^t |H_s|^2 ds \right|^{\frac{p}{2}} \right) \right)^{\frac{1}{q}}.$$

If we assume, in addition that the process H is bounded by a constant M , that is, instead of (ii) we impose:

(ii)' $M \geq |H_t| \geq \rho > 0$ for some constants ρ and M ,

then using the martingale exponential inequality we obtain

$$p_t(x) \leq c \frac{1}{\sqrt{t}} \exp\left(-\frac{|x|^2}{qM^2 t}\right). \quad (3.12)$$

3.2 Regularity of densities and composition of tempered distributions with elements of $\mathbb{D}^{-\infty}$

The results obtained in the last section for onedimensional random variables can be extended to the multidimensional case. Furthermore, under additional smoothness and integrability conditions one can show that the probability density of a random vector F is infinitely differentiable. The basic assumptions are introduced in the following definition of nondegenerate random vector.

Definition 3.2.1 *We will say that the random vector $F = (F^1, \dots, F^m) \in (\mathbb{D}^\infty)^m$ is nondegenerate if the matrix γ_F is invertible a.s. and*

$$(\det \gamma_F)^{-1} \in \bigcap_{p \geq 1} L^p(\Omega). \quad (3.13)$$

For a nondegenerate random vector the following integration by parts formula plays a basic role.

Proposition 3.2.1 *Let $F = (F^1, \dots, F^m) \in (\mathbb{D}^\infty)^m$ be a nondegenerate random vector in the sense of Definition 3.2.1, let $G \in \mathbb{D}^\infty$ and let $g \in C_p^\infty(\mathbb{R}^m)$. Then $(\det \gamma_F)^{-1} \in \mathbb{D}^\infty$ and for any multi-index $\alpha \in \{1, \dots, m\}^k$, $k \geq 1$, there exists an element $H_\alpha(F, G) \in \mathbb{D}^\infty$ such that:*

$$E[(\partial_\alpha g)(F)G] = E[g(F)H_\alpha(F, G)]. \quad (3.14)$$

Moreover the elements $H_\alpha(F, G)$ are recursively given by:

$$\begin{aligned} H_{(i)}(F, G) &= \sum_{j=1}^m \delta \left(G(\gamma_F^{-1})^{ij} D F^j \right), \\ H_\alpha(F, G) &= H_{\alpha_k}(F, H_{(\alpha_1, \dots, \alpha_{k-1})}(F, G)). \end{aligned}$$

Proof: Let us first show that $(\det \gamma_F)^{-1} \in \mathbb{D}^\infty$. For any $N \geq 1$ we have $(\det \gamma_F + \frac{1}{N})^{-1} \in \mathbb{D}^\infty$, because the random variable $(\det \gamma_F + \frac{1}{N})^{-1}$ can be written as the composition of $\det \gamma_F$ with a function in $C_p^\infty(\mathbb{R})$. It is not difficult to show using (3.13) that $\{(\det \gamma_F + \frac{1}{N})^{-1}, N \geq 1\}$ is a Cauchy sequence in the norms $\|\cdot\|_{k,p}$ for all k, p , and we obtain the desired result. Also $\gamma_F^{-1} \in \mathbb{D}^\infty(\mathbb{R}^{m \times m})$. By the chain rule we have

$$D[g(F)] = \sum_{i=1}^m (\partial_i g)(F) D F^i,$$

hence,

$$\langle D[g(F)], D F^j \rangle_H = \sum_{i=1}^m (\partial_i g)(F) \gamma_F^{ij},$$

and as a consequence,

$$(\partial_i g)(F) = \sum_{j=1}^m \langle D[g(F)], D F^j \rangle_H (\gamma_F^{-1})^{ji}.$$

Finally, taking expectations in the above equality and introducing the adjoint of the operator D we get

$$E[G(\partial_i g)(F)] = E[g(F)H_{(i)}(F, G)],$$

where $H_{(i)}(F, G) = \sum_{j=1}^m \delta(G(\gamma_F^{-1})^{ij} DF^j)$. We complete the proof by a recurrence argument. $\square$

Proposition 3.2.2 *For any $p > 1$, and for any multi-index α there exist a constant $C(p, \alpha)$, natural numbers n_1, n_2 , and indices k, d, d', b, b' , depending also on p and α such that*

$$\|H_\alpha(F, G)\|_p \leq C(p, \alpha) \left(\|\gamma_F^{-1}\|_k^{n_1} \|F\|_{d,b}^{n_2} \|G\|_{d',b'} \right).$$

Proof: This estimate is an immediate consequence of the continuity of the operator δ from $\mathbb{D}^{k+1,p}$ into $\mathbb{D}^{k,p}$, Hölder's inequality for the $\|\cdot\|_{k,p}$ -norms, and the equality:

$$D[(\gamma_F^{-1})^{ij}] = - \sum_{k,l=1}^m (\gamma_F^{-1})^{ik} (\gamma_F^{-1})^{jl} D[\gamma_F^{kl}].$$

$\square$

From Proposition 3.2.1 it follows that the density of a smooth and nondegenerate random vector is infinitely differentiable. We recall that $\mathcal{S}(\mathbb{R}^m)$ is the space of infinitely differentiable functions $f: \mathbb{R}^m \rightarrow \mathbb{R}$ such that for any $k \geq 1$, and for all multi-index $\beta \in \{1, \dots, m\}^j$ one has $\sup_{x \in \mathbb{R}^m} |x|^k |\partial_\beta f(x)| < \infty$.

Corollary 3.2.1 *Let $F = (F^1, \dots, F^m) \in (\mathbb{D}^\infty)^m$ be a nondegenerate random vector in the sense of Definition 3.2.1. Then the density of F belongs to the Schwartz space $\mathcal{S}(\mathbb{R}^m)$, and*

$$p(x) = E[\mathbf{1}_{\{F > x\}} H_{(1,2,\dots,m)}(F, 1)], \quad (3.15)$$

where

$$H_{(1,2,\dots,m)}(F, 1) = \delta((\gamma_F^{-1} DF)^m \delta((\gamma_F^{-1} DF)^{m-1} \dots \delta((\gamma_F^{-1} DF)^1) \dots).$$

Proof: Consider the multi-index $\alpha = (1, 2, \dots, m)$. From (3.14) we obtain, for any function $\psi \in C_0^\infty(\mathbb{R}^m)$,

$$E[(\partial_\alpha \psi)(F)] = E[\psi(F) H_\alpha(F, 1)].$$

By Fubini's theorem we can write

$$E[(\partial_\alpha \psi)(F)] = \int_{\mathbb{R}^m} (\partial_\alpha \psi)(x) E[\mathbf{1}_{\{x < F\}} H_\alpha(F, 1)] dx.$$

We can take $\partial_\alpha \psi = \mathbf{1}_B$, where B is a bounded Borel subset of $\mathbb{R}^m$, and in this way we deduce the absolute continuity of the law of F and the expression (3.15) for its density. Moreover, for any multi-index β we obtain

$$\partial_\beta p(x) = E[\partial_\beta(\mathbf{1}_{\{x < F\}}) H_\alpha(F, 1)] = E[\mathbf{1}_{\{x < F\}} H_\beta(F, H_\alpha(F, 1))],$$

and the density of F is infinitely differentiable. Finally in order to show that the density belongs to $\mathcal{S}(\mathbb{R}^m)$ we need to prove that

$$\sup_{x \in \mathbb{R}^m} x_j^{2k} |E[\mathbf{1}_{\{x < F\}} H_\beta(F, H_\alpha(F, 1))]| < \infty,$$

for all $j = 1, \dots, m$. If $x_j > 0$ we have

$$x_j^{2k} |E[\mathbf{1}_{\{x < F\}} H_\beta(F, H_\alpha(F, 1))]| \leq E[|F^j|^{2k} |H_\beta(F, H_\alpha(F, 1))|] < \infty.$$

If $x_j < 0$, then we use the alternative expression for the density

$$p(x) = E[\prod_{i \neq j} \mathbf{1}_{\{x_i < F^i\}} \mathbf{1}_{\{x_j > F^j\}} H_\alpha(F, 1)],$$

and we deduce a similar estimate. $\square$

Let F be an m -dimensional random vector. The probability density of F at a point x can be formally defined as the generalized expectation $E[\delta_x(F)]$, where δ_x denotes the Dirac function at x . The expression $E[\delta_x(F)]$ can be defined as the coupling $\langle \delta_x(F), 1 \rangle$, provided we show that $\delta_x(F)$ is an element of $\mathbb{D}^{-\infty}$. The Dirac function δ_x is a measure, and more generally we can define the composition $T(F)$ of a Schwartz distribution $T \in \mathcal{S}'(\mathbb{R}^m)$ with a random variable in $\mathbb{D}^\infty$ the result being a distribution in $\mathbb{D}^{-\infty}$. This approach was introduced by Watanabe in [107]. Furthermore, the differentiability of the mapping $x \rightarrow \delta_x(F)$ from $\mathbb{R}^m$ into some Sobolev space $\mathbb{D}^{-k,p}$ will provide an alternative proof of the smoothness of the density.

Let us describe the main steps of this approach. We introduce the following sequence of seminorms on the Schwartz space $\mathcal{S}(\mathbb{R}^m)$:

$$\|\phi\|_{2k} = \|(1 + |x|^2 - \Delta)^k \phi\|_\infty, \quad \phi \in \mathcal{S}(\mathbb{R}^m),$$

for $k \in \mathbb{Z}$. Let $\mathcal{S}_{2k}$, $k \in \mathbb{Z}$ be the completion of $\mathcal{S}(\mathbb{R}^m)$ by the seminorm $\|\cdot\|_{2k}$. Then we have

$$\mathcal{S}_{2k+2} \subset \mathcal{S}_{2k} \subset \dots \subset \mathcal{S}_2 \subset \mathcal{S}_0 \subset \mathcal{S}_{-2} \subset \dots \subset \mathcal{S}_{-2k} \subset \mathcal{S}_{-2k-2},$$

and $\mathcal{S}_0 = \hat{\mathcal{C}}(\mathbb{R}^m)$ is the space of continuous functions on $\mathbb{R}^m$ which vanish at infinity. Moreover, $\bigcap_{k \geq 1} \mathcal{S}_{2k} = \mathcal{S}(\mathbb{R}^m)$, and $\bigcup_{k \geq 1} \mathcal{S}_{-2k} = \mathcal{S}'(\mathbb{R}^m)$.

Proposition 3.2.3 *Let $F = (F^1, \dots, F^m) \in (\mathbb{D}^\infty)^m$ be a nondegenerate random vector in the sense of Definition 3.2.1. For any $k \geq 1$, $p > 1$, there exists a constant $C(p, k, F)$ such that for any $\phi \in \mathcal{S}(\mathbb{R}^m)$ we have*

$$\|\phi(F)\|_{-2k,p} \leq C(p, k, F) \|\phi\|_{-2k}.$$

Proof: By a duality principle we have

$$\|\phi(F)\|_{-2k,p} = \sup\{E(\phi(F)G), \|G\|_{2k,q} \leq 1\},$$

where q is the conjugate of p . Now using the expression (3.14) we can write

$$\begin{aligned} E(\phi(F)G) &= E[((1 + |x|^2 - \Delta)^k (1 + |x|^2 - \Delta)^{-k} \phi)(F)G] \\ &= E[(1 + |x|^2 - \Delta)^{-k} \phi(F) \eta_{2k}], \end{aligned}$$

where $\eta_{2k} \in \mathbb{D}^\infty$ verifies

$$C(p, k, F) := \sup\{E(|\eta_{2k}|), \|G\|_{2k,q} \leq 1\} < \infty,$$

and this completes the proof. $\square$

As a consequence, the mapping $\phi \rightarrow \phi(F)$ can be extended uniquely to a continuous linear mapping $T \rightarrow T \circ F$ from $\mathcal{S}_{-2k}$ to $\mathbb{D}^{-2k,p}$ for every $p > 1$ and $k = 0, 1, 2, \dots$. In particular for every Schwartz distribution $T \in \mathcal{S}'(\mathbb{R}^m)$, we can define a Wiener distribution $T \circ F \in \mathbb{D}^{-\infty}$. Actually

$$T \circ F \in \bigcup_{k=1}^{\infty} \bigcap_{p>1} \mathbb{D}^{-2k,p}.$$

We have that for any fixed point $x \in \mathbb{R}^m$, the Dirac function belongs to $\mathcal{S}_{-2k}$ if $k > \frac{m}{2}$, and the mapping $x \rightarrow \delta_x$ is $2j$ times continuously differentiable from $\mathbb{R}^m$ to $\mathcal{S}_{-2k-2j}$. This implies that we can define the “composition” $\delta_x(F)$ as an element of $\mathbb{D}^{-2k,p}$ for any $p > 1$ and any integer k such that $k > \frac{m}{2}$. The density of the random vector F is then given by $\langle \delta_x(F), 1 \rangle$.

Moreover, for any random variable $G \in \mathbb{D}^{\infty}$ and for any x such that $p(x) \neq 0$, we have

$$\langle \delta_x(F), G \rangle = p(x) E[G | F = x].$$

3.3 The case of diffusion processes

We can apply the previous results to derive the smoothness of the density for solutions to stochastic differential equations. This provides probabilistic arguments to study heat kernels. Suppose that $\{W(t), t \geq 0\}$ is a d -dimensional Brownian motion defined on the canonical probability space $\Omega = C_0(\mathbb{R}_+; \mathbb{R}^d)$. Let $A_j : \mathbb{R}^m \rightarrow \mathbb{R}^m$, $j = 0, \dots, d$ a system of C^{∞} functions with bounded derivatives of all orders, and consider the following stochastic Stratonovich differential equation on $\mathbb{R}^m$:

$$\begin{cases} dX_t = \sum_{j=1}^d A_j(X_t) \circ dW_t^j + A_0(X_t) dt, \\ X_0 = x_0 \end{cases} \quad (3.16)$$

One can show that $X_t \in (\mathbb{D}^{\infty})^m$ for all $t \geq 0$. The following nondegeneracy condition assures that for each $t > 0$ the random vector X_t is nondegenerate:

- (H) There exists an integer $k_0 \geq 0$ such that the vector space spanned by the vector fields

$$[A_{j_k}, [A_{j_{k-1}}, [\dots [A_{j_1}, A_{j_0}]] \dots]], \quad 0 \leq k \leq k_0,$$

where $j_0 \in \{1, 2, \dots, d\}$, $j_i \in \{0, 1, 2, \dots, d\}$ if $1 \leq i \leq k$, at point x_0 has dimension m .

In the above hypothesis $[A, B]$ denotes the Lie bracket of the differentiable functions $A, B : \mathbb{R}^m \rightarrow \mathbb{R}^m$, defined as

$$[A, B] = \sum_{i=1}^m \left(A^i \frac{\partial B}{\partial x_i} - B^i \frac{\partial A}{\partial x_i} \right).$$

Theorem 3.3.1 *Suppose that the above condition (H) is satisfied. Then there exists a positive integer ν depending only on k_0 , and for each $p > 1$ a positive constant $c(p, x_0)$ such that*

$$\left\| (\det \gamma_{X_t})^{-1} \right\|_p \leq ct^{-\nu},$$

for all $t > 0$.

This theorem is a consequence of the precise estimates obtained by Kusuoka and Stroock in [57].

3.4 L^p estimates of the density and applications

The stochastic calculus of variations can be used to establish Krylov-type estimates which are a useful tool in deriving existence and uniqueness of a solution to stochastic differential equations and to partial stochastic differential equations whose diffusion coefficient is nondegenerate and the drift is not smooth. Let us first establish a preliminary result similar to Lemma 3.1.1.

Lemma 3.4.1 *Let α, β be two positive real numbers such that $\frac{1}{\alpha} + \frac{1}{\beta} < 1$. Let F be an m -dimensional random variable ($m > 1$) whose components belong to the space $\mathbb{D}^{2,\alpha}$ and $E(|(\gamma_F^{-1})^{ii}|^\beta) < \infty$, $i = 1, \dots, m$. Then for any nonnegative and measurable function $f : \mathbb{R}^m \rightarrow \mathbb{R}$ we have:*

$$E[f(F)] \leq c_{\alpha,\beta,m} \|f\|_m \sup_i \left(E(\sqrt{(\gamma_F^{-1})^{ii}}) + \|D^2 F\|_{L^\alpha(\Omega; H \otimes H \otimes \mathbb{R}^m)} \|(\gamma_F^{-1})^{ii}\|_\beta \right).$$

Proof: We can assume that the function f is continuous and with compact support. Let ψ_ϵ be an approximation of the identity in $\mathbb{R}^m$. We can write

$$\begin{aligned} E[f(F)] &= \lim_{\epsilon \downarrow 0} \int_{\mathbb{R}^m} E[\psi_\epsilon(x - F)] f(x) dx \\ &\leq \|f\|_m \limsup_{\epsilon \downarrow 0} \left(\int_{\mathbb{R}^m} |E[\psi_\epsilon(x - F)]|^{\frac{m}{m-1}} dx \right)^{\frac{m-1}{m}} \\ &\leq \|f\|_m \limsup_{\epsilon \downarrow 0} \prod_{i=1}^m \left(\int_{\mathbb{R}^m} |E[\partial_i \psi_\epsilon(x - F)]| dx \right)^{\frac{1}{m}}. \end{aligned}$$

Here we have used the Gagliardo-Nirenberg inequality which says that for any function f in the space $C_0^\infty(\mathbb{R}^m)$ one has

$$\|f\|_{L^{\frac{m}{m-1}}} \leq \prod_{i=1}^m \|\partial_i f\|_{L^1}^{\frac{1}{m}}.$$

Applying Proposition 3.2.1 we can write

$$E[\partial_i \psi_\epsilon(x - F)] = E[\psi_\epsilon(x - F) H_i(F, 1)], \quad (3.17)$$

where

$$H_i(F, 1) = \delta \left(\sum_{j=1}^m D F^j (\gamma_F^{-1})^{ij} \right).$$

Now, using (3.17) and (2.22) we obtain, for any $p > 1$

$$\begin{aligned} E[f(F)] &\leq \|f\|_m \prod_{i=1}^m (E(|H_i(F, 1)|))^{\frac{1}{m}} \\ &\leq c_p \|f\|_m \prod_{i=1}^m \left(\|E(\sum_{j=1}^m DF^j(\gamma_F^{-1})^{ij})\|_H + \right. \\ &\quad \left. + \|D(\sum_{j=1}^m DF^j(\gamma_F^{-1})^{ij})\|_{L^p(\Omega; H \otimes H)} \right)^{\frac{1}{m}}. \end{aligned}$$

We conclude the proof choosing p in such a way that $\frac{1}{p} = \frac{1}{\alpha} + \frac{1}{\beta}$, applying Hölder's inequality, and using the relations

$$\left\| \sum_{j=1}^m DF^j(\gamma_F^{-1})^{ij} \right\|_H^2 = (\gamma_F^{-1})^{ii},$$

and

$$\begin{aligned} \|D(\sum_{j=1}^m DF^j(\gamma_F^{-1})^{ij})\|_{H \otimes H} &\leq \left\| \sum_{j=1}^m D^2 F^j(\gamma_F^{-1})^{ij} \right\|_{H \otimes H} \\ &+ \left\| \sum_{j,k,s=1}^m DF^j \otimes (\gamma_F^{-1})^{ik} D(\langle DF^k, DF^s \rangle_H) (\gamma_F^{-1})^{sj} \right\|_{H \otimes H}. \end{aligned}$$

□

Lemma 3.4.1 allows us to deduce Krylov-type estimates for solutions to stochastic differential equations. Let $\{W_t, t \in [0, 1]\}$ be an m -dimensional Brownian motion. Consider the m -dimensional stochastic differential equation

$$\begin{cases} dX_t = f(X, t)dt + \sum_{i=1}^m g_i(X, t)dW_t^i, & t \in [0, 1], \\ X_0 = x_0, \end{cases} \quad (3.18)$$

where $f, g_i : C([0, 1]; \mathbb{R}^m) \times [0, 1] \rightarrow \mathbb{R}^m$, $i = 1, \dots, m$, are progressively measurable functions verifying the following conditions

- (a) $|f(x, t)| \leq K$ for some constant K .
- (b) g is twice Fréchet differentiable in the first variable, with uniformly bounded first and second order derivatives, and $g(0, t)$ is bounded.
- (c) There exist positive constants c_1 and c_2 such that

$$c_1 |\theta|^2 \leq |g(x, t)\theta|^2 \leq c_2 |\theta|^2,$$

for all $\theta \in \mathbb{R}^m$, $x \in C([0, 1]; \mathbb{R}^m)$, and $t \in [0, 1]$.

We will denote this equation by $\text{Eq}(f, g)$. By a solution to this equation we will mean a continuous and adapted process that satisfies the corresponding stochastic integral equation.

Proposition 3.4.1 *Let $g_i : C([0, 1]; \mathbb{R}^m) \times [0, 1] \rightarrow \mathbb{R}^m$, $i = 1, \dots, m$, be measurable functions verifying conditions (b) and (c), and let $\{F_t, t \in [0, 1]\}$ be an m -dimensional progressively measurable process bounded by K . We denote by $\{X_t, t \in [0, 1]\}$ the solution of the following equation (denoted in the sequel by $Eq(F, g)$):*

$$X_t = x_0 + \int_0^t F_s ds + \sum_{i=1}^m \int_0^t g_i(X, s) dW_s^i, \quad t \in [0, 1]. \quad (3.19)$$

Then for any measurable function $h : \mathbb{R}^m \times [0, 1] \rightarrow \mathbb{R}$, and for all $p > m$, $\gamma > 1$, we have

$$E \left(\int_0^1 |h(X_t, t)| dt \right) \leq C \left(\int_0^1 \left(\int_{\mathbb{R}} |h(x, t)|^p dx \right)^\gamma dt \right)^{\frac{1}{p\gamma}}, \quad (3.20)$$

where the constant C depends on p , γ , K , and the coefficient g .

Proof: Define a new probability $\tilde{P}$ by $\frac{d\tilde{P}}{dP} = Z$, where

$$Z = \exp \left(- \int_0^1 F_t^T (g^{-1})^T(X, t) dW_t - \frac{1}{2} \int_0^1 |(g^{-1})(X, t) F_t|^2 dt \right).$$

By Girsanov's theorem

$$\tilde{W}_t = W_t + \int_0^t (g^{-1})(X, s) F_s ds$$

is a Brownian motion under $\tilde{P}$. Thus, under $\tilde{P}$ the process X has the same law as the process Y solution of

$$\begin{cases} dY_t = g(Y, t) dW_t, & t \in [0, 1], \\ Y_0 = x_0, \end{cases} \quad (3.21)$$

By Hölder's inequality

$$\begin{aligned} E \left(\int_0^1 |h(X_t, t)| dt \right) &\leq (\tilde{E} Z^{-\alpha})^{\frac{1}{\alpha}} \left(\tilde{E} \int_0^1 |h(X_t, t)|^\beta dt \right)^{\frac{1}{\beta}} \\ &= C(\alpha, K, g) \left(E \int_0^1 |h(Y_t, t)|^\beta dt \right)^{\frac{1}{\beta}} \end{aligned}$$

for every $\alpha, \beta > 1$ such that $\frac{1}{\alpha} + \frac{1}{\beta} = 1$. The random vector Y_t belongs to the space $\mathbb{D}^{2,p}$ for all $p > 1$ (see Lemma 3.4.2 below). When $m = 1$ one can use the estimate on the density of Y_t obtained in (3.12). Suppose $m > 1$. Applying Lemma 3.4.1 to the random variable Y_t we obtain if $\frac{1}{\alpha'} + \frac{1}{\beta'} = 1$

$$\begin{aligned} E \left(\int_0^1 |h(Y_t, t)|^\beta dt \right) &\leq c_{m, \alpha', \beta'} \left(\int_0^1 \left(\int_{\mathbb{R}^m} |h(y, t)|^{m\beta} dy \right)^\alpha dt \right)^{\frac{1}{\alpha}} \\ &\quad \times \sup_i \left(E \left(\sqrt{(\gamma_{Y_t}^{-1})^{ii}} \right) + \|D^2 Y_t\|_{\alpha'} \|(\gamma_{Y_t}^{-1})^{ii}\|_{\beta'} \right) dt. \end{aligned}$$

The inequality (3.25) allows to control the term $\|D^2 Y_t\|_{\alpha'}$ by a constant times t . On the other hand, using hypothesis (c) one can show (a sketch of the proof is given below) that

$$E[(\gamma_{Y_t}^{-1})^{ii}]^p < a_3 t^{-p} \quad (3.22)$$

for all $p \geq \frac{1}{2}$. Hence,

$$\begin{aligned} E \left(\int_0^1 |h(Y_t, t)|^\beta dt \right) &\leq c_{m, \alpha', \beta'} \int_0^1 \left(\int_{\mathbb{R}^m} |h(y, t)|^{m\beta} dy \right)^{\frac{1}{m}} \left(\frac{1}{\sqrt{t}} + c \right) dt \\ &\leq c'_{m, \alpha', \beta'} \int_0^1 \left(\int_{\mathbb{R}^m} |h(y, t)|^{m\beta} dy \right)^{\frac{2+\epsilon}{m}} dt. \end{aligned}$$

This allows to complete the proof of the Proposition.

Sketch of the proof of (3.22): First we write

$$E((\gamma_{Y_t}^{-1})^{ii})^p \leq c_{m,p} \left[E(\|DY_t\|_H^{4p(m-1)}) \right]^{\frac{1}{2}} \left[E((\det \gamma_{Y_t})^{-2p}) \right]^{\frac{1}{2}}.$$

From (3.24) we get that the first factor in the right-hand side of the above equality is bounded by a constant times $t^{p(m-1)}$. In order to estimate the second factor we write

$$\det \gamma_{Y_t} \geq \inf_{|v|=1} (v^T \gamma_{Y_t} v)^m = \inf_{|v|=1} \|\langle v, DY_t \rangle\|_H^{2m}.$$

From (3.28) we get

$$\langle v, D_s^j Y_t \rangle = \langle v, g_j(Y, s) \rangle + \langle v, \int_s^t \langle g'(Y, r), (D_s^j Y) \rangle dW_r \rangle, \quad j = 1, \dots, m. \quad (3.23)$$

Consequently, we deduce, with the convention of summation over repeated indices, for any $h < 1$,

$$\begin{aligned} \|\langle v, DY_t \rangle\|_H^2 &= \int_0^t \sum_{k=1}^m |D_s^k(v_j Y_t^j)|^2 ds \\ &\geq \frac{1}{2} \int_{t(1-h)}^t \sum_{k=1}^m |v_j g_k^j(Y, s)|^2 ds \\ &\quad - \int_{t(1-h)}^t \sum_{k=1}^m \left| v_j \int_s^t \langle (g')_i^j(Y, r), D_s^k Y \rangle dW_r^i \right|^2 ds \\ &\geq \frac{c_1 t h}{2} - \int_{t(1-h)}^t |J(s)|^2 ds, \end{aligned}$$

where

$$J^j(s) = \int_s^t \langle (g')_i^j(Y, r), D_s^k Y \rangle dW_r^i.$$

Now, using the same technique as in the proof of (3.7) we can show that

$$E \left(\inf_{|v|=1} \|\langle v, DY_t \rangle\|_H^{-4pm} \right) \leq E \left(\left| \frac{c_1 t h}{2} - \int_{t(1-h)}^t |J(s)|^2 ds \right|^{-2pm} \right) \leq a_4 t^{-2pm},$$

and this allows us to conclude the proof. $\square$

Remark: In the preceding proposition we can take $\gamma = 1$ when $m = 2$.

Lemma 3.4.2 *Let $\{Y_t, t \in [0, 1]\}$ be the solution to equation $Eq(0, g)$ where the function g satisfies hypothesis (b). Then $Y_t \in \bigcap_{p>1} \mathbb{D}^{2,p}$, and*

$$E(\|DY_t\|_H^p) < a_1 t^{\frac{p}{2}}, \quad (3.24)$$

$$E(\|D^2 Y_t\|_{H \otimes H}^p) < a_2 t^p, \quad (3.25)$$

for some constants a_1 and a_2 .

Proof: We will denote by g' and by g'' the Fréchet derivatives of the function g . We introduce the sequence of Picard approximations defined recursively by $Y^0 = x_0$, and

$$Y_t^{n+1} = x_0 + \int_0^t g(Y^n, s) dW_s, \quad n \geq 0.$$

Then for each n the random variable Y_t^{n+1} belongs to $\cap_{p>1} \mathbb{D}^{1,p}$, and

$$D_s^j Y_t^{n+1} = g_j(Y^n, s) + \int_s^t \langle g'(Y^n, r), (D_s^j Y^n) \rangle dW_r, \quad j = 1, \dots, m,$$

where D^j denotes the derivative with respect to the j th component of the Brownian motion. That is, $D_s^j W_t^k = \mathbf{1}_{\{s \leq t\}} \delta_{jk}$. We have

$$\lim_n E[\|Y^n - Y\|_\infty^p] = 0,$$

for all $p > 1$, and using Gronwall's lemma we deduce

$$\sup_n \sup_{s \in [0,1]} E[\|D_s Y^{n+1}\|_\infty^p] < \infty, \quad (3.26)$$

and

$$\sup_n \sup_{s, r \in [0,1]} E[\|D_{s,r}^2 Y^{n+1}\|_\infty^p] < \infty, \quad (3.27)$$

for all $p > 1$. This implies that $Y_t \in \cap_{p>1} \mathbb{D}^{2,p}$. Finally we use the chain rule (properly extended to $C([0,1])$ -valued functions) in order to compute the derivative of $g(Y, t)$, and we obtain:

$$D_s^j Y_t = g_j(Y, s) + \int_s^t \langle g'(Y, r), (D_s^j Y) \rangle dW_r, \quad j = 1, \dots, m. \quad (3.28)$$

Using Burkholder's inequality we can show (3.24) from the expression (3.28). Similar computations can be done for the second derivative. $\square$

Notice that the inequality (3.20) also holds for a process X such for each $t \in [0, 1]$, X_t is the almost sure limit of a sequence of random variables X_t^n such that X^n solves an equation of the form (3.19) with coefficients g and F_n , where F_n is bounded by K .

We can now to derive a convergence criterion for solutions to equation (3.19):

Corollary 3.4.1 *Let $h_n : \mathbb{R}^m \times [0, 1] \rightarrow \mathbb{R}$ be a sequence of measurable functions uniformly bounded by a constant K_1 and converging a.e. to h . Consider a sequence of processes X^n solutions of equations $Eq(F_n, g)$, where g satisfies conditions (b) and (c) and F_n is a progressively measurable process bounded by K . Then*

$$\lim_n E \left(\int_0^1 |h_n(X_t^n, t) - h(X_t, t)| dt \right) = 0.$$

Proof: Set

$$J_n := E \left(\int_0^1 |h_n(X_t^n, t) - h(X_t, t)| dt \right).$$

Fix $\eta > 0$. Let ψ be a nonnegative smooth function with support included in $[-1, 1]$, $\psi(0) = 1$ and bounded by 1. Choose $R > 0$ such that

$$E \left(\int_0^1 \left(1 - \psi \left(\frac{X_t}{R} \right) \right) dt \right) < \eta.$$

Consider a continuous function $g_{R,\eta}$ on $\mathbb{R} \times [0, 1]$ bounded by $2K_1$ such that $g(x, t) = 0$ for all $|x| > R$ and for a fixed $p > m$

$$\int_{[0,1]} \left(\int_{-R}^R |h(x, t) - g_{R,\eta}(x, t)|^p dx \right)^\gamma dt \leq \eta^{p\gamma},$$

where $p > m$ and $\gamma > 1$ are the exponents appearing in Proposition 3.4.1. Consider the following decomposition

$$J_n = J_n^1 + J_n^2 + J_n^3 + J_n^4,$$

where

$$\begin{aligned} J_n^1 &= E \left(\int_0^1 |h_n(X_t^n, t) - h(X_t^n, t)| dt \right) \\ J_n^2 &= E \left(\int_0^1 |h(X_t^n, t) - g_{R,\eta}(X_t^n, t)| dt \right) \\ J_n^3 &= E \left(\int_0^1 |h(X_t, t) - g_{R,\eta}(X_t, t)| dt \right) \\ J_n^4 &= E \left(\int_0^1 |g_{R,\eta}(X_t^n, t) - g_{R,\eta}(X_t, t)| dt \right). \end{aligned}$$

We write

$$\begin{aligned} J_n^1 &= E \left(\int_0^1 \psi \left(\frac{X_t^n}{R} \right) |h_n(X_t^n, t) - h(X_t^n, t)| dt \right) \\ &+ E \left(\int_0^1 \left(1 - \psi \left(\frac{X_t^n}{R} \right) \right) |h_n(X_t^n, t) - h(X_t^n, t)| dt \right), \end{aligned}$$

and we make similar decompositions for the terms J_n^2 and J_n^3 . In this way we obtain, using Proposition 3.4.1,

$$\begin{aligned} \limsup_n J_n &\leq C \left(\limsup_n \left(\int_0^1 \left(\int_{\mathbb{R}} \psi \left(\frac{x}{R} \right) |h_n(x, t) - h(x, t)|^p dx \right)^\gamma dt \right)^{\frac{1}{p\gamma}} \right. \\ &\quad + 2 \left(\int_0^1 \left(\int_{-R}^R |h(x, t) - g_{R,\eta}(x, t)|^p dx \right)^\gamma dt \right)^{\frac{1}{p\gamma}} \\ &\quad + 6K_1 E \left(\int_0^1 \left(1 - \psi \left(\frac{X_t}{R} \right) \right) dt \right) \\ &\quad \left. + \limsup_n E \left(\int_0^1 |g_{R,\eta}(X_t^n, t) - g_{R,\eta}(X_t, t)| dt \right) \right). \end{aligned}$$

Hence,

$$\limsup_n J_n \leq C\eta.$$

□

Corollary 3.4.2 *Let $g_i : C([0, 1]; \mathbb{R}^m) \times [0, 1] \rightarrow \mathbb{R}$, $1 \leq i \leq m$, be measurable functions satisfying conditions (b) and (c) of Proposition 3.4.1. Consider sequences of*

progressively measurable processes $F, F_n : [0, 1] \times \Omega \rightarrow \mathbb{R}^m$, and measurable functions $f, f_n : [0, 1] \times \mathbb{R}^m \rightarrow \mathbb{R}^m$ such that

$$\begin{aligned}\lim_n F_n(t) &= F(t), \quad dt \otimes dP - a.e., \\ \lim_n f_n(t, x) &= f(t, x), \quad dt \otimes dx - a.e.. \end{aligned}$$

Suppose that for each $n \geq 1$ equation $\text{Eq}(F_n, g)$ admits a solution X^n , and for each $t \in [0, 1]$, X_t^n converges a.s. to a process X_t . Then X solves equation $\text{Eq}(F, g)$.

Proof: It suffices to pass to the limit each term in the equation

$$X_t^n = x_0 + \int_0^t F_n(s) ds + \sum_{i=1}^m \int_0^t g_i(X^n, s) dW_s^i.$$

□

Applying the preceding results we can derive existence and uniqueness results for equations of the form $\text{Eq}(f, g)$ where f is a measurable and bounded function of X_t and g is a smooth (up to the second order) nondegenerate function. To do this one usually makes use of comparison theorems for stochastic differential equations, and for this reason, one has to consider particular type of equations. In order to illustrate this method we will describe the onedimensional case.

Proposition 3.4.2 *Suppose that $f : \mathbb{R} \times [0, 1] \rightarrow \mathbb{R}$ is a measurable function bounded by K and $g : \mathbb{R} \times [0, 1] \rightarrow \mathbb{R}$ is twice continuously differentiable, with bounded derivatives and such that $0 < c_1 \leq |g(x, t)| \leq c_2 < \infty$. Then equation (3.18) has a unique solution.*

Proof of the existence: Let ρ be a smooth nonnegative function with compact support in $\mathbb{R}$ such that $\int_{\mathbb{R}} \rho(x) dx = 1$. Define

$$\tilde{f}_j(x, t) = j \int_{\mathbb{R}} f(z, t) \rho(j(x - z)) dz.$$

Moreover, let

$$f_{n,k} = \bigwedge_{j=n}^k \tilde{f}_j, \quad n \leq k$$

and

$$f_n = \bigwedge_{j=n}^{\infty} \tilde{f}_j.$$

Clearly, $f_{n,k}(x, t)$ is Lipschitz in x uniformly with respect to t , $f_{n,k}(x, t) \downarrow f_n(x, t)$ as $k \uparrow \infty$, and $f_n(x, t) \uparrow f(x, t)$ as $n \uparrow \infty$, dx -a.e. for each t . For each $n \geq k$ equation $\text{Eq}(f_{n,k}, g)$ has a unique solution $X_{n,k}$. From the comparison theorem for stochastic differential equations the sequence $\{X_{n,k}, k = n, n+1, \dots\}$ is decreasing. Hence it has a limit

$$X_n = \lim_{k \rightarrow \infty} X_{n,k}.$$

Since $X_{n,k}$ is bounded above by the solution of $\text{Eq}(K, g)$ and below by the solution of $\text{Eq}(-K, g)$, we can apply Corollary 3.4.1 and deduce that

$$\lim_k \int_0^t f_{n,k}(X_{n,k}(t), t) dt = \int_0^t f_n(X_n(t), t) dt, \quad \text{a.s..}$$

On the other hand, by the continuity of the function g we get

$$\lim_k \int_0^t g(X_{n,k}(t), t) dW_t = \int_0^t g(X_n(t), t) dW_t, \quad \text{a.s.}$$

Hence the process X_n is a solution to $\text{Eq}(f_n, g)$.

Proof of the uniqueness: The proof of the uniqueness has been inspired by the work [36]. Let X denote the solution constructed as the limit of the sequence X_n , and let Z be another solution. We can write with the above notations

$$dZ_t = \tilde{F}_j(t)dt + \tilde{f}_j(Z_t, t)dt + g(Z_t, t)dW_t,$$

where

$$\tilde{F}_j(t) = f(Z_t, t) - \tilde{f}_j(Z_t, t), \quad j \geq 1.$$

Let us denote by $Z_{n,k,-}$ and by $Z_{n,k,+}$ the solutions of the following equations

$$dZ_{n,k,-}(t) = \left(\bigwedge_{j=n}^k \tilde{F}_j(t) \right) \wedge 0 dt + f_{n,k}(Z_{n,k,-}(t), t)dt + g(Z_{n,k,-}(t), t)dW_t,$$

and

$$dZ_{n,k,+}(t) = \left(\bigvee_{j=n}^k \tilde{F}_j(t) \right) \vee 0 dt + f_{n,k}(Z_{n,k,+}(t), t)dt + g(Z_{n,k,+}(t), t)dW_t.$$

By the comparison theorem we have

$$Z_{n,k,-} \leq X_{n,k} \leq Z_{n,k,+},$$

and

$$Z_{n,k,-} \leq Z \leq Z_{n,k,+},$$

Moreover, the sequences $\{Z_{n,k,-}, k \geq n\}$ and $\{Z_{n,k,+}, k \geq n\}$ are decreasing in k . Hence, they converge to some limits $\{Z_{n,-}, k \geq n\}$ and $\{Z_{n,+}, k \geq n\}$, respectively, which are increasing in n . Let us denote by $\{Z_-, k \geq n\}$ and $\{Z_+, k \geq n\}$ the limit of these sequences. We have

$$Z_- \leq X, Z \leq Z_+. \quad (3.29)$$

The coefficients $\bigwedge_{j=n}^k \tilde{F}_j(t)$ and $f_{n,k}$ converge almost everywhere to 0 and f , respectively, as k and n tend to infinity. Consequently, applying Corollary 3.4.2 we get that the processes Z_- and Z_+ solve (3.18). But from Girsanov's theorem, the law of (3.18) is unique. Hence, from (3.29) we deduce that $Z_- = X = Z = Z_+$ which completes the proof of the uniqueness. $\square$

This approach has been used to derive existence and uniqueness of the solution and approximation of the implicit approximation scheme for the unidimensional heat equation perturbed by a space-time white noise. Let us describe the results obtained for these equations.

Let $\{W(s, t) = W([0, s] \times [0, t]), s, t, \in [0, 1]\}$ be a two-parameter Wiener process. Let us consider the stochastic partial differential equation

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} + f(t, x, u(t, x)) + g(t, x, u(t, x)) \frac{\partial^2 W}{\partial t \partial x} \quad (3.30)$$

with Dirichlet boundary conditions

$$u(t, 0) = u(t, 1) = 0, \quad t \in [0, 1]; \quad (3.31)$$

and with initial condition

$$u(0, x) = u_0(x), \quad x \in [0, 1], \quad (3.32)$$

where u_0 is a continuous function on $[0, 1]$ vanishing at 0 and 1. The coefficients $f(t, x, r)$ and $g(t, x, r)$ are locally bounded Borel functions mapping $[0, 1]^2 \times \mathbb{R}$ into $\mathbb{R}$.

The existence and uniqueness of the strong solution is well known when f and g satisfy the linear growth condition and are Lipschitz in r . Consider the following hypotheses on the coefficients of Eq. (3.30):

(H1) g has a Lipschitz continuous derivative, it has linear growth, and it satisfies the nondegeneracy condition $g^2 \geq \varepsilon > 0$. f satisfies the one-sided linear growth condition $rf(t, x, r) \leq C(1 + r^2)$.

Under hypothesis (H) the existence and uniqueness of the solution for equation (3.30) has been established in [5]. The main tool used in this paper is an L^p estimate, obtained using Malliavin calculus, similar to Eq. (3.20). This existence and uniqueness result is improved in [36], where the smoothness condition on g is replaced by the local Lipschitzness of g in r .

Consider the implicit approximation scheme for the equation (3.30). For every integer $n \geq 1$ we construct a random field u^n in the following way:

$$\begin{aligned} u^n(0, x) &= u_0(x) \\ u^n(t_{i+1}^n, x) &= \left(I - \frac{1}{n}\Delta\right)^{-1} u^n(t_i^n, x) \\ &\quad + \int_{t_i^n}^{t_{i+1}^n} G_n(x, y) f(s, x, u^n(t_i^n, x)) ds \\ &\quad + \int_{t_i^n}^{t_{i+1}^n} \int_0^1 G_n(x, y) g(s, y, u^n(t_i^n, y)) W(ds, dy), \quad i \geq 0, \end{aligned}$$

where $t_i^n := \frac{i}{n}$, $0 \leq i \leq n$, G_n is the kernel of the operator $\left(I - \frac{1}{n}\Delta\right)^{-1}$, and $\Delta := \frac{\partial^2}{\partial x^2}$.

When $t \in (t_i^n, t_{i+1}^n)$ we define $u^n(t, x)$ by the polygonal approximation

$$u^n(t, x) = u^n(t_i^n, x) + n(t - t_i^n) (u^n(t_{i+1}^n, x) - u^n(t_i^n, x)).$$

The sequence u_n is called an implicit approximation scheme for the equation (3.30) because we can write:

$$\begin{aligned} u^n(t_{i+1}^n) &= u^n(t_i^n) + \frac{1}{n}\Delta u^n(t_{i+1}^n) + \int_{t_i^n}^{t_{i+1}^n} f(s, u^n(t_i^n)) ds \\ &\quad + \int_{t_i^n}^{t_{i+1}^n} \int_0^1 g(s, u^n(t_i^n)) W(ds, dy). \end{aligned}$$

Under hypothesis (H) one can show (see [37]) that

$$\lim_n P \left(\sup_{x,t \in [0,1]} |u^n(t,x) - u(t,x)| > \epsilon \right) = 0,$$

for all $\epsilon > 0$. The main ingredient of the proof is the following L^p estimate for the density $p_{t,x}^n$ of the law of $u^n(t,x)$. We obtain it using Lemma 3.1.1:

$$\sup_n \int_0^1 \int_0^1 \int_{\mathbb{R}} p_{t,x}^n(r)^\alpha \varphi(t,x) \, dr \, dx \, dt < \infty$$

for any function $\varphi \in C^\infty((0,1)^2)$ with compact support, and for all $\alpha > 1$.

Bibliographical notes: For an m -dimensional nondegenerate random vector F , Watanabe has precised the order (s,p) of the negative Sobolev space $\mathbb{D}^{s,p}$ which contains the distribution $\delta_x(F)$ (cf. [109]). The asymptotic expansion of $X_t(\epsilon\omega)$ using Malliavin calculus has been studied in [108].

Chapter 4

Support theorems

In this chapter we apply the stochastic calculus of variations to study the properties of the support of the law of a random vector. We also discuss the characterization of the support of the law using the so-called skeleton approximation, and the application of the Malliavin calculus to the proof of Varadhan-type estimates.

4.1 Properties of the support

Given a random variable $F : \Omega \rightarrow S$ with values on a Polish space S , the topological support of the probability distribution of F is defined as the set of points $x \in S$ such that $P(d(F, x) < \epsilon) > 0$ for all $\epsilon > 0$.

The connected property of the support of the law of a finite-dimensional random variable vector was established by Fang [28]:

Proposition 4.1.1 *Let $F \in (\mathbb{D}^{1,p})^m$ for some $p > 1$. Then the topological support of the law of F is a closed connected subset of $\mathbb{R}^m$.*

Proof: Suppose that $\text{supp } P \circ F^{-1}$ is not connected. There exists two nonempty disjoint closed sets A and B such that $\text{supp } P \circ F^{-1} = A \cup B$.

For each integer $M \geq 2$ let $\psi_M : \mathbb{R}^m \rightarrow \mathbb{R}$ be an infinitely differentiable function such that $0 \leq \psi_M \leq 1$, $\psi_M(x) = 0$ if $|x| \geq M$, $\psi_M(x) = 1$ if $|x| \leq M - 1$, and $\sup_{x,M} |\nabla \psi_M(x)| < \infty$.

Set $A_M = A \cap \{|x| \leq M\}$ and $B_M = B \cap \{|x| \leq M\}$. For M large enough we have $A_M \neq \emptyset$ and $B_M \neq \emptyset$, and we can find an infinitely differentiable function f_M such that $0 \leq f_M \leq 1$, $f_M = 1$ in a neighborhood of A_M , and $f_M = 0$ in a neighborhood of B_M .

The sequence $(f_M \psi_M)(F)$ converges a.s. and in L^p to $\mathbf{1}_{\{F \in A\}}$ as M tends to infinity. On the other hand, we have

$$\begin{aligned} D[(f_M \psi_M)(F)] &= \sum_{i=1}^m (\psi_M \partial_i f_M)(F) DF^i + \sum_{i=1}^m (f_M \partial_i \psi_M)(F) DF^i \\ &= \sum_{i=1}^m (f_M \partial_i \psi_M)(F) DF^i. \end{aligned}$$

Hence,

$$\sup_M \|D[(f_M \psi_M)(F)]\|_H \leq \sum_{i=1}^m \sup_M \|\partial_i \psi_M\|_\infty \|DF^i\|_H \in L^p.$$

By Lemma 2.4.2 we obtain that $\mathbf{1}_{\{F \in A\}}$ belongs to $\mathbb{D}^{1,p}$ and, due to Lemma 1.4.2, this is contradictory because $0 < P(F \in A) < 1$. $\square$

As a consequence, the support of the law of a random variable $F \in \mathbb{D}^{1,p}$, $p > 1$ is a closed interval. The next results provides sufficient conditions for the density of F to be nonzero in the interior of the support.

Proposition 4.1.2 *Let $F \in \mathbb{D}^{1,p}$, $p > 2$, and suppose that F possesses a locally Lipschitz density $p(x)$. Let a be a point in the interior of the support of the law of F . Then $p(a) > 0$.*

Proof: Suppose $p(a) = 0$. Set $r = \frac{2p}{p+2} > 1$. By Lemma 1.4.2 we know that $\mathbf{1}_{\{F > a\}} \notin \mathbb{D}^{1,r}$ because $0 < P(F > a) < 1$. Fix $\epsilon > 0$ and set

$$\varphi_\epsilon(x) = \int_{-\infty}^x \frac{1}{2\epsilon} \mathbf{1}_{[a-\epsilon, a+\epsilon]}(y) dy.$$

Then $\varphi_\epsilon(F)$ converges to $\mathbf{1}_{\{F > a\}}$ in $L^r(\Omega)$, as $\epsilon \downarrow 0$. Moreover, $\varphi_\epsilon(F) \in \mathbb{D}^{1,r}$ and

$$D(\varphi_\epsilon(F)) = \frac{1}{2\epsilon} \mathbf{1}_{[a-\epsilon, a+\epsilon]}(F) DF.$$

We have

$$E(\|D(\varphi_\epsilon(F))\|_H^r) \leq (E(\|DF\|_H^p))^{\frac{2}{p+2}} \left(\frac{1}{(2\epsilon)^2} \int_{a-\epsilon}^{a+\epsilon} p(x) dx \right)^{\frac{p}{p+2}}.$$

The local Lipschitz property of p implies that $p(x) \leq K|x - a|$, and we obtain

$$E(\|D(\varphi_\epsilon(F))\|_H^r) \leq (E(\|DF\|_H^p))^{\frac{2}{p+2}} 2^{-r} K^{\frac{p}{p+2}}.$$

By Lemma 2.4.2 this implies $\mathbf{1}_{\{F > a\}} \in \mathbb{D}^{1,r}$ and we are in contradiction. $\square$

The following example shows that, unlike the one-dimensional case, in dimension $m > 1$ the density of a nondegenerate random vector can vanish in the interior of the support.

Example: Let h_1 and h_2 be two orthonormal elements of H , and define

$$\begin{aligned} X_1 &= \arctan W(h_1), \\ X_2 &= \arctan W(h_2). \end{aligned}$$

Then $X_1, X_2 \in \mathbb{D}^\infty$ and

$$DX_i = (1 + W(h_i)^2)^{-1} h_i,$$

for $i = 1, 2$, and

$$\det \gamma_X = (1 + W(h_1)^2)^{-2} (1 + W(h_2)^2)^{-2},$$

where γ_X denotes the Malliavin matrix of the nondegenerate random vector $X = (X_1, X_2)$. Notice that the support of the law of X is the rectangle $[-\frac{\pi}{2}, \frac{\pi}{2}]^2$, and

the density of X is strictly positive in the interior of the support. Now consider the following vector

$$\begin{aligned} Y_1 &= (X_1 + \frac{3\pi}{2}) \cos(2X_2 + \pi) \\ Y_2 &= (X_1 + \frac{3\pi}{2}) \sin(2X_2 + \pi). \end{aligned}$$

We have that $Y \in (\mathbb{D}^\infty)^2$ and

$$\det \gamma_Y = 4(X_1 + \frac{3\pi}{2})^2 (1 + W(h_1)^2)^{-2} (1 + W(h_2)^2)^{-2}.$$

This implies that Y is a nondegenerate vector. Its support is the set $\{(x, y) : \pi^2 \leq x^2 + y^2 \leq 4\pi^2\}$, and the density of Y vanishes on the points (x, y) in the support such that $\pi < y < 2\pi$, and $x = 0$.

For a smooth and nondegenerate random vector when the density vanishes, then all its partial derivatives also vanish (see Ben Arous and Léandre [9]).

Proposition 4.1.3 *Let $F = (F^1, \dots, F^m) \in (\mathbb{D}^\infty)^m$ be a nondegenerate random vector in the sense of Definition 3.2.1, and denote its density by $p(x)$. Then $p(x) = 0$ implies $\partial_\alpha p(x) = 0$ for any multiindex α .*

Proof: Suppose that $p(x) = 0$. We know that $p(x) = \langle \delta_x(F), 1 \rangle$. We have that $\langle \delta_x(F), R \rangle \geq 0$ for any smooth random variable R because

$$\langle \delta_x(F), R \rangle = \lim_{\epsilon \downarrow 0} E(\psi_\epsilon(x - F)R),$$

where ψ_ϵ is an approximation of the identity. So, from Lemma 2.4.3 there exists a measure η_x on the σ -field $\mathcal{G}$ generated by $\{W(e_i), i \geq 1\}$ such that $\langle \delta_x(F), R \rangle = \int_\Omega R d\eta_x$ for any $\mathcal{G}$ -measurable and smooth random variable R . Notice that $p(x) = \eta_x(\Omega)$. Therefore, $\eta_x = 0$, which implies that $\delta_x(F) = 0$ as an element of $\mathbb{D}^{-\infty}$. For any multiindex α we have

$$\partial_\alpha p(x) = \partial_\alpha \langle \delta_x(F), 1 \rangle = \langle (\partial_\alpha \delta_x)(F), 1 \rangle.$$

Hence, it suffices to show that $(\partial_\alpha \delta_x)(F)$ vanishes. Suppose first that $\alpha = (i)$. We can write

$$D(\delta_x(F)) = \sum_{i=1}^m (\partial_i \delta_x)(F) D F^i,$$

as elements of $\mathbb{D}^{-\infty}$, which implies

$$(\partial_i \delta_x)(F) = \sum_{j=1}^m \langle D[\delta_x(F)], D F^j \rangle_H (\gamma_F^{-1})^{ji} = 0,$$

because $D[\delta_x(F)] = 0$. The general case follows by recurrence. $\square$

In the case of a diffusion processes, the characterization of the support of the law is obtained by means of the notion of skeleton. A general notion of skeleton is provided by the following definition.

In the sequel we will assume that $\{\Pi^N, N \geq 1\}$ is a sequence of orthogonal projections on H of finite-dimensional range, which converges strongly to the identity. If $\{e_1, \dots, e_N\}$ is an orthonormal basis of the image on Π^N we set $\Pi^N W = \sum_{i=1}^N W(e_i) e_i$.

Definition 4.1.1 Let $F : \Omega \rightarrow S$ be a random variable taking values on a Polish space (S, d) . We will say that a measurable function $\Phi : H \rightarrow S$ is a skeleton of F if the following two conditions are satisfied:

(i) For all $\epsilon > 0$ we have

$$\lim_N P \left\{ d(\Phi(\Pi^N W), F) > \epsilon \right\} = 0.$$

(ii) For all $h \in H$, there exists a sequence of measurable transformations $T_N^h : \Omega \rightarrow \Omega$ such that $P \circ (T_N^h)^{-1}$ is absolutely continuous with respect to P , and for every $\epsilon > 0$

$$\limsup_N P \left\{ d(F \circ T_N^h, \Phi(h)) < \epsilon \right\} > 0.$$

Proposition 4.1.4 Suppose that Φ is a skeleton of F in the sense of Definition 4.1.1. Then the support of the law of F is the closure in S of the set $\{\Phi(h), h \in H\}$.

Proof:

Step 1: Let us first show that condition (i) implies the inclusion

$$\text{supp}(P \circ F^{-1}) \subset \overline{\Phi(H)}.$$

It suffices to show that $P(F \in \overline{\Phi(H)}) = 1$, and this follows from

$$P \left\{ d(F, \overline{\Phi(H)}) \leq \epsilon \right\} \geq P \left\{ d(\Phi(\Pi^N W), F) \leq \epsilon \right\} \rightarrow 1,$$

as N tends to infinity. Letting ϵ tend to zero yields the result.

Step 2: We have to check that for each $h \in H$ and each $\epsilon > 0$,

$$P \left\{ d(F, \Phi(h)) < \epsilon \right\} > 0.$$

Since $P \circ (T_N^h)^{-1}$ is absolutely continuous with respect to P it suffices to show that

$$P \left\{ d(F \circ T_N^h, \Phi(h)) < \epsilon \right\} > 0$$

for some $N \geq 1$, and this follows from condition (ii). □

Remarks: Both parts of the proof are independent, in the sense that Step 1 uses only assumption (i) and Step 2 uses (ii), and we could have taken different skeletons Φ_1 and Φ_2 in both assumptions. Also we can replace H by a dense subspace H_0 such that $\Pi^N W$ takes values in H_0 , and $\Pi^N W$ by a sequence of random variables ξ_N taking values on H_0 .

4.2 Strict positivity of the density and skeleton

Under stronger hypotheses, and assuming that F is a finite-dimensional random variable, one can characterize the set of points where the density is strictly positive in terms of the skeleton. The following propositions are devoted to this problem. More precisely we want to establish the equivalence between the following conditions:

- (a) The density p of a random vector F satisfies $p(y) > 0$ at some point $y \in \mathbb{R}^m$.
- (b) Fix $y \in \mathbb{R}^m$. There exists an element $h \in H$ such that $\Phi(h) = y$ and $\det \gamma_\Phi(h) > 0$, where $\Phi : H \rightarrow \mathbb{R}^m$ is a differentiable mapping and $\gamma_\Phi^{ij} = \langle D\Phi^i, D\Phi^j \rangle_H$.

Proposition 4.2.1 *Let $F = (F^1, \dots, F^m)$ be a nondegenerate vector in $(\mathbb{D}^\infty)^m$. Suppose that $\Phi : H \rightarrow \mathbb{R}^m$ is an infinitely differentiable function such that Φ and all its derivatives have polynomial growth (i.e., $\|D^{(k)}\Phi(h)\| \leq C(1 + \|h\|_H^{\nu(k)})$, for all $h \in H$, $k \geq 0$). We will also assume that the following condition holds:*

(H1) $\lim_{N \rightarrow \infty} \Phi(\Pi^N W) = F$, in the norm $\|\cdot\|_{k,p}$ for all $k \geq 0, p > 1$.

Then for each $y \in \mathbb{R}^m$ (a) implies (b).

Proof: Fix $y \in \mathbb{R}^m$. We assume that $p(y) = E[\delta_y(F)] > 0$. For every $M \geq 1$ we consider a function $\alpha_M \in C^\infty(\mathbb{R})$ such that $0 \leq \alpha_M \leq 1$, $\alpha_M(x) = 0$ if $|x| \leq \frac{1}{M}$, and $\alpha_M(x) = 1$ if $|x| \geq \frac{2}{M}$. The fact that F is nondegenerate implies that

$$\lim_{M \rightarrow \infty} \alpha_M(\det \gamma_F) = 1,$$

in the norm $\|\cdot\|_{k,p}$ for all k, p . Consequently,

$$0 < E[\delta_y(F)] = \lim_{M \rightarrow \infty} E[\delta_y(F) \alpha_M(\det \gamma_F)],$$

and we can find a positive integer M such that

$$E[\delta_y(F) \alpha_M(\det \gamma_F)] > 0. \quad (4.1)$$

Our assumptions imply that $\Phi(\Pi^N W) \in \mathbb{D}^\infty$ for every N , and for every $k \geq 1$ we have

$$D^k(\Phi(\Pi^N W)) = (\Pi^N)^{\otimes k} (D^k \Phi)(\Pi^N W).$$

We have

$$E[\delta_y(F) \alpha_M(\det \gamma_F)] = \lim_{N \rightarrow \infty} E[\delta_y(\Phi(\Pi^N W)) \alpha_M(\det \gamma_{\Phi(\Pi^N W)})] \quad (4.2)$$

in all the norms $\|\cdot\|_{k,p}$. This convergence follows from hypothesis (H1) and the following integration by parts formulas (see (3.14)):

$$\begin{aligned} E[\delta_y(F) \alpha_M(\det \gamma_F)] &= E[\mathbf{1}_{\{y < F\}} H_{(1,2,\dots,m)}(F, \alpha_M(\det \gamma_F))] \\ E[\delta_y(\Phi(\Pi^N W)) \alpha_M(\det \gamma_{\Phi(\Pi^N W)})] &= E[\mathbf{1}_{\{y < \Phi(\Pi^N W)\}} \\ &\quad \times H_{(1,2,\dots,m)}(\Phi(\Pi^N W), \alpha_M(\det \gamma_{\Phi(\Pi^N W)})]. \end{aligned}$$

Therefore, from (4.1) and (4.2) we can find a positive integer N such that

$$E[\delta_y(\Phi(\Pi^N W)) \alpha_M(\det \gamma_{\Phi(\Pi^N W)})] > 0.$$

Now consider a function $\beta_K \in C^\infty(\mathbb{R})$ such that $0 \leq \beta_K \leq 1$, $\beta_K(x) = 0$ if $|x| \geq K$, and $\beta_K(x) = 1$ if $|x| \leq K - 1$. Then $\beta_K(\|\Pi^N W\|_H^2)$ converges to 1 in $\mathbb{D}^\infty$, and we can find an integer $K \geq 1$ such that

$$E[\delta_y(\Phi(\Pi^N W)) \alpha_M(\det \gamma_{\Phi(\Pi^N W)}) \beta_K(\|\Pi^N W\|_H^2)] > 0.$$

This implies that

$$P \left\{ |\Phi(\Pi^N W) - y| < \epsilon, \det \gamma_{\Phi(\Pi^N W)} \geq \frac{1}{M}, \|\Pi^N W\|_H^2 \leq K \right\} > 0, \quad (4.3)$$

for every $\epsilon > 0$.

Notice that

$$\det \gamma_{\Phi(\Pi^N W)} \leq \det \gamma_{\Phi(\Pi^N W)},$$

because for every $t \in \mathbb{R}^m$ we have

$$\begin{aligned} & \sum_{i,j=1}^m t^i \langle D(\Phi^i(\Pi^N W)), D(\Phi^j(\Pi^N W)) \rangle_H t^j \\ &= \sum_{i,j=1}^m t^i \langle \Pi^N [(D\Phi^i)(\Pi^N W)], \Pi^N [D(\Phi^j)(\Pi^N W)] \rangle_H t^j \\ &= \|\Pi^N (\sum_{i=1}^m t^i (D\Phi^i)(\Pi^N W))\|_H^2 \leq \|\sum_{i=1}^m t^i (D\Phi^i)(\Pi^N W)\|_H^2 \\ &= \sum_{i,j=1}^m t^i \langle (D\Phi^i)(\Pi^N W), (D\Phi^j)(\Pi^N W) \rangle_H t^j. \end{aligned}$$

Hence, (4.3) implies

$$P \left\{ |\Phi(\Pi^N W) - y| < \epsilon, \det \gamma_{\Phi(\Pi^N W)} \geq \frac{1}{M}, \|\Pi^N W\|_H^2 \leq K \right\} > 0. \quad (4.4)$$

Finally, from (4.4) we can find a sequence of elements $h_k \in \text{Im}(\Pi^N)$ such that $|\Phi(h_k) - y| < \frac{1}{k}$ for every k , $\|h_k\|_H^2 \leq K$, and $\det \gamma_{\Phi(h_k)} \geq \frac{1}{M}$. Bounded and closed sets in the image of Π^N are compact, so we can select subsequence converging to some element $h \in H$ which verifies the desired properties. $\square$

In order to formulate and prove the converse implication (that is, that (b) implies (a)) we need some preliminaries. We will denote by $B_a(x)$ the ball of $\mathbb{R}^m$ with center x and radius $a > 0$. The following lemma is a somewhat quantitative version of the classical inverse function theorem. For a function $g : B_1(0) \rightarrow \mathbb{R}^m$ which is twice continuous differentiable we will denote by $\|g\|_{C^2}$ the norm

$$\|g\|_{C^2} = \sup_{z \in B_1(0)} \{|g(z)| + |g'(z)| + |g''(z)|\}. \quad (4.5)$$

Lemma 4.2.1 *For each $\beta > 1$ there exist constants $c_\beta \in (0, \frac{1}{\beta})$, $\delta_\beta > 0$ such that any mapping $g : B_1(0) \rightarrow \mathbb{R}^m$ verifying $g(0) = 0$,*

$$\|g\|_{C^2} \leq \beta, \quad \text{and} \quad |\det g'(0)| \geq \frac{1}{\beta}$$

is diffeomorphic from $B_{c_\beta}(0)$ into a neighborhood of $B_{\delta_\beta}(0)$.

Consider a random vector $F \in (\mathbb{ID}^\infty)^m$. Given m elements $h_1, \dots, h_m \in H$ and $z \in \mathbb{R}^m$ we define the shifted Gaussian process

$$(T_z W)(h) = W(h) + \sum_{j=1}^m z_j \langle h, h_j \rangle_H, \quad h \in H. \quad (4.6)$$

By an elementary change of probability argument we know that for any (integrable or nonnegative) random variable F we have

$$E(F) = E(F(T_z W)J_z), \quad (4.7)$$

where

$$J_z = \exp \left(- \sum_{j=1}^m z_j W(h_j) - \frac{1}{2} \left\| \sum_{j=1}^m z_j h_j \right\|_H^2 \right).$$

We set $\underline{h} = (h_1, \dots, h_m)$, and note that $T_z W$ and J_z depend on $\underline{h}$. We also set, given $p > m$ and $k \geq 0$

$$\mathcal{R}_{\underline{h},k,p} F = \int_{\{|z| \leq 1\}} \left\| (D^k F)(T_z W) \right\|_{H^{\otimes k}}^p dz.$$

Proposition 4.2.2 *Consider a nondegenerate random vector $F \in (\mathbb{ID}^\infty)^m$. Let $\Phi : H \rightarrow \mathbb{R}^m$ be a C^1 mapping. Suppose that the following condition holds:*

(H2) *For all $h \in H$, there exists a sequence of measurable transformations $T_N^h : \Omega \rightarrow \Omega$ such that $P \circ (T_N^h)^{-1}$ is absolutely continuous with respect to P , and for every $\epsilon > 0$,*

$$\begin{aligned} \lim_N P \left\{ |F \circ T_N^h - \Phi(h)| > \epsilon \right\} &= 0 \\ \lim_N P \left\{ \|(DF) \circ T_N^h - (D\Phi)(h)\|_{H^m} > \epsilon \right\} &= 0 \\ \lim_{M \rightarrow \infty} \sup_N P \left\{ (\mathcal{R}_{D\Phi(h),k,p} F) \circ T_N^h > M \right\} &= 0 \end{aligned}$$

for some $p > m$, and for $k = 0, 1, 2, 3$.

Then (b) implies (a).

Proof: We fix a point $x \in \mathbb{R}^m$ and an element $h \in H$ such that $\Phi(h) = x$ and $\det \gamma_\Phi(h) > 0$. Consider the elements of H given by $h_j = (D\Phi^j)(h)$, $j = 1, \dots, m$. Using these elements we can introduce the random mapping $g : \mathbb{R}^m \rightarrow \mathbb{R}^m$ defined by $g(z) = F(T_z W) - F$, where $T_z W$ is defined in (4.6). Notice that the random function g has an infinitely differentiable version and for each multi-index $\alpha = (\alpha_1, \dots, \alpha_k) \in \{1, 2, \dots, m\}^k$ its derivatives are given by

$$(\partial_\alpha g)(z) = \left\langle (D^k F)(T_z W), h_{\alpha_1} \otimes \dots \otimes h_{\alpha_k} \right\rangle_{H^{\otimes k}}. \quad (4.8)$$

By Sobolev's inequality, if $p > m$ is the exponent appearing in hypothesis (H2), we have

$$\sup_{|z| \leq 1} |g(z)| \leq c_p \left(\int_{\{|z| \leq 1\}} |g(z)|^p dz + \int_{\{|z| \leq 1\}} \sum_{j=1}^m |\partial_j g(z)|^p dz \right)^{\frac{1}{p}}. \quad (4.9)$$

Consequently, using (4.8) and (4.9) we can estimate the norm (4.5) in the following way

$$\|g\|_{C^2} \leq c_p \left(\sum_{k=0}^3 \int_{\{|z| \leq 1\}} \sum_{|\alpha| = k} |\partial_\alpha g(z)|^p dz \right)^{\frac{1}{p}} \leq G,$$

where G is the random variable defined by

$$G = c_p \left(|F|^p + \sum_{k=0}^3 \left(\sum_{j=1}^m \|h_j\|_H^p \right)^k \int_{\{|z| \leq 1\}} \|(D^k F)(T_z W)\|_{H^{\otimes k}}^p dz \right)^{\frac{1}{p}}.$$

For any $\beta > 0$ we will denote by α_β a continuous function such that $0 \leq \alpha_\beta \leq 1$, $\alpha_\beta(x) = 0$ if $|x| \leq \frac{1}{\beta}$, and $\alpha_\beta(x) = 1$ if $|x| \geq \frac{2}{\beta}$. Also k_β will be a continuous function such that $0 \leq k_\beta \leq 1$, $k_\beta(x) = 0$ if $|x| \geq \beta$, and $k_\beta(x) = 1$ if $|x| \leq \beta - 1$. Set

$$H_\beta = k_\beta(G) \alpha_\beta(|\det \langle DF^i, D\Phi^j \rangle_H|).$$

Suppose that $\rho : \mathbb{R}^m \rightarrow \mathbb{R}$ is a strictly positive continuous function such that $\int_{\mathbb{R}^m} \rho(z) dz = 1$. Let $f : \mathbb{R}^m \rightarrow \mathbb{R}$ be a nonnegative continuous and bounded function. We can write, applying (4.7)

$$\begin{aligned} E[f(F)] &= \int_{\mathbb{R}^m} E[f(F)] \rho(z) dz \\ &= \int_{\mathbb{R}^m} E[f(F(T_z W)) J_z] \rho(z) dz \\ &\geq E \left[H_\beta \int_{\mathbb{R}^m} f(F(T_z W)) J_z \rho(z) dz \right] \end{aligned}$$

Fix $\beta > 1$ and let c_β and δ_β the constants provided by Lemma 4.2.1. We can apply this lemma to the function $g(z) = F(T_z W) - F$ (notice that this function verifies the hypotheses of the lemma if $H_\beta \neq 0$, because $\|g\|_{C^2} \leq G$, and $\det g'(0) = \det \langle DF^i, D\Phi^j \rangle_H$). Hence, making the change of variable $y = g(z)$ we obtain

$$\begin{aligned} &E \left[H_\beta \int_{\mathbb{R}^m} f(F(T_z W)) J_z \rho(z) dz \right] \\ &\geq E \left[H_\beta \int_{\{|z| \leq c_\beta\}} f(g(z) + F) J_z \rho(z) dz \right] \\ &\geq E \left[H_\beta \int_{\{|y| \leq \delta_\beta\}} f(y + F) J_{g^{-1}(y)} \rho(g^{-1}(y)) |\det \partial_j g^i(g^{-1}(y))| dy \right]. \end{aligned}$$

From the above inequalities we deduce

$$p(x) \geq E \left[H_\beta \mathbf{1}_{\{|F-x| \leq \delta_\beta\}} J_{g^{-1}(x-F)} \rho(g^{-1}(x-F)) |\det \partial_j g^i(g^{-1}(x-F))| \right]$$

Notice that if $H_\beta \neq 0$ and $|x - F| \leq \delta_\beta$ then

$$J_{g^{-1}(x-F)} \rho(g^{-1}(x-F)) |\det \partial_j g^i(g^{-1}(x-F))| > 0.$$

Hence, in order to deduce $p(x) > 0$ it only remains to show that we can choose β in such a way that

$$P \left\{ |F - x| \leq \delta_\beta, G \leq \beta, |\det \langle DF^i, D\Phi^j \rangle_H| \geq \frac{1}{\beta} \right\} > 0.$$

By the absolute continuity of $P \circ (T_N^h)^{-1}$ with respect to P it suffices to show that

$$P \left\{ |F \circ T_N^h - x| \leq \delta_\beta, G \circ T_N^h \leq \beta, |\det \langle (DF^i) \circ T_N^h, D\Phi^j \rangle_H| \geq \frac{1}{\beta} \right\} > 0$$

for some $N \geq 1$. Finally we can write

$$\begin{aligned} & P \left\{ |F \circ T_N^h - x| \leq \delta_\beta, G \circ T_N^h \leq \beta, |\det \langle (DF^i) \circ T_N^h, D\Phi^j \rangle_H| \geq \frac{1}{\beta} \right\} \\ & \geq 1 - P \left\{ |F \circ T_N^h - x| > \delta_\beta \right\} - P \left\{ G \circ T_N^h > \beta \right\} \\ & \quad - P \left\{ |\det \langle (DF^i) \circ T_N^h, D\Phi^j \rangle_H| \geq \frac{1}{\beta} \right\}, \end{aligned}$$

and letting N tend to infinity and using hypothesis (H2) we complete the proof. $\square$

4.3 Skeleton and support for diffusion processes

Let us now describe the application of the above results to the case of a diffusion process. In the sequel we will assume that the underlying Gaussian process is a d -dimensional Brownian motion $\{W(t), t \in [0, 1]\}$, defined in the canonical probability space $\Omega = C_0([0, 1], \mathbb{R}^d)$. Moreover, for each positive integer N , and for any element $h \in H = L^2([0, 1]; \mathbb{R}^d)$ we set

$$\Pi^N(h) = \sum_{i=1}^{2^N-1} 2^N \left(\int_{[(i-1)2^{-N}, i2^{-N}]} h(s) ds \right) \mathbf{1}_{(i2^{-N}, (i+1)2^{-N})}.$$

With this definition $(\Pi^N W)(\omega) = \dot{\omega}^N$, where for any continuous function $\omega \in \Omega$ we denote by ω^N the element of Ω such that on each interval $(i2^{-N}, (i+1)2^{-N})$, $1 \leq i \leq 2^N - 1$, has a constant derivative equal to $(\omega(i2^{-N}) - \omega((i-1)2^{-N}))2^N$.

We will denote by i the isometry between H and the Cameron-Martin space H^1 . That is, for each $h \in H$ we have $i(h)(t) = \int_0^t h(s) ds$.

Let $A_j, B : \mathbb{R}^m \rightarrow \mathbb{R}^m$, $j = 1, \dots, d$ a system of functions such that A_j is of class C_b^2 (i.e., the partial derivatives of first and second order are continuous and bounded), and B is Lipschitz. Consider the following stochastic differential equation on $\mathbb{R}^m$:

$$X_t = x_0 + \int_0^t B(X_s) ds + \sum_{j=1}^d \int_0^t A_j(X_s) dW_s^j. \quad (4.10)$$

Let $\Phi(h)$ be the solution of

$$\Phi(h)_t = x_0 + \int_0^t A_0(\Phi(h)_s) ds + \sum_{j=1}^d \int_0^t A_j(\Phi(h)_s) h_s^j ds, \quad (4.11)$$

where

$$A_0^i = B_0^i - \frac{1}{2} \sum_{k=1}^m \sum_{j=1}^d \partial_k A_j^i A_j^k.$$

Notice that A_0 drift of the diffusion process X_t if we write the stochastic differential equation (4.10) in the Stratonovich form.

Then one can show that the mapping $\Phi : H \rightarrow C([0, 1]; \mathbb{R}^m)$ is a skeleton of the process X in the sense of Definition 4.1.1. That is, the following convergences hold ([71]):

$$\lim_N P \left(\sup_{t \in [0, 1]} |\Phi(\Pi^N W)_t - X_t| > \epsilon \right) = 0,$$

and

$$\lim_N P \left(\sup_{t \in [0,1]} |X_t \circ T_N^h - \Phi(h)_t| > \epsilon \right) = 0,$$

where $T_N^h(\omega) = \omega - \omega^N + i(h)$. Actually, the above convergences hold if we replace $C([0,1]; \mathbb{R}^m)$ by the space $C^\alpha([0,1]; \mathbb{R}^m)$ of α -Hölder continuous functions equipped with the α -Hölder norm

$$\|x\|_\alpha = \sup_{t \in [0,1]} |x_t| + \sup_{s \neq t} \frac{|x_t - x_s|}{|t - s|^\alpha},$$

for $\alpha \in [0, \frac{1}{2})$.

If the coefficients A_j , $j = 0, \dots, m$ are of class C^∞ with bounded partial derivatives of all orders, then, for any fixed $t \in [0, 1]$ the convergences of $\Phi(\Pi^N W)_t$ to X_t and $X_t \omega - \omega^N + i(h)$ to $\Phi(h)_t$ holds in the topology of $\mathbb{D}^\infty$. Hence, if the nondegeneracy condition (H) holds then we can apply Propositions 4.2.1 and 4.2.2 to the random vector X_t for all $t \in (0, 1]$ and we deduce the following characterization of the points where the density of X_t is strictly positive (see [11]):

Proposition 4.3.1 *The density p_t of the diffusion process X_t satisfies $p_t(y) > 0$ at some point $y \in \mathbb{R}^m$ if and only if there exists an element $h \in H$ such that $\Phi(h)_t = y$ and $\det \langle D\Phi^i(h)_t, D\Phi^j(h)_t \rangle_H > 0$.*

Example: Consider the following example (cf. [1]). Suppose $\alpha(x)$, $\beta(t, x)$, $\gamma(x, y)$ are smooth functions with bounded derivatives of all orders such that

$$\begin{aligned} \alpha(x) &= x^2 \quad \text{if } x \in [-1, 1], \\ \beta(t, x) &\neq 0 \quad \text{iff } (t, x) \in [0, \frac{1}{2}) \times (-1, 1), \\ \gamma(x, y) &\neq 0 \quad \text{iff } (x, y) \notin [-1, 0] \times [-1, 1]. \end{aligned}$$

Consider the process $\{X_t, t \in [0, 1]\}$ solution to the equation

$$\begin{aligned} dX_t^1 &= \gamma(X_t^1, X_t^2) \circ dW_t^2 + \alpha(X_t^2) dt \\ dX_t^2 &= \beta(t, X_t^2) \circ dW_t^1 + \gamma(X_t^1, X_t^2) \circ dW_t^3 \\ X_0 &= 0. \end{aligned}$$

Hörmander's condition is satisfied in this example, and, applying the techniques of the Malliavin calculus for the case of time dependent coefficients (cf. [31]), one can show that the vector (X_1^1, X_1^2) is nondegenerate. We have $\{\Phi_1(h), h \in H\} = \mathbb{R}^2$, and the support of the law of X_1 is $\mathbb{R}^2$. However at the point 0 the density vanish because the unique element $h \in H$ such that $\Phi_1(h) = 0$ is $h_0 = 0$, and $\det \langle D\Phi_1^i(h_0), D\Phi_1^j(h_0) \rangle = 0$.

4.4 Varadhan estimates

Fix a small parameter $\epsilon \in [0, 1]$, and consider the solution X_t^ϵ to the stochastic differential equation

$$X_t^\epsilon = x + \int_0^t B(X_s^\epsilon) ds + \epsilon \sum_{j=1}^d \int_0^t A_j(X_s^\epsilon) dW_s^j. \quad (4.12)$$

That is, $X_t^\epsilon = X_t(\epsilon W)$. Suppose that the coefficients A_i and B are infinitely differentiable with bounded derivatives of all orders, and Hörmander's condition (H) holds. We can introduce the following functions on $\mathbb{R}^m$, which depend on the skeleton $\Phi_t(h)$:

$$d^2(y) = \inf_{\Phi_1(h)=y} \|h\|_H^2,$$

and

$$d_R^2(y) = \inf_{\Phi_1(h)=y, \det \gamma_{\Phi_1}(h) > 0} \|h\|_H^2.$$

Then the following theorem holds (cf. Léandre [58], Ben Arous and Léandre [11] and Léandre and Russo [59]):

Theorem 4.4.1 *Let us denote by $p_\epsilon(y)$ the density of X_1^ϵ . Then*

$$\liminf_{\epsilon \downarrow 0} 2\epsilon^2 \log p_\epsilon(y) \geq -d_R^2(y), \quad (4.13)$$

and

$$\limsup_{\epsilon \downarrow 0} 2\epsilon^2 \log p_\epsilon(y) \leq -d^2(y). \quad (4.14)$$

Moreover, if $\inf_{\Phi_1(h)=y, \det \gamma_{\Phi_1}(h) > 0} \det \gamma_{\Phi_1}(h) > 0$ then

$$\lim_{\epsilon \downarrow 0} 2\epsilon^2 \log p_\epsilon(y) = -d_R^2(y). \quad (4.15)$$

We will sketch the proof of this theorem which provides the asymptotic behaviour for the density of a perturbed dynamical system when the noise is small. In order to show the limit results stated in the theorem we will consider a more general case which contains the diffusion case as a particular example.

In the sequel we will work in the framework of an arbitrary Gaussian family $\{W(h), h \in H\}$. Let us first prove the minoration inequality.

Proposition 4.4.1 *Consider a family $\{F_\epsilon, 0 < \epsilon \leq 1\}$ of nondegenerate random vectors, and a function $\Phi \in C_p^\infty(H; \mathbb{R}^m)$ such that:*

$$\lim_{\epsilon \downarrow 0} \frac{1}{\epsilon} \left(F_\epsilon \left(W + \frac{h}{\epsilon} \right) - \Phi(h) \right) = Z(h),$$

in the topology of $\mathbb{D}^\infty$, for each $h \in H$, where $Z(h)$ is an m -dimensional random vector in the first Wiener chaos with variance $\gamma_\Phi(h)$. Define

$$d_R^2(y) = \inf_{\Phi(h)=y, \det \gamma_\Phi(h) > 0} \|h\|_H^2, \quad y \in \mathbb{R}^m.$$

Then

$$\liminf_{\epsilon \downarrow 0} 2\epsilon^2 \log p_\epsilon(y) \geq -d_R^2(y). \quad (4.16)$$

Notice that the hypothesis of Proposition 4.4.1 implies

$$\lim_{\epsilon \downarrow 0} F_\epsilon \left(W + \frac{h}{\epsilon} \right) = \Phi(h),$$

in the topology of $\mathbb{D}^\infty$, for each $h \in H$.

Proof: Fix $y \in \mathbb{R}^m$, and $\eta > 0$. If $d_R^2(y) = \infty$ there is nothing to say. Suppose $d_R^2(y) < \infty$. Let $h \in H$ be such that $\Phi(h) = y$, $\det \gamma_\Phi(h) > 0$, and $\|h\|_H^2 \leq d_R^2(y) + \eta$. Let $f \in C_0^\infty(\mathbb{R}^m)$. By Girsanov's theorem

$$E(f(F_\epsilon)) = e^{-\frac{\|h\|_H^2}{2\epsilon^2}} E\left(f(F_\epsilon(W + \frac{h}{\epsilon}))e^{-\frac{W(h)}{\epsilon}}\right).$$

Consider a function $\chi \in C^\infty(\mathbb{R})$, $0 \leq \chi \leq 1$, such that $\chi(t) = 0$ if $t \notin [-2\eta, 2\eta]$, and $\chi(t) = 1$ if $t \in [-\eta, \eta]$. Then, if $f \geq 0$, we have

$$E(f(F_\epsilon)) \geq e^{-\frac{\|h\|_H^2 + 4\eta}{2\epsilon^2}} E\left(\chi(\epsilon W(h))f(F_\epsilon(W + \frac{h}{\epsilon}))\right).$$

This implies that

$$2\epsilon^2 \log p_\epsilon(y) \geq -(\|h\|_H^2 + 4\eta) + 2\epsilon^2 \log E\left(\chi(\epsilon W(h))\delta_y(F_\epsilon(W + \frac{h}{\epsilon}))\right).$$

Hence, it suffices to show that

$$\lim_{\epsilon \downarrow 0} \epsilon^2 \log E\left(\chi(\epsilon W(h))\delta_y(F_\epsilon(W + \frac{h}{\epsilon}))\right) = 0.$$

We have

$$E\left(\chi(\epsilon W(h))\delta_y(F_\epsilon(W + \frac{h}{\epsilon}))\right) = \epsilon^{-m} E\left(\chi(\epsilon W(h))\delta_0\left(\frac{F_\epsilon(W + \frac{h}{\epsilon}) - \Phi(h)}{\epsilon}\right)\right).$$

Then, using hypothesis (ii) the expectation

$$E\left(\chi(\epsilon W(h))\delta_0\left(\frac{F_\epsilon(W + \frac{h}{\epsilon}) - \Phi(h)}{\epsilon}\right)\right)$$

converges, as ϵ tends to zero, to $E(\delta_0(Z(h)))$, and this completes the proof of the proposition. $\square$

The majoration estimate requires large deviation assumptions:

Proposition 4.4.2 *Consider a family $\{F_\epsilon, 0 < \epsilon \leq 1\}$ of nondegenerate random vectors, and a function $\Phi \in C_p^\infty(H; \mathbb{R}^m)$ such that:*

- (i') $\sup_{\epsilon \in (0,1]} \|F_\epsilon\|_{k,p} < \infty$, for each $k \geq 1$ and $p > 1$.
- (ii') $\|(\gamma_{F_\epsilon})^{-1}\|_k \leq \epsilon^{-N(k)}$ for any integer $k \geq 1$.
- (iii) The family $\{F_\epsilon, 0 < \epsilon \leq 1\}$ satisfies a large deviation principle with rate function

$$d^2(y) = \inf_{\Phi(h)=y} \|h\|_H^2, \quad y \in \mathbb{R}^m.$$

Then

$$\limsup_{\epsilon \downarrow 0} 2\epsilon^2 \log p_\epsilon(y) \leq -d^2(y). \quad (4.17)$$

Proof: Fix a point $y \in \mathbb{R}^m$ and consider a function $\chi \in C_0^\infty(\mathbb{R}^m)$, $0 \leq \chi \leq 1$ such that χ is equal to one in a neighborhood of y . The density of F_ϵ at point y is given by

$$p_\epsilon(y) = E(\chi(F_\epsilon)\delta_y(F_\epsilon)).$$

Using Proposition 3.2.1 we can write

$$\begin{aligned} E(\chi(F_\epsilon)\delta_y(F_\epsilon)) &= E\left(\mathbf{1}_{\{F_\epsilon > y\}} H_{(1,2,\dots,m)}(F_\epsilon, \chi(F_\epsilon))\right) \\ &\leq E(|H_{(1,2,\dots,m)}(F_\epsilon, \chi(F_\epsilon))|) = E(|H_{(1,2,\dots,m)}(F_\epsilon, \chi(F_\epsilon))| \mathbf{1}_{\{F_\epsilon \in \text{supp}\chi\}}) \\ &\leq (P(F_\epsilon \in \text{supp}\chi))^{\frac{1}{q}} \|H_{(1,2,\dots,m)}(F_\epsilon, \chi(F_\epsilon))\|_p, \end{aligned}$$

where $\frac{1}{p} + \frac{1}{q} = 1$. By Proposition 3.2.2 we know that

$$\|H_{(1,2,\dots,m)}(F_\epsilon, \chi(F_\epsilon))\|_p \leq C(p) \|\gamma_{F_\epsilon}^{-1}\|_k \|F_\epsilon\|_{d,b} \|\chi(F_\epsilon)\|_{d',b'},$$

for some constants b, d, b', d' . Thus, hypothesis (ii') implies that

$$\lim_{\epsilon \downarrow 0} \epsilon^2 \log \|H_{(1,2,\dots,m)}(F_\epsilon, \chi(F_\epsilon))\|_p = 0.$$

Finally, the large deviation principle for F_ϵ ensures that for ϵ small enough we have

$$(P(F_\epsilon \in \text{supp}\chi))^{\frac{1}{q}} \leq e^{-\frac{1}{2q\epsilon^2} (\inf_{y \in \text{supp}\chi} d^2(y))}.$$

□

Proposition 4.4.3 Consider a family $\{F_\epsilon, 0 < \epsilon \leq 1\}$ of nondegenerate random vectors, and a function $\Phi \in C_p^\infty(H; \mathbb{R}^m)$ such that:

- (i') $\sup_{\epsilon \in (0,1]} \|F_\epsilon\|_{k,p} < \infty$, for each $k \geq 1$ and $p > 1$.
- (ii') $\|(\gamma_{F_\epsilon})^{-1}\|_k \leq \epsilon^{-N(k)}$ for any integer $k \geq 1$.
- (iii') The family $\{(F_\epsilon, \gamma_{F_\epsilon}), 0 < \epsilon \leq 1\}$ satisfies a large deviation principle with rate function

$$\lambda^2(y, a) = \inf_{\Phi(h)=y, \gamma_\Phi(h)=a} \|h\|_H^2, \quad (y, a) \in \mathbb{R}^m \times \mathbb{R}^{m^2}.$$
- (iv) $\lim_{N \rightarrow \infty} \Phi(\epsilon \Pi^N W) = F_\epsilon$, in the norm $\|\cdot\|_{k,p}$ for all $k \geq 0$, $p > 1$, where $\{\Pi^N, N \geq 1\}$ is a sequence of orthogonal projections of H of finite-dimensional range strongly convergent to the identity.

Define $d_R^2(y) = \inf_{\Phi(h)=y, \det \gamma_\Phi(h) > 0} \|h\|_H^2$, $y \in \mathbb{R}^m$ and suppose that

$$\gamma := \inf_{\Phi(h)=y, \det \gamma_\Phi(h) > 0} \det \gamma_\Phi(h) > 0.$$

Then

$$\limsup_{\epsilon \downarrow 0} 2\epsilon^2 \log p_\epsilon(y) \leq -d_R^2(y). \quad (4.18)$$

Proof: The proof is similar to that of Proposition 4.4.2. Consider a function $g \in C^\infty(\mathbb{R})$, $0 \leq g \leq 1$, such that $g(u) = 1$ if $|u| < \frac{1}{4}\gamma$, and $g(u) = 0$ if $|u| > \frac{1}{2}\gamma$. Set $G_\epsilon = g(\det \gamma_{F_\epsilon})$. With the notations of the proof of Proposition 4.4.2 we have

$$E(\chi(F_\epsilon)\delta_y(F_\epsilon)) = E(G_\epsilon\chi(F_\epsilon)\delta_y(F_\epsilon)) + E((1 - G_\epsilon)\chi(F_\epsilon)\delta_y(F_\epsilon)).$$

We have $E(G_\epsilon\chi(F_\epsilon)\delta_y(F_\epsilon)) = 0$, because, otherwise, using condition (i') and applying the method of the proof of Proposition 4.2.1, one can find an element $\epsilon h \in H$ such that $\Phi(\epsilon h) = y$ and $0 < \det \gamma_\Phi(\epsilon h) < \frac{\gamma}{2}$, and this is in contradiction with the definition of γ .

Proceeding as in the Proposition 4.4.2 we obtain

$$\begin{aligned} E((1 - G_\epsilon)\chi(F_\epsilon)\delta_y(F_\epsilon)) &= E\left(\mathbf{1}_{F_\epsilon > y} H_{(1,2,\dots,m)}(F_\epsilon, (1 - G_\epsilon)\chi(F_\epsilon))\right) \\ &\leq E(|H_{(1,2,\dots,m)}(F_\epsilon, (1 - G_\epsilon)\chi(F_\epsilon))|) \\ &= E(|H_{(1,2,\dots,m)}(F_\epsilon\chi(F_\epsilon))\mathbf{1}_{F_\epsilon \in \text{supp}\chi, \det \gamma_{F_\epsilon} \geq \frac{1}{4}\gamma}|) \\ &\leq \left(P\left(F_\epsilon \in \text{supp}\chi, \det \gamma_{F_\epsilon} \geq \frac{1}{4}\gamma\right)\right)^{\frac{1}{q}} \|H_{(1,2,\dots,m)}(F_\epsilon, \chi(F_\epsilon))\|_p. \end{aligned}$$

Finally, hypotheses (ii') and (iii') imply that for any $q > 1$ we have

$$\begin{aligned} \limsup_{\epsilon \downarrow 0} 2\epsilon^2 \log p_\epsilon(y) &\leq \exp\left(-\frac{1}{2q\epsilon^2} \left(\inf_{\Phi(h) \in \text{supp}\chi, \det \gamma_\Phi(h) \geq \frac{1}{4}\gamma} \|h\|_H^2\right)\right) \\ &\leq \exp\left(-\frac{1}{2q\epsilon^2} \left(\inf_{y \in \text{supp}\chi} d_R^2(y)\right)\right). \end{aligned}$$

□

When the random variable F_ϵ is X_1^ϵ , conditions (i),(ii),(iii), (iv), (i'), (ii') and (iii') are satisfied, and in this way one can show Proposition 4.4.1. More precisely, condition (ii') is proved by Kusuoka and Stroock in [57], the large deviation principle for both the diffusion X_t^ϵ and the Malliavin matrix $\gamma_{X_t^\epsilon}$ are known, and the convergences (i), (ii) and (iv) are easy to check.

Bibliographical notes The ideas used in this chapter in order to characterize the support of the law of a diffusion go back to the works of Mackevičius [61] and Gyöngy [35]. The support theorem for the Hölder norms has been studied in [8], [71] and [38]. The results on the skeleton are based on the main references [1] and [11]. The characterization of the support of the law of hyperbolic stochastic partial differential equations has been established by Millet and Sanz-Solé in [70], and the characterization of the points of positive density for the solution to such equations has been done by these authors in [72]. The support theorem for parabolic stochastic partial differential equations is proved by Bally, Millet, and Sanz-Solé in [6].

Chapter 5

Anticipating stochastic calculus

We have seen in Chapter 1 that the divergence operator δ on the classical Wiener space is an extension of the Itô stochastic integral, in the sense that the class L_a^2 of square integrable and adapted process is included in $\text{Dom } \delta$, and the operator δ restricted to L_a^2 coincides with the Itô integral. Actually, the operator δ coincides with an extension of the Itô integral introduced by Skorohod in [98] using the Wiener chaos expansion. One can develop a stochastic calculus for both the Skorohod integral and the generalized Stratonovich integral. In this chapter we will present the basic facts of the stochastic calculus for anticipating integrals.

5.1 Skorohod integral processes

In this section we will assume that $W = \{W(t), t \in [0, 1]\}$ is a Brownian motion defined on the canonical probability space $(\Omega, \mathcal{F}, P)$. In order to simplify the presentation we will assume that W is one-dimensional, but most of the results can be easily generalized to the multidimensional case.

We will say that a square integrable process $u = \{u(t), 0 \leq t \leq 1\}$ is Skorohod integrable if u belongs to $\text{Dom } \delta$, and we will write

$$\delta(u) = \int_0^1 u_t dW_t.$$

It may happen that $u \in \text{Dom } \delta$, but $u \mathbf{1}_{[0,t]}$ is not Skorohod integrable for some $t \in [0, 1]$.

Example: Consider the process

$$u_t = \mathbf{1}_{[0, \frac{1}{2}]}(t) \mathbf{1}_{\{W_1 > 0\}} - \mathbf{1}_{(\frac{1}{2}, 1]}(t) \mathbf{1}_{\{W_1 > 0\}}.$$

The process u is Skorohod integrable and

$$\delta(u) = (2W_{\frac{1}{2}} - W_1) \mathbf{1}_{\{W_1 > 0\}}. \quad (5.1)$$

Indeed, if we denote by φ_k a smooth nonnegative function such that $0 \leq \varphi_k \leq 1$, $\varphi_k(x) = 0$ for $x \leq 0$, and $\varphi_k(x) = 1$ for $x \geq \frac{1}{k}$, then the sequence of processes

$$u^k(t) = \mathbf{1}_{[0, \frac{1}{2}]}(t) \varphi_k(W_1) - \mathbf{1}_{(\frac{1}{2}, 1]}(t) \varphi_k(W_1),$$

converges in $L^2([0, 1] \times \Omega)$ to u , and

$$\delta(u^k) = (2W_{\frac{1}{2}} - W_1) \varphi_k(W_1)$$

converges in $L^2(\Omega)$ to the right-hand side of (5.1). Hence, u is Skorohod integrable and (5.1) holds. Nevertheless the process $u\mathbf{1}_{[0, \frac{1}{2}]}$ does not belong to $\text{Dom } \delta$. In fact, the divergence of this process is the element of $\mathbb{D}^{-\infty}$ given by

$$\delta(u\mathbf{1}_{[0, \frac{1}{2}]}) = W_{\frac{1}{2}}\mathbf{1}_{\{W_1 > 0\}} - \frac{1}{2}\delta_0(W_1).$$

Let us denote by $\mathbb{L}^s$ the set of processes u such that $u\mathbf{1}_{[0, t]}$ is Skorohod integrable for any $t \in [0, 1]$. Notice that the space $\mathbb{D}^{1,2}(H)$ is included into $\mathbb{L}^s$. Suppose that u belongs to $\mathbb{L}^s$ and define

$$X(t) = \delta(u\mathbf{1}_{[0, t]}) = \int_0^t u_s dW_s. \quad (5.2)$$

The process $X = \{X_t, t \in [0, 1]\}$ is not adapted, and its increments satisfy the following orthogonality property:

Lemma 5.1.1 *For any process $u \in \mathbb{L}^s$ we have*

$$E\left(\int_s^t u_r dW_r \middle| \mathcal{F}_{[s, t]^c}\right) = 0, \quad (5.3)$$

for all $s < t$, where, as usual, $\mathcal{F}_{[s, t]^c}$ denotes the σ -field generated by the increments of the Brownian motion in the complement of the interval $[s, t]$.

Proof: To show (5.3) it suffices to take an arbitrary $\mathcal{F}_{[s, t]^c}$ -measurable random variable F belonging to the space $\mathbb{D}^{1,2}$, and check that

$$E\left(F \int_s^t u_r dW_r\right) = E\left(\int_0^1 u_r D_r F \mathbf{1}_{[s, t]}(r) dr\right) = 0,$$

which holds due to the duality relation (1.7) and Lemma 1.5.1. $\square$

The Skorohod integral process X is continuous in L^2 for any $u \in \mathbb{L}^s$. However, there exist processes u in $\mathbb{L}^s$ such that the indefinite integral $\int_0^t u_s dW_s$ does not have a continuous version.

Example: Consider the process

$$u(t) = \text{sign}(W_1 - t) \exp\left(W_t - \frac{t}{2}\right).$$

This process belongs to the space $\mathbb{L}^s$, and

$$X_t := \int_0^t u_s dW_s = \text{sign}(W_1 - t) \exp\left(W_t - \frac{t}{2}\right) - \text{sign} W_1. \quad (5.4)$$

As a consequence, the process X has a discontinuity on a point t such that $W_1 = t$. We can show (5.4) by means of a duality argument. In fact, let $G \in \mathcal{S}$ be a smooth random variable. Consider the family of transformations of the Wiener space given by $(T_t W)_s = W_s + s \wedge t$, $t \in [0, 1]$. Notice that if $G = g(W(h_1), \dots, W(h_n))$, $g \in C_b(\mathbb{R}^n)$ then

$$\begin{aligned} \frac{d}{ds} G(T_s W) &= \frac{d}{ds} g\left(W(h_1) + \int_0^s h_1(r) dr, \dots, W(h_n) + \int_0^s h_n(r) dr\right) \\ &= \sum_{i=1}^n (\partial_i g) \left(W(h_1) + \int_0^s h_1(r) dr, \dots, W(h_n) + \int_0^s h_n(r) dr\right) h_i(s) \\ &= (D_s G)(T_s W). \end{aligned}$$

Using Girsanov's theorem we can write

$$\begin{aligned}
 E\left(\int_0^t D_s G u_s ds\right) &= E\left(\int_0^t D_s G \operatorname{sign}(W_1 - s) e^{W_s - \frac{s}{2}} ds\right) \\
 &= E\left(\int_0^t (D_s G)(T_s W) \operatorname{sign}(W_1) ds\right) = E(G(T_t W) \operatorname{sign} W_1 - G \operatorname{sign} W_1) \\
 &= E\left(G\left(\operatorname{sign}(W_1 - t) \exp\left(W_t - \frac{t}{2}\right) - \operatorname{sign} W_1\right)\right),
 \end{aligned}$$

and (5.4) follows.

A useful tool in proving the existence of a continuous version for stochastic processes is the real analysis lemma of Garsia, Rodemich and Rumsey [32]. Let us recall this lemma:

Lemma 5.1.2 *Let $p, \Psi : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ be continuous and strictly increasing functions vanishing at zero and such that $\lim_{t \uparrow \infty} \Psi(t) = \infty$. Suppose that $\phi : \mathbb{R}^m \rightarrow \mathbb{B}$ is a measurable function with values in a separable Banach space $(\mathbb{B}, \|\cdot\|)$. Let $B_K = \{x \in \mathbb{R}^m, |x| \leq K\}$. Suppose that*

$$\Gamma = \int_{B_K} \int_{B_K} \Psi\left(\frac{\|\phi(t) - \phi(s)\|}{p(|t - s|)}\right) ds dt < \infty.$$

Then, there is a set $N \subset B_K$ of measure zero such that, for all $s, t \in B_K - N$,

$$\|\phi(t) - \phi(s)\| \leq 8 \int_0^{2|t-s|} \Psi^{-1}(\lambda_m \Gamma u^{-2m}) p(du),$$

where λ_m is a universal constant depending only on m .

Now suppose that $X = \{X(t), t \in B_K\}$ is a stochastic process with values on $\mathbb{B}$, such that the following estimate holds:

$$E(\|X(t) - X(s)\|^\gamma) \leq H|t - s|^\alpha$$

for some constants $H > 0$, $\gamma > 0$, $\alpha > m$, and for all $s, t \in B_K$. Then taking $\Psi(x) = x^\gamma$ and $p(x) = x^{\frac{d+2m}{\gamma}}$, with $0 < d < \alpha - m$, one obtains for all $s, t \in B_K - N_\omega$,

$$\|X(t) - X(s)\|^\gamma \leq C_1 |t - s|^d \Gamma, \quad (5.5)$$

where N_ω is a subset of B_K of zero Lebesgue measure depending on ω , and $E(\Gamma) \leq HC_2$ for some constants C_1 and C_2 depending only on γ, d , and m . In particular, the above inequality implies that the process X has a continuous version (this is the classical Kolmogorov continuity criterion).

For every $p > 1$ and any positive integer k we will denote by $\mathbb{L}^{k,p}$ the space $L^p([0, 1]; \mathbb{D}^{k,p})$. Notice that $\mathbb{L}^{1,2} = \mathbb{D}^{1,2}(H)$, and $L^p([0, 1]; \mathbb{D}^{1,p}) \subset \mathbb{D}^{1,p}(H)$ for $p \geq 2$.

Proposition 5.1.1 *Let u be a process in the class $\mathbb{L}^{1,2}$. Suppose that $E \int_0^1 \|Du_t\|_H^p dt < \infty$ for some $p > 2$. Then the integral process $\{\int_0^t u_s dW_s, 0 \leq t \leq 1\}$ has a continuous version.*

Proof: We can assume that $E(u_t) = 0$ for each $t \in [0, 1]$ because the Gaussian process $\int_0^t E(u_s) dW_s$ has a continuous version. Applying the estimate (2.22) we obtain

$$E(|X_t - X_s|^p) \leq c_p E \left(\left| \int_s^t \int_0^1 (D_\theta u_r)^2 d\theta dr \right|^{\frac{p}{2}} \right) \leq c_p |t-s|^{\frac{p}{2}-1} \int_s^t E \left(\left| \int_0^1 (D_\theta u_r)^2 d\theta \right|^{\frac{p}{2}} \right) dr.$$

Set

$$A_r = \left| \int_0^1 (D_\theta u_r)^2 d\theta \right|^{\frac{p}{2}}.$$

Fix an exponent $2 < \alpha < 1 + \frac{p}{2}$, and assume that p is close to 2. Applying Fubini's theorem we can write

$$\begin{aligned} E \left(\int_0^1 \int_0^1 \frac{|X_t - X_s|^p}{|t-s|^\alpha} ds dt \right) &\leq c_p E \left(\int_0^1 \int_0^1 |t-s|^{\frac{p}{2}-1-\alpha} \int_s^t A_r dr ds dt \right) \\ &= \frac{2c_p}{\frac{p}{2}-\alpha} E \left(\int_{\{s < r\}} (|1-s|^{\frac{p}{2}-\alpha} - |r-s|^{\frac{p}{2}-\alpha}) A_r dr ds \right) \\ &= \frac{2c_p}{(\alpha - \frac{p}{2})(\frac{p}{2} + 1 - \alpha)} \int_0^1 (r^{\frac{p}{2}+1-\alpha} + |1-r|^{\frac{p}{2}+1-\alpha} - 1) E(A_r) dr < \infty. \end{aligned}$$

Hence, the random variable defined by

$$\Gamma = \int_0^1 \int_0^1 \frac{|X_t - X_s|^p}{|t-s|^\alpha} ds dt$$

is finite almost surely, and by Lemma 5.1.2 we obtain

$$|X_t - X_s| \leq c_{p,\alpha} \Gamma^{\frac{1}{p}} |t-s|^{\frac{\alpha-2}{p}},$$

for some constant $c_{p,\alpha}$. □

Note that the above proposition implies that for a process u in the space $\mathbb{L}_{\text{loc}}^{1,p}$, with $p > 2$, the Skorohod integral $\int_0^t u_s dW_s$ has a continuous version. Furthermore, if $u \in \mathbb{L}^{1,p}$, $p > 2$, we have

$$E \left(\sup_{t \in [0,1]} \left| \int_0^t u_s dW_s \right|^p \right) < \infty.$$

The next result will show the existence of a nonzero quadratic variation for the indefinite Skorohod integral.

Theorem 5.1.1 *Suppose that u is a process of the space $\mathbb{L}_{\text{loc}}^{1,2}$. Then*

$$\sum_{i=0}^{n-1} \left(\int_{t_i}^{t_{i+1}} u_s dW_s \right)^2 \rightarrow \int_0^1 u_s^2 ds, \quad (5.6)$$

in probability, as $|\pi| \rightarrow 0$, where π runs over all finite partitions $\{0 = t_0 < t_1 < \dots < t_n = 1\}$ of $[0, 1]$, and $|\pi| = \sup_{0 \leq i \leq n-1} |t_{i+1} - t_i|$. Moreover, the convergence is in $L^1(\Omega)$ if u belongs to $\mathbb{L}^{1,2}$.

Proof: We will describe the details of the proof only for the case $u \in \mathbb{L}^{1,2}$. The general case would be deduced by an easy argument of localization.

For any process u in $\mathbb{L}^{1,2}$ and for any partition $\pi = \{0 = t_0 < t_1 < \dots < t_n = 1\}$ we define

$$V^\pi(u) = \sum_{i=0}^{n-1} \left(\int_{t_i}^{t_{i+1}} u_s dW_s \right)^2.$$

Suppose that u and v are two processes in $\mathbb{L}^{1,2}$. Then we have

$$\begin{aligned} E(|V^\pi(u) - V^\pi(v)|) &\leq \left(E \sum_{i=0}^{n-1} \left(\int_{t_i}^{t_{i+1}} (u_s - v_s) dW_s \right)^2 \right)^{\frac{1}{2}} \\ &\quad \times \left(E \sum_{i=0}^{n-1} \left(\int_{t_i}^{t_{i+1}} (u_s + v_s) dW_s \right)^2 \right)^{\frac{1}{2}} \\ &\leq \|u - v\|_{\mathbb{L}^{1,2}} \|u + v\|_{\mathbb{L}^{1,2}}. \end{aligned} \quad (5.7)$$

Therefore, it suffices to show the result for a class of processes u which is dense in $\mathbb{L}^{1,2}$. So we can assume that

$$u_t = \sum_{j=0}^{m-1} F_j \mathbf{1}_{(s_j, s_{j+1}]},$$

where for each j , F_j is a smooth and bounded random variable, and $0 = s_0 < \dots < s_m = 1$. We can assume that the partition π contains the points $\{s_0, \dots, s_m\}$, because

$$\lim_{|\pi| \downarrow 0} E(|V^{\pi \vee \{s_0, \dots, s_m\}}(u) - V^\pi(u)|) = 0.$$

If π contains the points s_j we can write

$$\begin{aligned} V^\pi(u) &= \sum_{j=0}^{m-1} \sum_{\{i: s_j \leq t_i < s_{j+1}\}} \left(F_j (W(t_{i+1}) - W(t_i)) - \int_{t_i}^{t_{i+1}} D_s F_j ds \right)^2 \\ &= \sum_{j=0}^{m-1} \left[\sum_{\{i: s_j \leq t_i < s_{j+1}\}} F_j^2 (W(t_{i+1}) - W(t_i))^2 \right. \\ &\quad \left. - 2(W(t_{i+1}) - W(t_i)) \int_{t_i}^{t_{i+1}} D_s F_j ds + \left(\int_{t_i}^{t_{i+1}} D_s F_j ds \right)^2 \right]. \end{aligned}$$

Using the properties of the quadratic variation of the Brownian motion this converges in $L^1(\Omega)$ to

$$\sum_{j=0}^{m-1} F_j^2 (s_{j+1} - s_j) = \int_0^1 u_s^2 ds,$$

as $|\pi|$ tends to zero. □

As a consequence of these results, if $u \in \mathbb{L}_{\text{loc}}^{1,p}$, $p > 2$, is a process such that the Skorohod integral $\int_0^t u_s dW_s$ has bounded variation paths, then $u = 0$.

Let $X \in \mathbb{L}^{k,p}$, $p > 1$, and let $q \in [1, p]$. We will denote by $D^+ X$ (resp. $D^- X$) the element of $L^q([0, 1] \times \Omega)$ defined by

$$\lim_n \int_0^1 \sup_{s < t \leq (s + \frac{1}{n}) \wedge 1} E(|D_s X_t - (D^+ X)_s|^q) ds = 0. \quad (5.8)$$

(resp.

$$\lim_n \int_0^1 \sup_{(s-\frac{1}{n}) \vee 0 \leq t < s} E(|D_s X_t - (D^- X)_s|^q) ds = 0 \quad (5.9)$$

provided that these limit exist. We will denote by $\mathbb{L}_{q^+}^{k,p}$ (resp. $\mathbb{L}_{q^-}^{k,p}$) the class of processes in $\mathbb{L}^{k,p}$ such that the limit (5.8) (resp. (5.9)) exist. For each $p > 1$, and $q \in [1, p]$ we set $\mathbb{L}_q^{k,p} = \mathbb{L}_{q^+}^{k,p} \cap \mathbb{L}_{q^-}^{k,p}$. For $u \in \mathbb{L}_q^{k,p}$ we will write $(\nabla X)_t = (D^+ X)_t + (D^- X)_t$.

Notice that a process X of the form

$$X_t = X_0 + \int_0^t u_s dW_s + \int_0^t v_s ds,$$

where $X_0 \in \mathbb{D}^{1,2}$, $u \in \mathbb{L}^{2,2}$, and $v \in \mathbb{L}^{1,2}$ belongs to the class $\mathbb{L}_2^{1,2}$. Indeed, we have

$$D_s X_t = u_s \mathbf{1}_{\{s \leq t\}} + D_s X_0 + \int_0^t D_s v_r dr + \int_0^t D_s u_r dW_r,$$

and this implies that $(D^+ X)_t$ and $(D^- X)_t$ exist and

$$(D^+ X)_t = u_t + D_t X_0 + \int_0^t D_t v_r dr + \int_0^t D_t u_r dW_r,$$

$$(D^- X)_t = D_t X_0 + \int_0^t D_t v_r dr + \int_0^t D_t u_r dW_r.$$

In fact, we have

$$\begin{aligned} & \lim_n \int_0^1 \sup_{s < t \leq (s+\frac{1}{n}) \wedge 1} E(|D_s X_t - (D^+ X)_s|^2) ds \\ & \leq \frac{1}{n} \int_0^1 \int_s^{(s+\frac{1}{n}) \wedge 1} E(|D_s v_r|^2) dr ds + \int_0^1 \int_s^{(s+\frac{1}{n}) \wedge 1} E(|D_s u_r|^2) dr ds \\ & \quad + \int_0^1 \int_0^1 \int_s^{(s+\frac{1}{n}) \wedge 1} E(|D_\theta D_s u_r|^2) dr ds d\theta, \end{aligned}$$

and this converges to zero as n tends to infinity.

One can show a change-of-variables formula similar to Itô formula for Skorohod integral processes.

Theorem 5.1.2 *Consider a process of the form $X_t = X_0 + \int_0^t u_s dW_s + \int_0^t v_s ds$, where $X_0 \in \mathbb{D}_{\text{loc}}^{1,2}$, $u \in (\mathbb{L}^{2,2} \cap \mathbb{L}^{1,4})_{\text{loc}}$, and $v \in \mathbb{L}_{\text{loc}}^{1,2}$. Let $F : \mathbb{R} \rightarrow \mathbb{R}$ be a twice continuously differentiable function. Then we have*

$$F(X_t) = F(X_0) + \int_0^t F'(X_s) dX_s + \frac{1}{2} \int_0^t F''(X_s) u_s^2 ds + \int_0^t F''(X_s) (D^- X)_s u_s ds. \quad (5.10)$$

Notice that if the process X is adapted then $(D^- X)_s$ vanishes, and we obtain the classical Itô formula.

Proof: Suppose that $(\Omega^{n,1}, X_0^n)$, $(\Omega^{n,2}, u^n)$ and $(\Omega^{n,3}, v^n)$ are localizing sequences for X_0 , u , and v , respectively. For each positive integer k let ψ_k be a smooth function such that $0 \leq \psi_k \leq 1$, $\psi_k(x) = 0$ if $|x| \geq k+1$, and $\psi_k(x) = 1$ if $|x| \leq k$. Define

$$u_t^{n,k} = u_t^n \psi_k \left(\int_0^1 (u_s^n)^2 ds \right).$$

Set $X_t^{n,k} = X_0^n + \int_0^t u_s^{n,k} dW_s + \int_0^t v_s^n ds$, and consider the family of sets

$$G^{n,k} = \Omega^{n,1} \cap \Omega^{n,2} \cap \Omega^{n,3} \cap \left\{ \sup_{t \in [0,1]} |X_t| \leq k \right\} \cap \left\{ \int_0^1 (u_s^n)^2 ds \leq k \right\}.$$

Then it suffices to show the result for the processes X_0^n , $u^{n,k}$, and v^n , and for the function $F^n := F\psi_n$. In this way we can assume that $X_0 \in \mathbb{D}^{1,2}$, $u \in \mathbb{L}^{2,2} \cap \mathbb{L}^{1,4}$, $v \in \mathbb{L}^{1,2}$, $\int_0^1 u_s^2 ds \leq k$, and that the functions F , F' and F'' are bounded.

Set $t_i^n = \frac{it}{2^n}$, $0 \leq i \leq 2^n$. Applying Taylor development up to the second order we obtain

$$F(X_t) = F(X_0) + \sum_{i=0}^{2^n-1} F'(X(t_i^n))(X(t_{i+1}^n) - X(t_i^n)) + \sum_{i=0}^{2^n-1} \frac{1}{2} F''(\bar{X}_i)(X(t_{i+1}^n) - X(t_i^n))^2, \quad (5.11)$$

where $\bar{X}_i$ denotes a random intermediate point between $X(t_i^n)$ and $X(t_{i+1}^n)$. Now the proof will be decomposed in several steps.

Step 1. Let us show that

$$\sum_{i=0}^{2^n-1} F''(\bar{X}_i)(X(t_{i+1}^n) - X(t_i^n))^2 \rightarrow \int_0^t F''(X_s) u_s^2 ds, \quad (5.12)$$

in probability (actually in $L^1(\Omega)$), as n tends to infinity.

The increment $(X(t_{i+1}^n) - X(t_i^n))^2$ can be decomposed into

$$\left(\int_{t_i^n}^{t_{i+1}^n} u_s dW_s \right)^2 + \left(\int_{t_i^n}^{t_{i+1}^n} v_s ds \right)^2 + 2 \left(\int_{t_i^n}^{t_{i+1}^n} u_s dW_s \right) \left(\int_{t_i^n}^{t_{i+1}^n} v_s ds \right).$$

Using the continuity of $F''(X_t)$ we can show that the contribution of the last two terms to the limit (5.12) is zero. Therefore, it suffices to show that

$$\sum_{i=0}^{2^n-1} F''(\bar{X}_i) \left(\int_{t_i^n}^{t_{i+1}^n} u_s dW_s \right)^2 \rightarrow \int_0^t F''(X_s) u_s^2 ds. \quad (5.13)$$

Suppose that $n \geq m$, and for any $i = 1, \dots, 2^n$ let us denote by $t_i^{(m)}$ the point of the m th partition which is closer to t_i^n from the left. Then we have

$$\begin{aligned} & \left| \sum_{i=0}^{2^n-1} F''(\bar{X}_i) \left(\int_{t_i^n}^{t_{i+1}^n} u_s dW_s \right)^2 - \int_0^t F''(X_s) u_s^2 ds \right| \\ & \leq \left| \sum_{i=0}^{2^n-1} [F''(\bar{X}_i) - F''(X(t_i^{(m)}))] \left(\int_{t_i^n}^{t_{i+1}^n} u_s dW_s \right)^2 \right| \\ & \quad + \left| \sum_{j=0}^{2^m-1} F''(X(t_j^m)) \sum_{i: t_i^n \in [t_j^m, t_{j+1}^m)} \left[\left(\int_{t_i^n}^{t_{i+1}^n} u_s dW_s \right)^2 - \int_{t_i^n}^{t_{i+1}^n} u_s^2 ds \right] \right| \\ & \quad + \left| \sum_{j=0}^{2^m-1} F''(X(t_j^m)) \int_{t_j^m}^{t_{j+1}^m} u_s^2 ds - \int_0^t F''(X_s) u_s^2 ds \right| \\ & = b_1 + b_2 + b_3. \end{aligned}$$

The term b_3 can be bounded by

$$\sup_{|s-r| \leq t2^{-m}} |F''(X_s) - F''(X_r)| \int_0^t u_s^2 ds,$$

which converges to zero as m tends to infinity by the continuity of the process X_t . In the same way the term b_1 is bounded by

$$\sup_{|s-r| \leq t2^{-m}} |F''(X_s) - F''(X_r)| \sum_{i=0}^{2^n-1} \left(\int_{t_i^n}^{t_{i+1}^n} u_s dW_s \right)^2,$$

which again tends to zero in probability as m tends to infinity uniformly in n . Finally, the term b_2 converges to zero as n tends to infinity, for any fixed m , due to Theorem 5.1.1.

Step 2. Clearly the term

$$\sum_{i=0}^{2^n-1} F'(X(t_i^n)) \left(\int_{t_i^n}^{t_{i+1}^n} v_s ds \right)$$

converges a.s. and in $L^1(\Omega)$ to $\int_0^t F'(X_s) v_s ds$, as n tends to infinity.

Step 3. From property (1.12) of the Skorohod integral we deduce

$$\begin{aligned} F'(X(t_i^n)) \int_{t_i^n}^{t_{i+1}^n} u_s dW_s &= \int_{t_i^n}^{t_{i+1}^n} F'(X(t_i^n)) u_s dW_s + \int_{t_i^n}^{t_{i+1}^n} D_s [F'(X(t_i^n))] u_s ds \\ &= \int_{t_i^n}^{t_{i+1}^n} F'(X(t_i^n)) u_s dW_s + F''(X(t_i^n)) \int_{t_i^n}^{t_{i+1}^n} D_s X(t_i^n) u_s ds \\ &= c_1 + c_2. \end{aligned}$$

Notice that the process $F'(X_t)u_t$ belongs to $\mathbb{L}^{1,2}$ because $u \in \mathbb{L}^{2,2} \cap \mathbb{L}^{1,4}$, $v \in \mathbb{L}^{1,2}$, and the processes $F'(X_t)$, $F''(X_t)$, and $\int_0^1 u_s^2 ds$ are uniformly bounded. In fact, we have

$$\begin{aligned} D_s [F'(X_t)u_t] &= F''(X_t) \left(u_s \mathbf{1}_{\{s \leq t\}} + D_s X_0 + \int_0^t D_s u_r dW_r + \int_0^t D_s v_r dr \right) u_t \\ &\quad + F'(X_t) D_s u_t, \end{aligned}$$

and all the terms in the right-hand side of the above expression are square integrable. For the second term we use the duality relationship of the Skorohod integral:

$$\begin{aligned} &\int_0^1 \int_0^1 E \left(\left| u_t \int_0^t D_s u_r dW_r \right|^2 \right) ds dt \\ &= E \left\{ \int_0^1 \int_0^1 \int_0^t D_s u_r \left[2u_t D_r u_t \left(\int_0^t D_s u_\theta dW_\theta \right) + u_t^2 D_s u_r \right. \right. \\ &\quad \left. \left. + u_t^2 \left(\int_0^t D_r D_s u_\theta dW_\theta \right) \right] dr ds dt \right\} \\ &\leq cE \left((\|Du\|_{L^2([0,1]^2)}^2 + \|D^2u\|_{L^2([0,1]^3)}^2) \right). \end{aligned}$$

Then, for any smooth random variable $G \in \mathcal{S}$ we have

$$\lim_n E \left(G \sum_{i=0}^{2^n-1} \int_{t_i^n}^{t_{i+1}^n} F'(X(t_i^n)) u_s dW_s \right) = E \left(G \int_0^t F'(X_s) u_s dW_s \right). \quad (5.14)$$

This convergence is easily checked by duality and in this way obtain the convergence of $E(Gc_1)$ to $E(G \int_0^t F'(X_s)u_s dW_s)$. One can also show the convergence in L^2 of c_1 to $\int_0^t F'(X_s)u_s dW_s$.

On the other hand, the term c_2 converges in L^1 to $\int_0^t F''(X_s)(D^-X)_s u_s ds$. In fact, we have

$$\begin{aligned} & \left| \sum_{i=0}^{2^n-1} F''(X(t_i^n)) \int_{t_i^n}^{t_{i+1}^n} D_s X(t_i^n) u_s ds - \int_0^t F''(X_s)(D^-X)_s u_s ds \right| \\ & \leq \left| \sum_{i=0}^{2^n-1} F''(X(t_i^n)) \int_{t_i^n}^{t_{i+1}^n} [D_s X(t_i^n) - (D^-X)_s] u_s ds \right| \\ & \quad + \left| \sum_{i=0}^{2^n-1} \int_{t_i^n}^{t_{i+1}^n} [F''(X(t_i^n)) - F''(X_s)] (D^-X)_s u_s ds \right|. \end{aligned}$$

Consequently, we obtain

$$\begin{aligned} & E \left(\left| \sum_{i=0}^{2^n-1} F''(X(t_i^n)) \int_{t_i^n}^{t_{i+1}^n} D_s X(t_i^n) u_s ds - \int_0^t F''(X_s)(D^-X)_s u_s ds \right| \right) \\ & \leq \|F''\|_\infty \left\{ \int_0^1 E(u_s^2) ds \int_0^1 \sup_{(s-2^{-n})^+ \leq t \leq s} E(|D_s X_t - (D^-X)_s|^2) ds \right\}^{\frac{1}{2}} \\ & \quad + E \left(\sup_{|s-t| \leq 2^{-n}} |F''(X_t) - F''(X_s)| \int_0^1 |(D^-X)_s u_s| ds \right), \end{aligned}$$

which converges to zero as n tends to infinity. $\square$

Remarks:

1. Notice that for a process $u \in \mathbb{L}^{2,2}$, the condition $E(\int_0^1 \|Du_t\|^2 dt) < \infty$ is only required to insure that X has a continuous version, and to show that $F'(X_t)u_t$ belongs to $\mathbb{L}^{1,2}$. One can show the Itô formula under different type of hypotheses. More precisely, one can impose some conditions on X and modify the assumptions on X_0 , u , and v . For instance, one can assume either

- (a) $u \in \mathbb{L}_{\text{loc}}^{1,4}$, $v \in L^1([0, 1])$ a.s., and $X \in (\mathbb{L}_{2^-}^{1,2} \cap \mathbb{L}^{1,4})_{\text{loc}}$, or
- (b) $u \in (\mathbb{L}^{1,2} \cap L^\infty([0, 1] \times \Omega))_{\text{loc}}$, $v \in L^1([0, 1])$ a.s., $X \in \mathbb{L}_{2^-, \text{loc}}^{1,2}$, and X possesses a continuous version.

2. Suppose that X , u , v are stochastic processes verifying one of the hypotheses (a) or (b). Then if there is a set $G \in \mathcal{F}$ such that for almost all $\omega \in G$ we have

$$X_t = X_0 + \int_0^t u_s dW_s + \int_0^t v_s ds, \quad t \in [0, 1], \quad (5.15)$$

then (5.10) holds a.s. on G . In fact, it suffices to check that

$$\begin{aligned} & \sum_{i=0}^{2^n-1} F'(X(t_i^n)) \left(\int_{t_i^n}^{t_{i+1}^n} u_s dW_s + \int_{t_i^n}^{t_{i+1}^n} v_s ds \right) \\ & + \sum_{i=0}^{2^n-1} \frac{1}{2} F''(\bar{X}_i) \left(\int_{t_i^n}^{t_{i+1}^n} u_s dW_s + \int_{t_i^n}^{t_{i+1}^n} v_s ds \right)^2, \end{aligned}$$

converges in probability as $|\pi|$ tends to zero to the right-hand side of (5.10).

3. Using the operator ∇ we can write Eq. (5.10) in the following way:

$$F(X_t) = F(X_0) + \int_0^t F'(X_s) dX_s + \frac{1}{2} \int_0^t F''(X_s) (\nabla X)_s u_s ds. \quad (5.16)$$

The following result contains a multidimensional and local version of the change-of-variables formula for the Skorohod integral. We will make the usual convention of summation over repeated indices.

Theorem 5.1.3 *Let $W = \{W_t, t \in [0, 1]\}$ be an N -dimensional Wiener process. Suppose that $X_0^i \in \mathbb{D}^{1,2}$, $u^{ij} \in \mathbb{L}^{2,2} \cap \mathbb{L}^{1,4}$, and $v^i \in \mathbb{L}^{1,2}$, $1 \leq i \leq M$, $1 \leq j \leq N$, are processes such that on a set $G \in \mathcal{F}$ we have a.s.*

$$X_t^i = X_0^i + \int_0^t u_s^{ij} dW_s^j + \int_0^t v_s^i ds, \quad 0 \leq t \leq 1.$$

Let $F : \mathbb{R}^M \rightarrow \mathbb{R}$ be a twice continuously differentiable function. Then we have on G a.s.

$$\begin{aligned} F(X_t) &= F(X_0) + \int_0^t (\partial_i F)(X_s) dX_s^i + \frac{1}{2} \int_0^t (\partial_i \partial_j F)(X_s) u_s^{ik} u_s^{jk} ds \\ &\quad + \int_0^t (\partial_i \partial_j F)(X_s) ((D^k)^- X^j)_s u_s^{ik} ds, \end{aligned}$$

where D^k denotes the derivative with respect to the k th component of the Wiener process.

5.2 Extended Stratonovich integral

In the sequel, π will denote a partition of the interval $[0, 1]$ of the form $\pi = \{0 = t_0 < t_1 < \dots < t_n = 1\}$. The mesh of π is defined by $|\pi| = \sup_i (t_{i+1} - t_i)$. Given a measurable process $u = \{u_t, t \in [0, 1]\}$ such that $\int_0^1 |u_t| dt < \infty$ a.s., we can introduce the following Riemann-type sums

$$S^\pi = \sum_{i=0}^{n-1} \frac{1}{t_{i+1} - t_i} \left(\int_{t_i}^{t_{i+1}} u_s ds \right) (W(t_{i+1}) - W(t_i)).$$

Notice that

$$S^\pi = \int_0^1 u_s W_s^\pi ds,$$

where

$$W^\pi = \sum_{i=0}^{n-1} \left(\frac{W(t_{i+1}) - W(t_i)}{t_{i+1} - t_i} \right) \mathbf{1}_{[t_i, t_{i+1})}.$$

Definition 5.2.1 *We say that a measurable process $u = \{u_t, t \in [0, 1]\}$ such that $\int_0^1 |u_t| dt < \infty$ a.s. is Stratonovich integrable if the family S^π converges in probability as $|\pi| \downarrow 0$, and in this case the limit will be denoted by $\int_0^1 u_t \circ dW_t$.*

We know that if the process u is a continuous semimartingale, then the Stratonovich integral of u exists and it coincides with the Itô integral plus the correction term $\frac{1}{2}\langle u, W \rangle_t$. Let us now describe the relationship between the extended Stratonovich integral and the Skorohod integral. A basic result on the approximation of the Skorohod integral by Riemann-type sums is the following. For any process $u \in L^2([0, 1] \times \Omega)$ we define

$$\hat{S}^\pi = \sum_{i=0}^{n-1} \frac{1}{t_{i+1} - t_i} \left(\int_{t_i}^{t_{i+1}} E(u_s | \mathcal{F}_{[t_i, t_{i+1}]^c} ds) \right) (W(t_{i+1}) - W(t_i)).$$

Notice each term in the summation is the product of two independent factors. Furthermore, from Lemma 1.5.2 the processes

$$\hat{u}^\pi(t) = \sum_{i=0}^{n-1} \frac{1}{t_{i+1} - t_i} \left(\int_{t_i}^{t_{i+1}} E(u_s | \mathcal{F}_{[t_i, t_{i+1}]^c} ds) \right) \mathbf{1}_{(t_i, t_{i+1}]}(t)$$

are Skorohod integrable, and

$$\hat{S}^\pi = \delta(\hat{u}^\pi).$$

Proposition 5.2.1 *Let $u \in L^2([0, 1] \times \Omega)$. If there exist a sequence of partitions $\{\pi(n)\}$ of $[0, 1]$ whose mesh tends to zero such that $\hat{S}^{\pi(n)}$ converges in $L^2(\Omega)$ as n tends to infinity, then u is Skorohod integrable, and*

$$\delta(u) = \lim_n \hat{S}^{\pi(n)}.$$

Moreover, this convergence always holds if $u \in \mathbb{L}^{1,2}$.

Proof: The first part of the proof follows from the fact that the operator δ is closed and $\hat{u}_n^\pi$ converges to u in $L^2([0, 1] \times \Omega)$. The second part follows from the convergence of $\hat{u}^\pi$ to u as $|\pi|$ tends to zero, in the norm of $\mathbb{L}^{1,2}$ (see [76] for the details of the proof). $\square$

Concerning the extended Stratonovich integral we can show the following result:

Theorem 5.2.1 *Let $u \in \mathbb{L}_{1,\text{loc}}^{1,2}$. Then u is Stratonovich integrable and*

$$\int_0^1 u_t \circ dW_t = \int_0^1 u_t dW_t + \frac{1}{2} \int_0^1 (\nabla u)_t dt. \quad (5.17)$$

Proof: By a localization argument we can assume that $u \in \mathbb{L}_1^{1,2}$. We have, for any partition π on $[0, 1]$

$$\delta(u^\pi) = S^\pi - \sum_{i=0}^{n-1} \frac{1}{t_{i+1} - t_i} \int_{t_i}^{t_{i+1}} \int_{t_i}^{t_{i+1}} D_s u_t ds dt, \quad (5.18)$$

where

$$u^\pi(t) = \sum_{i=0}^{n-1} \frac{1}{t_{i+1} - t_i} \left(\int_{t_i}^{t_{i+1}} u_s ds \right) \mathbf{1}_{(t_i, t_{i+1}]}(t).$$

The processes u^π converge to u as $|\pi|$ tends to zero in the norm of $\mathbb{L}^{1,2}$. Hence, $\delta(u^\pi)$ converges to $\delta(u)$ in $L^2(\Omega)$ as $|\pi| \downarrow 0$. Therefore, it suffices to show the following convergences:

$$\lim_{|\pi| \downarrow 0} E \left(\left| \sum_{i=0}^{n-1} \frac{1}{t_{i+1} - t_i} \int_{t_i}^{t_{i+1}} dt \int_t^{t_{i+1}} (D_t u_s) ds - \frac{1}{2} \int_0^1 (D^+ u)_t dt \right| \right) = 0, \quad (5.19)$$

$$\lim_{|\pi| \downarrow 0} E \left(\left| \sum_{i=0}^{n-1} \frac{1}{t_{i+1} - t_i} \int_{t_i}^{t_{i+1}} dt \int_t^t (D_t u_s) ds - \frac{1}{2} \int_0^1 (D^- u)_t dt \right| \right) = 0. \quad (5.20)$$

We will only show (5.19). We can write

$$\begin{aligned} & E \left(\left| \sum_{i=0}^{n-1} \frac{1}{t_{i+1} - t_i} \int_{t_i}^{t_{i+1}} dt \int_t^{t_{i+1}} (D_t u_s) ds - \frac{1}{2} \int_0^1 (D^+ u)_t dt \right| \right) \\ & \leq E \left(\left| \sum_{i=0}^{n-1} \frac{1}{t_{i+1} - t_i} \int_{t_i}^{t_{i+1}} \left(\int_t^{t_{i+1}} [D_t u_s - (D^+ u)_t] ds \right) dt \right| \right) \\ & \quad + E \left(\left| \sum_{i=0}^{n-1} \int_{t_i}^{t_{i+1}} \frac{t_{i+1} - t}{t_{i+1} - t_i} (D^+ u)_t dt - \frac{1}{2} \int_0^1 (D^+ u)_t dt \right| \right) \\ & \leq \int_0^1 \sup_{t \leq s \leq (t+|\pi|) \wedge 1} E(|D_t u_s - (D^+ u)_t|) dt \\ & \quad + E \left(\left| \int_0^1 (D^+ u)_t \left(\sum_{i=0}^{n-1} \frac{t_{i+1} - t}{t_{i+1} - t_i} \mathbf{1}_{(t_i, t_{i+1}]}(t) - \frac{1}{2} \right) dt \right| \right). \end{aligned}$$

The first summand in the above converges to zero due to the definition of the class $\mathbb{L}_{1+}^{1,2}$. For the second term we will use the convergence of the functions $\sum_{i=0}^{n-1} \frac{t_{i+1} - t}{t_{i+1} - t_i} \mathbf{1}_{(t_i, t_{i+1}]}(t)$ to the constant $\frac{1}{2}$ in the weak topology of $L^2([0, 1])$. This weak convergence implies that

$$\int_0^1 (D^+ u)_t \left(\sum_{i=0}^{n-1} \frac{t_{i+1} - t}{t_{i+1} - t_i} \mathbf{1}_{(t_i, t_{i+1}]}(t) - \frac{1}{2} \right) dt$$

converges a.s. to zero as $|\pi| \rightarrow 0$. Finally, the convergence in $L^1(\Omega)$ follows by dominated convergence, using the definition of the space $\mathbb{L}_{1+}^{1,2}$. $\square$

Itô's formula for the Skorohod integral allows us to deduce a change-of-variables formula for the Stratonovich integral. Let us first introduce the following classes of processes:

Set $\mathbb{L}_S^{2,4} = \{u \in \mathbb{L}_1^{2,4} : \nabla u \in \mathbb{L}^{1,2}, u \text{ is continuous in } L^2(\Omega)\}$.

Theorem 5.2.2 *Let F be a real-valued, twice continuously differentiable function. Consider a process of the form $X_t = X_0 + \int_0^t u_s \circ dW_s + \int_0^t v_s ds$, where $X_0 \in \mathbb{D}_{\text{loc}}^{1,2}$, $u \in \mathbb{L}_{S,\text{loc}}^{2,4}$ and $v \in \mathbb{L}_{\text{loc}}^{1,2}$. Then we have*

$$F(X_t) = F(X_0) + \int_0^t F'(X_s) v_s ds + \int_0^t [F'(X_s) u_s] \circ dW_s. \quad (5.21)$$

Proof: As in the proof of the change-of-variables formula for the Skorohod integral we can assume that the functions F , F' , and F'' are bounded, $\int_0^1 u_s^2 ds$ is bounded,

$X_0 \in \mathbb{D}^{1,2}$, $u \in \mathbb{L}_S^{2,4}$, and $v \in \mathbb{L}^{1,2}$. We know that the process X_t has the following decomposition:

$$X_t = X_0 + \int_0^t u_s dW_s + \int_0^t v_s ds + \frac{1}{2} \int_0^t (\nabla u)_s ds.$$

This process verifies the assumptions of Theorem 5.1.2. Consequently, we can apply Itô's formula to X and obtain

$$\begin{aligned} F(X_t) &= F(X_0) + \int_0^t F'(X_s) v_s ds + \frac{1}{2} \int_0^t F'(X_s) (\nabla u)_s ds \\ &\quad + \int_0^t F'(X_s) u_s dW_s + \int_0^t F''(X_s) (D^- X)_s u_s ds + \frac{1}{2} \int_0^t F''(X_s) u_s^2 ds. \end{aligned}$$

The process $\{F'(X_t)u_t, t \in [0, 1]\}$ belongs to $\mathbb{L}_1^{1,2}$. In fact, notice first that as in the proof of Theorem 5.1.2, the boundedness of F' , F'' , $\int_0^1 u_s^2 ds$, and the fact that $u \in \mathbb{L}^{2,4}$, $v, \nabla u \in \mathbb{L}^{1,2}$ and $X_0 \in \mathbb{D}^{1,2}$ imply that this process belongs to $\mathbb{L}^{1,2}$ and

$$D_s[F'(X_t)u_t] = F'(X_t)D_s u_t + F''(X_t)u_t D_s X_t.$$

On the other hand, using that $u \in \mathbb{L}_1^{1,2}$, u is continuous in L^2 , and $X \in \mathbb{L}_2^{1,2}$ we deduce that $\{F'(X_t)u_t, t \in [0, 1]\}$ belongs to $\mathbb{L}_1^{1,2}$ and that

$$(\nabla(F'(X)u))_t = F'(X_t)(\nabla u)_t + F''(X_t)u_t(\nabla X)_t.$$

Hence, applying Theorem 5.2.1 we can write

$$\begin{aligned} \int_0^t [F'(X_s)u_s] \circ dW_s &= \int_0^t F'(X_s)u_s dW_s + \frac{1}{2} \int_0^t (\nabla(F'(X)u))_s ds \\ &= \int_0^t F'(X_s)u_s dW_s + \frac{1}{2} \int_0^t F'(X_s)(\nabla u)_s ds + \frac{1}{2} \int_0^t F''(X_s)u_s(\nabla X)_s ds. \end{aligned}$$

Finally, notice that

$$(\nabla X)_t = 2(D^- X)_t + u_t.$$

This completes the proof of the theorem. $\square$

The above theorem is still valid if we assume $X \in \mathbb{L}_2^{1,4}$, $u \in (\mathbb{L}_1^{1,4} \cap C^2([0, 1], L^2(\Omega)))$, and $v \in L^1([0, 1])$ a.s. In the next theorem we state a multidimensional version of the change-of-variables formula for the Stratonovich integral, using these type of hypotheses.

Theorem 5.2.3 *Let $W = \{W_t, t \in [0, 1]\}$ be an N -dimensional Wiener process. Suppose that we have processes $X^i \in \mathbb{L}_{2-}^{1,4}$, $u^{ij} \in \mathbb{L}_1^{1,4} \cap C^2([0, 1], L^2(\Omega))$, and $v^i \in L^1([0, 1])$ a.s., $1 \leq i \leq M$, $1 \leq j \leq N$, such that on some set $G \in \mathcal{F}$ a.s. we have*

$$X_t^i = X_0^i + \int_0^t u_s^{ij} \circ dW_s^j + \int_0^t v_s^i ds, \quad 0 \leq t \leq 1.$$

Let $F : \mathbb{R}^M \rightarrow \mathbb{R}$ be a twice continuously differentiable function. Then we have on G

$$F(X_t) = F(X_0) + \int_0^t (\partial_i F)(X_s) u_s^{ij} \circ dW_s^j + \int_0^t (\partial_i F)(X_s) v_s^i ds.$$

5.3 Substitution formulas

In this section we will consider the following problem. Suppose that $u = \{u_t(x), 0 \leq t \leq 1\}$ is a stochastic process parametrized by $x \in \mathbb{R}^m$, which is square integrable and adapted for each $x \in \mathbb{R}^m$. For each x we can define the Itô integral

$$\int_0^1 u_t(x) dW_t.$$

Assume now that the resulting random field is a.s. continuous in x , and let F be an m -dimensional random variable. Then we can evaluate the stochastic integral at $x = F$, that is, we can define the random variable

$$\int_0^1 u_t(x) dW_t|_{x=F}. \quad (5.22)$$

A natural question is under which conditions is the nonadapted process $\{u_t(F), 0 \leq t \leq 1\}$ Skorohod integrable, and what is the relationship between the Skorohod integral of this process and the random variable defined by (5.22). A similar question can be asked for the extended Stratonovich integral.

Let us first consider the following preliminary result.

Lemma 5.3.1 *Suppose that $\{Y_n(\theta), \theta \in \mathbb{R}^m\}$, $n \geq 1$ is a sequence of processes which converges in probability to a random field $\{Y(\theta), \theta \in \mathbb{R}^m\}$ for each $\theta \in \mathbb{R}^m$. Suppose that*

$$E(|Y_n(\theta) - Y_n(\theta')|^p) \leq c_{p,K} |\theta - \theta'|^\alpha, \quad (5.23)$$

for all $|\theta|, |\theta'| \leq K$, $n \geq 1$, $K > 0$ and for some constants $p > 0$ and $\alpha > m$. Then, for any m -dimensional random variable F one has

$$\lim_n Y_n(F) = Y(F),$$

in probability. Moreover, the convergence is in L^p if F is bounded.

Proof: Fix $K > 0$. Replacing F by $F_K := F \mathbf{1}_{\{|F| \leq K\}}$ we can assume that F is bounded by K . Fix $\epsilon > 0$ and consider a random variable F_ϵ which takes finitely many values and such that $|F_\epsilon| \leq K$ and $\|F - F_\epsilon\|_\infty \leq \epsilon$. We can write

$$|Y_n(F) - Y(F)| \leq |Y_n(F) - Y_n(F_\epsilon)| + |Y_n(F_\epsilon) - Y(F_\epsilon)| + |Y(F_\epsilon) - Y(F)|.$$

Take $0 < m' < \alpha - m$. By (5.5) there exist constants C_1 , C_2 , and random variables Γ , Γ_n such that

$$\begin{aligned} |Y_n(\theta) - Y_n(\theta')|^p &\leq C_1 |\theta - \theta'|^{m'} \Gamma_n, \\ E(\Gamma_n) &\leq C_2 c_{p,K}, \end{aligned}$$

and

$$\begin{aligned} |Y(\theta) - Y(\theta')|^p &\leq C_1 |\theta - \theta'|^{m'} \Gamma, \\ E(\Gamma) &\leq C_2 c_{p,K}. \end{aligned}$$

Notice that Eq. (5.23) is also satisfied by the random field $Y(\theta)$. Hence,

$$E(|Y_n(F) - Y(F)|^p) \leq c_p \left(2C_1 C_2 c_{p,K} \epsilon^{m'} + E(|Y_n(F_\epsilon) - Y(F_\epsilon)|^p) \right),$$

and the desired convergence follows by taking first the limit as $n \rightarrow \infty$ and then the limit as $\epsilon \downarrow 0$. $\square$

Consider a random field $u = \{u_t(x), 0 \leq t \leq 1, x \in \mathbb{R}^m\}$ satisfying the following conditions:

(h1) For each $x \in \mathbb{R}^m$, and $t \in [0, 1]$, $u_t(x)$ is $\mathcal{F}_t$ -measurable.

(h2) There exist constants $p \geq 2$ and $\alpha > m$ such that

$$E(|u_t(x) - u_t(x')|^p) \leq C_{t,K} |x - x'|^\alpha,$$

for all $|x|, |x'| \leq K$, $K > 0$, where $\int_0^1 C_{t,K} dt < \infty$. Moreover $\int_0^1 E(|u_t(0)|^2) dt < \infty$.

Notice that under the above conditions for each $t \in [0, 1]$ the random field $\{u_t(x), x \in \mathbb{R}^m\}$ possesses a continuous version, and the Itô integral $\int_0^1 u_t(x) dW_t$ possess a continuous version in (t, x) . In fact, for all $|x|, |x'| \leq K$, $K > 0$, we have

$$\begin{aligned} E \left(\sup_{t \in [0,1]} \left| \int_0^t (u_s(x) - u_s(x')) dW_s \right|^p \right) &\leq c_p E \left(\left| \int_0^1 |u_s(x) - u_s(x')|^2 ds \right|^{\frac{p}{2}} \right) \\ &\leq c_p \left(\int_0^1 C_{t,K} dt \right) |x - x'|^\alpha. \end{aligned}$$

The following theorem provides the relationship between the evaluated integral $\int_0^1 u_t(x) dW_t|_{x=F}$ and the Skorohod integral $\delta(u(F))$. We will need the following hypothesis which is stronger that **(h2)**:

(h3) For each (t, ω) the mapping $x \rightarrow u_t(x)$ is continuously differentiable, and for each $K > 0$.

$$\int_0^1 E \left(\sup_{|x| \leq K} |u'_t(x)|^q \right) dt < \infty,$$

where u' denotes the gradient of u and $q \geq 4$, $q > m$. Moreover $\int_0^1 E(|u_t(0)|^2) dt < \infty$.

Theorem 5.3.1 Suppose that $u = \{u_t(x), 0 \leq t \leq 1, x \in \mathbb{R}^m\}$ is a random field satisfying conditions (h1) and (h3). Let $F : \Omega \rightarrow \mathbb{R}^m$ be a bounded random variable such that $F^i \in \mathbb{D}^{1,4}$ for $1 \leq i \leq m$. Then the composition $u(F) = \{u_t(F), 0 \leq t \leq 1\}$ belongs to the domain of δ and

$$\delta(u(F)) = \int_0^1 u_t(x) dW_t \Big|_{x=F} - \sum_{j=1}^m \int_0^1 (\partial_j u_t)(F) D_t F^j dt. \quad (5.24)$$

Proof: Consider the approximation of the process u given by

$$u_t^n(x) = \sum_{i=1}^{n-1} n \left(\int_{\frac{i-1}{n}}^{\frac{i}{n}} u_s(x) ds \right) \mathbf{1}_{(\frac{i}{n}, \frac{i+1}{n}]}(t). \quad (5.25)$$

Notice that $u_t(F)$ belongs to $L^2([0, 1] \times \Omega)$ and the sequence of processes $u_t^n(F)$ converges to $u_t(F)$ in $L^2([0, 1] \times \Omega)$. Indeed, if F is bounded by K we have

$$\begin{aligned} E \left(\int_0^1 u_t(F)^2 dt \right) &\leq \int_0^1 E \left(\sup_{|x| \leq K} |u_t(x)|^2 dt \right) \\ &\leq 2 \int_0^1 E(|u_t(0)|^2) dt + 2K^2 \int_0^1 E \left(\sup_{|x| \leq K} |u'_t(x)|^2 \right) dt < \infty. \end{aligned}$$

The convergence of $u^n(F)$ to $u(F)$ is obtained by first approximating $u(F)$ by an element of $C([0, 1]; L^2(\Omega))$.

From Lemma 1.5.3 we deduce that $u_t^n(F)$ belongs to $\text{Dom } \delta$ and

$$\begin{aligned} \delta(u^n(F)) &= \sum_{i=1}^{n-1} n \left(\int_{\frac{i-1}{n}}^{\frac{i}{n}} u_s(F) ds \right) \left(W\left(\frac{i+1}{n}\right) - W\left(\frac{i}{n}\right) \right) \\ &\quad - \sum_{i=1}^{n-1} \sum_{j=1}^m n \left(\int_{\frac{i-1}{n}}^{\frac{i}{n}} (\partial_j u_s)(F) ds \right) \left(\int_{\frac{i}{n}}^{\frac{i+1}{n}} D_r F^j dr \right). \end{aligned} \quad (5.26)$$

The Itô stochastic integrals $\int_0^1 u_t^n(x) dW_t$ satisfy

$$E \left(\left| \int_0^1 u_t^n(x) dW_t - \int_0^1 u_t^n(x') dW_t \right|^q \right) \leq c_q |x - x'|^q.$$

Hence, by Lemma 5.3.1 the sequence $\int_0^1 u_t^n(x) dW_t \Big|_{x=F}$ converges in L^q to the random variable $\int_0^1 u_t(x) dW_t \Big|_{x=F}$. On the other hand, the second summand in the right-hand side of (5.26) converges to $\sum_{j=1}^m \int_0^1 (\partial_j u_t)(F) D_t F^j dt$ in L^2 , as it follows from the estimate

$$\begin{aligned} &\left| \int_0^1 (\partial_j u_t)(F) D_t F^j dt - \int_0^1 (\partial_j u_t^n)(F) D_t F^j dt \right| \\ &\leq \|D F^j\|_H \left(\int_0^1 |(\partial_j u_t)(F) - (\partial_j u_t^n)(F)|^2 dt \right)^{\frac{1}{2}}. \end{aligned}$$

The operator δ being closed the result follows. $\square$

The preceding theorem can be localized as follows:

Theorem 5.3.2 *Suppose that $u = \{u_t(x), 0 \leq t \leq 1, x \in \mathbb{R}^m\}$ is a random field that satisfies (h1) and (h3). Let $F : \Omega \rightarrow \mathbb{R}^m$ be a random variable such that $F^i \in \mathbb{D}_{\text{loc}}^{1,4}$ for $1 \leq i \leq m$. Then the composition $u(F)$ is locally in the domain of δ and (5.24) holds.*

Notice that in the above theorem the Skorohod integral $\delta(u(F))$ may depend on the particular localizing sequence, because we do not know if the operator δ is local in $\text{Dom } \delta$.

The Stratonovich integral also satisfies a commutativity relationship but in this case we do not need a complementary term. This fact is coherent with the the general behavior of the Stratonovich integral as an ordinary integral.

Let us consider the following condition which is stronger than (h2):

(h4) There exist constants $p \geq 2$ and $\alpha > m$ such that

$$\begin{aligned} E(|u_t(x) - u_t(x')|^p) &\leq c_K |x - x'|^\alpha, \\ E(|u_t(x) - u_t(x') - u_s(x) + u_s(x')|^p) &\leq c_K |x - x'|^\alpha |s - t|^{\frac{p}{2}}, \end{aligned}$$

for all $|x|, |x'| \leq K$, $K > 0$, and $s, t \in [0, 1]$. Moreover $\int_0^1 E(|u_t(0)|^2) dt < \infty$, and $t \rightarrow u_t(x)$ is continuous in L^2 for each x .

Theorem 5.3.3 *Let $u = \{u_t(x), 0 \leq t \leq 1, x \in \mathbb{R}^m\}$ be a random field satisfying hypothesis (h1) and (h4). Suppose that for each $x \in \mathbb{R}^m$ the Stratonovich integral $\int_0^1 u_t(x) \circ dW_t$ exists. Consider an arbitrary m -dimensional random variable F . Then $u(F)$ is Stratonovich integrable, and we have*

$$\int_0^1 u_t(F) \circ dW_t = \int_0^1 u_t(x) \circ dW_t \Big|_{x=F}. \quad (5.27)$$

Proof: Fix a partition $\pi = \{0 = t_0 < t_1 < \dots < t_n = 1\}$. We can write

$$\begin{aligned} S^\pi &= \sum_{i=0}^{n-1} \frac{1}{t_{i+1} - t_i} \left(\int_{t_i}^{t_{i+1}} u_s(F) ds \right) (W(t_{i+1}) - W(t_i)) \\ &= \sum_{i=0}^{n-1} u_{t_i}(F) (W(t_{i+1}) - W(t_i)) \\ &\quad + \sum_{i=0}^{n-1} \frac{1}{t_{i+1} - t_i} \left(\int_{t_i}^{t_{i+1}} [u_s(F) - u_{t_i}(F)] ds \right) (W(t_{i+1}) - W(t_i)) \\ &= a_1 + a_2. \end{aligned}$$

The continuity of $t \rightarrow u_t(x)$ in L^2 implies that for each $x \in \mathbb{R}^m$ the Riemann sums

$$\sum_{i=0}^{n-1} u_{t_i}(x) (W(t_{i+1}) - W(t_i))$$

converge in L^2 , as $|\pi|$ tends to zero, to the Itô integral $\int_0^1 u_t(x) dW_t$. Moreover, we have for $|x|, |x'| \leq K$,

$$E \left(\left| \sum_{i=0}^{n-1} [u_{t_i}(x) - u_{t_i}(x')] (W(t_{i+1}) - W(t_i)) \right|^p \right) \leq c_p c_K |x - x'|^\alpha.$$

Hence, by Lemma 5.3.1 we deduce that a_1 converges in probability to

$$\int_0^1 u_t(x) dW_t \Big|_{x=F}.$$

Now we will study the convergence of the term a_2 . For each fixed $x \in \mathbb{R}^m$, the sums

$$R^\pi(x) := \sum_{i=0}^{n-1} \frac{1}{t_{i+1} - t_i} \left(\int_{t_i}^{t_{i+1}} [u_s(x) - u_{t_i}(x)] ds \right) (W(t_{i+1}) - W(t_i))$$

converge in probability, as $|\pi|$ tends to zero, to the difference

$$V(x) := \int_0^1 u_t(x) \circ dW_t - \int_0^1 u_t(x) dW_t.$$

So it suffices to show that $R^\pi(F)$ converges in probability, as $|\pi|$ tends to zero, to $V(F)$. This convergence follows from Lemma 5.3.1 and the following estimations where $0 < \epsilon < 1$ verifies $\epsilon\alpha > m$, and $|x|, |x'| \leq K$,

$$\begin{aligned} & \|R^\pi(x) - R^\pi(x')\|_{p\epsilon} \\ & \leq \sum_{i=0}^{n-1} \frac{1}{t_{i+1} - t_i} \int_{t_i}^{t_{i+1}} \| [u_s(x) - u_{t_i}(x) - u_s(x') + u_{t_i}(x')] (W(t_{i+1}) - W(t_i)) \|_{p\epsilon} ds \\ & \leq c_{p,\epsilon} \sum_{i=0}^{n-1} |t_{i+1} - t_i|^{\frac{1}{2}} \sup_{s \in [t_i, t_{i+1}]} [E(|u_s(x) - u_{t_i}(x) - u_s(x') + u_{t_i}(x')|^p)]^{\frac{1}{p}} \\ & \leq c_{p,\epsilon} c_K |x - x'|^{\frac{\alpha}{p}}. \end{aligned}$$

□

It is also possible to establish substitution formulas similar to those appearing in theorems 5.3.1 and 5.3.3 for random fields $u_t(x)$ which are not adapted. In this case additional regularity assumptions are required. More precisely we need regularity in x of $D_s u_t(x)$ (for the Skorohod integral) and of $(\nabla u(x))_t$ for the Stratonovich integral.

One can also compute the differentials of processes of the form $F_t(X_t)$ where $\{X_t, t \in [0, 1]\}$ and $\{F_t(x), t \in [0, 1], x \in \mathbb{R}^m\}$ are generalized continuous semimartingales, i.e., they have a bounded variation component and an indefinite Skorohod (or Stratonovich) integral component. This type of change-of-variables formula are known as Itô-Ventzell formulas. Below we show a change-of-variables of this type which will be used in the next chapter (see [85]).

Let us first state the following differentiation rule (see Ocone and Pardoux [85, Lemma 2.3]) .

Lemma 5.3.2 *Let $F = (F^1, \dots, F^m)$ be a random vector whose components belong to $\mathbb{D}^{1,p}$, $p > 1$ and suppose that $|F| \leq K$. Consider a measurable random field $u = \{u(x), x \in \mathbb{R}^m\}$ with continuously differentiable paths, such that for any $x \in \mathbb{R}^m$, $u(x) \in \mathbb{D}^{1,r}$, $r > 1$, and the derivative $Du(x)$ has a continuous version as an H -valued process. Suppose that we have*

$$\begin{aligned} E \left(\sup_{|x| \leq K} [|u(x)|^r + \|Du(x)\|_H^r] \right) & < \infty, \\ E \left(\sup_{|x| \leq K} |u'(x)|^q \right) & < \infty, \end{aligned}$$

where $\frac{1}{q} + \frac{1}{p} = \frac{1}{r}$. Then the composition $G = u(F)$ belongs to $\mathbb{D}^{1,r}$, and we have

$$DG = \sum_{i=1}^k (\partial_i u)(F) DF^i + (Du)(F).$$

Sketch of the proof: Approximate the composition $u(F)$ by the integral

$$\int_{\mathbb{R}^m} u(x) \psi_\epsilon(F - x) dx,$$

where ψ_ϵ is an approximation of the identity. □

Let W be an N -dimensional Wiener process. Consider two stochastic processes of the form:

$$X_t^i = X_0^i + \int_0^t u_s^{ij} \circ dW_s^j + \int_0^t v_s^i ds, \quad 1 \leq i \leq m, \quad (5.28)$$

and

$$F_t(x) = F_0(x) + \int_0^t H_s^i(x) \circ dW_s^i + \int_0^t G_s(x) ds, \quad (5.29)$$

We introduce the following set of hypotheses:

- (A1) For all $1 \leq i \leq N$, $1 \leq j \leq m$, $u^{ij}, X^j \in \mathbb{L}_2^{1,4}$, $v^j \in L^1([0, 1])$ a.s., and the processes X and u are continuous and bounded by constants K and K_1 respectively.
- (A2) $x \rightarrow F_t(x)$ is of class C^2 for all (t, ω) ; the processes $F_t(x)$, $F_t'(x)$, and $F_t''(x)$ are continuous in (t, x) ; $F(x) \in \mathbb{L}^{1,2}$ and there exists a version of $D_s F_t(x)$ which is differentiable in x and for each fixed $s \in [0, 1]$ the functions $D_s F_t(x)$, $(D_s F_t)'(x)$ are continuous in the region $\{s \leq t \leq 1, |x| \leq K\}$ (resp. $\{0 \leq t \leq s, |x| \leq K\}$).
- (A3) For all $|x| \leq K$, $1 \leq i \leq N$, $H^i(x) \in \mathbb{L}^{1,2}$; the processes $H_s(x)$, and $H_s'(x)$ are continuous in (s, x) ; and there exists a version of $D_s H_t(x)$ which is differentiable in x and for each fixed $s \in [0, 1]$ the functions $D_s H_t(x)$, $(D_s H_t)'(x)$ are continuous in the region $\{s \leq t \leq 1, |x| \leq K\}$ (resp. $\{0 \leq t \leq s, |x| \leq K\}$).
- (A4) $x \rightarrow G_t(x)$ is continuous.
- (A5) The following estimates hold:

$$\begin{aligned} E \left(\sup_{|x| \leq K, t \in [0, 1]} \left(|F_t(x)|^4 + |F_t'(x)|^4 + |F_t''(x)|^4 \right. \right. \\ \left. \left. + |H_t(x)|^4 + |H_t'(x)|^4 + |G_t(x)|^4 \right) \right) < \infty, \\ \int_0^1 E \left(\sup_{|x| \leq K, t \in [0, 1]} \left(|D_s F_t(x)|^4 + |(D_s F_t)'(x)|^4 \right. \right. \\ \left. \left. + |D_s H_t(x)|^4 + |(D_s H_t)'(x)|^4 \right) \right) ds < \infty. \end{aligned}$$

Theorem 5.3.4 Assume that the processes X_t and $F_t(x)$ satisfy the above hypotheses. Then $\langle F_t'(X_t), u_t^i \rangle$ and $H_t^i(X_t)$ are elements of $\mathbb{L}_1^{1,2}$, $i = 1, \dots, m$, and

$$\begin{aligned} F_t(X_t) &= F_0(X_0) + \int_0^t \langle F_s'(X_s), u_s^i \rangle \circ dW_s^i + \int_0^t \langle F_s'(X_s), v_s \rangle ds \\ &\quad + \int_0^t H_s^i(X_s) \circ dW_s^i + \int_0^t G_s(X_s) ds. \end{aligned} \quad (5.30)$$

Moreover if (5.28) and (5.29) hold a.s. for all $t \in [0, 1]$ on some set $G \in \mathcal{F}$ then (5.30) holds for all $t \in [0, 1]$ a.s. on G .

Sketch of the proof: To simplify we will assume $N = 1$. The proof will be done in several steps.

Step 1: Consider an approximation of the identity ψ_ϵ on $\mathbb{R}^m$. For each $x \in \mathbb{R}^m$ we can apply the multidimensional change-of-variables for the Stratonovich integral (Theorem 5.2.3) to the process $F_t(x)\psi_\epsilon(X_t - x)$ and we obtain

$$\begin{aligned} F_t(x)\psi_\epsilon(X_t - x) &= F_0(x)\psi_\epsilon(X_0 - x) \\ &+ \int_0^t H_s(x)\psi_\epsilon(X_s - x) \circ dW_s + \int_0^t G_s(x)\psi_\epsilon(X_s - x) ds \\ &+ \int_0^t F_s(x)(\partial_i \psi_\epsilon)(X_s - x)u_s^i \circ dW_s + \int_0^t F_s(x)(\partial_i \psi_\epsilon)(X_s - x)v_s^i ds. \end{aligned}$$

If we express the above Stratonovich integrals in terms of the corresponding Skorohod integrals we obtain

$$\begin{aligned} F_t(x)\psi_\epsilon(X_t - x) &= F_0(x)\psi_\epsilon(X_0 - x) \\ &+ \int_0^t H_s(x)\psi_\epsilon(X_s - x)dW_s + \int_0^t [G_s(x) + \frac{1}{2}(\nabla H)_s(x)]\psi_\epsilon(X_s - x)ds \\ &+ \frac{1}{2} \int_0^t H_s(x)(\partial_i \psi_\epsilon)(X_s - x)(\nabla X^i)_s ds \\ &+ \int_0^t F_s(x)(\partial_i \psi_\epsilon)(X_s - x)u_s^i dW_s \\ &+ \frac{1}{2} \int_0^t (\nabla F)_s(x)(\partial_i \psi_\epsilon)(X_s - x)u_s^i ds + \int_0^t F_s(x)(\partial_i \psi_\epsilon)(X_s - x)(\nabla u^i)_s ds \\ &+ \int_0^t F_s(x)(\partial_j \partial_i \psi_\epsilon)(X_s - x)(\nabla X^j)_s u_s^i ds \\ &+ \int_0^t F_s(x)(\partial_i \psi_\epsilon)(X_s - x)v_s^i ds. \end{aligned}$$

Notice that all the terms appearing in the above expression have continuous versions in (t, x) . Grouping the Skorohod integral terms and the Lebesgue integral terms we can write

$$F_t(x)\psi_\epsilon(X_t - x) = F_0(x)\psi_\epsilon(X_0 - x) + \int_0^t \alpha_\epsilon(s, x)dW_s + \int_0^t \beta_\epsilon(s, x)ds.$$

Step 2: Consider the processes $H_s(X_s)$ and $\langle F'_s(X_s), u_s \rangle$. We claim that these processes belong to the space $\mathbb{L}^{1,2}$. In fact, first notice that for each x , $F'(x)$ belongs to $\mathbb{L}^{1,2}$ and $D_s(F'_t(x)) = (D_s F'_t)'(x)$. This follows from hypotheses (A2) and (A5). On the other hand, $X, u \in \mathbb{L}^{1,4}$. Then we can apply Lemma 5.3.2 (suitably extended to stochastic processes) to $H_t(x)$ and X_t , and also to $\langle F'_t(x), u_t \rangle$ and X_t taking into account the following estimates that

$$\begin{aligned} \int_0^1 E(|H_s(X_s)|^2)ds &\leq E \int_0^1 \sup_{|x| \leq K} |H_s(x)|^2 ds < \infty, \\ E \int_0^1 \sup_{|x| \leq K} \|DH_s(x)\|_H^2 ds &< \infty, \\ E \int_0^1 \sup_{|x| \leq K} |H'_s(x)|^4 ds &< \infty, \end{aligned}$$

$$\begin{aligned}
\int_0^1 E(|\langle F'_s(X_s), u_s \rangle|^2) ds &\leq \|u\|_\infty^2 E \int_0^1 \sup_{|x| \leq K} |F'_s(x)|^2 ds < \infty, \\
\int_0^1 E(|F'_s(X_s)|^2 \|Du_s\|_H^2) ds &\leq \left\{ E \int_0^1 \sup_{|x| \leq K} |F'_s(x)|^4 ds E \int_0^1 \|Du_s\|_H^4 ds \right\}^{\frac{1}{2}} < \infty, \\
\int_0^1 E(|F''_s(X_s)|^2 \|DX_s\|_H^2 u_s^2) ds &\leq \|u\|_\infty^2 \\
&\times \left\{ E \int_0^1 \sup_{|x| \leq K} |F''_s(x)|^4 ds E \int_0^1 \|DX_s\|_H^4 ds \right\}^{\frac{1}{2}} < \infty, \\
\int_0^1 E(\|(DF'_s)'(X_s)\|_H^2 u_s^2) ds &\leq \|u\|_\infty^2 E \int_0^1 \sup_{|x| \leq K} \|(DF'_s)'(x)\|_H^2 ds < \infty,
\end{aligned}$$

The derivatives of these processes are then given by

$$D_s[H_t(X_t)] = \langle H'_t(X_t), D_s X_t \rangle + (D_s H_t)(X_t),$$

and

$$D_s[\langle F'_t(X_t), u_t \rangle] = \langle F''_t(X_t), D_s X_t \otimes u_t \rangle + \langle (D_s F'_t)'(X_t), u_t \rangle + \langle F'_t(X_t), D_s u_t \rangle.$$

Then from our hypotheses it follows that the processes $H_s(X_s)$ and $\langle F'_s(X_s), u_s \rangle$ belong to $\mathbb{L}_1^{1,2}$, and

$$\begin{aligned}
(\nabla[H_t(X_t)])_t &= \langle H'_t(X_t), (\nabla X)_t \rangle + (\nabla H)_t(X_t), \\
(\nabla[\langle F'_t(X_t), u_t \rangle])_t &= \langle F''_t(X_t), (\nabla X)_t \otimes u_t \rangle + \langle (\nabla F'_t)'(X_t), u_t \rangle + \langle F'_t(X_t), (\nabla u)_t \rangle.
\end{aligned}$$

Step 3: We claim that

$$\int_{\mathbb{R}^m} \left(F_t(x) \psi_\epsilon(X_t - x) - F_0(x) \psi_\epsilon(X_0 - x) - \int_0^t \beta_\epsilon(s, x) ds \right) dx \quad (5.31)$$

converges in $L^1(\Omega)$ to the process

$$\begin{aligned}
\Phi_t &= F_t(X_t) - F_0(X_0) - \int_0^t \left[G_s(X_s) + \frac{1}{2} (\nabla H)_s(X_s) \right. \\
&\quad + \frac{1}{2} (\partial_i H)_s(X_s) (\nabla X^i)_s + \frac{1}{2} \partial_i (\nabla F)_s(X_s) u_s^i \\
&\quad + (\partial_i F)_s(X_s) (\nabla u^i)_s + (\partial_j \partial_i F)_s(X_s) (\nabla X^j)_s u_s^i \\
&\quad \left. + (\partial_i F)_s(X_s) v_s^i \right] ds.
\end{aligned}$$

In fact, the convergence for each (s, ω) follows from the continuity assumption on x of the processes $G_s(x)$, $(\nabla H)_s(x)$, $\partial_i (\nabla F)_s(x)$, and the fact that $H_s(x)$ is of class C^1 , and $F_s(x)$ is of class C^2 . The $L^1(\Omega)$ convergence follows by dominated convergence because we have

$$E \int_0^1 \sup_\epsilon \left(\int_{\mathbb{R}^m} |\beta_\epsilon(s, x)| dx \right) ds < \infty.$$

From Step 2 it follows that

$$\begin{aligned}\Phi_t &= F_t(X_t) - F_0(X_0) - \int_0^t [G_s(X_s) + \langle F'_s(X_s), v_s \rangle] ds \\ &\quad - \int_0^t [H_s(X_s) + \langle F'_s(X_s), u_s \rangle] \circ dW_s \\ &\quad + \int_0^t [H_s(X_s) + \langle F'_s(X_s), u_s \rangle] dW_s.\end{aligned}$$

Step 4: Finally, we have

$$\begin{aligned}\int_{\mathbb{R}^m} \alpha_\epsilon(s, x) dx &= \int_{\mathbb{R}^m} H_s(x) \psi_\epsilon(X_s - x) dx \\ &\quad + \int_{\mathbb{R}^m} F_s(x) (\partial_i \psi_\epsilon)(X_s - x) u_s^i dx \\ &= \int_{\mathbb{R}^m} [H_s(x) + (\partial_i F_s)(x) u_s^i] \psi_\epsilon(X_s - x) dx.\end{aligned}$$

The integrals $\int_{\mathbb{R}^m} \alpha_\epsilon(s, x) dx$ converge as ϵ tends to zero in the norm of $\mathbb{L}^{1,2}$ to $H_s(X_s) + \langle F'_s(X_s), u_s \rangle$. Hence, the Skorohod integral

$$\int_0^t \left(\int_{\mathbb{R}^m} \alpha_\epsilon(s, x) dx \right) dW_s \quad (5.32)$$

converges in $L^2(\Omega)$ to

$$\int_0^t [H_s(X_s) + \langle F'_s(X_s), u_s \rangle] dW_s.$$

By Fubini's theorem (5.32) coincides with expression (5.31). Consequently, we get

$$\Phi_t = \int_0^t [H_s(X_s) + \langle F'_s(X_s), u_s \rangle] dW_s,$$

which completes the proof. $\square$

Bibliographical notes: There are different methods to define stochastic integrals of non adapted integrands with respect to the Wiener process. One approach, developed by Ogawa [87, 88, 89] and Rosinski [94] is based on the orthogonal development of the integrand in $L^2([0, 1])$. Another possibility is to use Riemann sums that converge to the so-called forward integral, which is equal to $\delta(u) + \int_0^1 D_t^- u_t dt$. This stochastic integral is an extension of the Itô integral, and it follows the same change-of-variable rule. Berger and Mizel [13] introduced this integral in order to solve stochastic Volterra equations. Russo and Vallois [95] introduced the forward integral considering the approximation of the Brownian path by its convolution with a rectangular function. In [56] Kuo and Russek study anticipating stochastic integrals in the context of the white noise analysis. Different versions of the change-of-variables formula for the Skorohod integral can be found in Sevljakov [97], Hitsuda [43], Üstünel [105], and Sekiguchi and Shiota [96]. The indefinite Skorohod integral has properties similar to that of a semimartingale. In particular, one has established the existence of a continuous local time and a Tanaka's type formula in [47] and [48].

Chapter 6

Anticipating stochastic differential equations

The anticipating stochastic calculus developed in Chapter 5 can be used to formulate stochastic differential equations where the solutions are nonadapted processes. Such type of equations appear, for instance, when the initial condition is not independent of the Wiener process, or when we impose a condition relating the values of the process at the initial and final times.

6.1 Stochastic differential equations in the Stratonovich sense

In this section we will present examples of stochastic differential equations formulated using the extended Stratonovich integral. These equations can be solved taking into account that this integral satisfies the usual differentiation rules.

Stochastic differential equations with random initial condition

Suppose that $W = \{W(t), t \in [0, 1]\}$ is a d -dimensional Brownian motion on the canonical probability space $(\Omega, \mathcal{F}, P)$. Let $A_i : \mathbb{R}^m \rightarrow \mathbb{R}^m$, $0 \leq i \leq d$, be continuous functions. Consider the stochastic differential equation

$$X_t = X_0 + \sum_{i=1}^d \int_0^t A_i(X_s) \circ dW_s^i + \int_0^t A_0(X_s) ds, \quad t \in [0, 1]. \quad (6.1)$$

The initial condition is an arbitrary random variable $X_0 \in L^0(\Omega; \mathbb{R}^m)$.

Suppose that the coefficients A_i , $1 \leq i \leq d$, are continuously differentiable with bounded derivatives. Define $B = A_0 + \frac{1}{2} \sum_{k=1}^m \sum_{i=1}^d A_i^k \partial_k A_i$, and suppose moreover that B is Lipschitz.

Let $\{\varphi_t(x), t \in [0, 1]\}$ be the solution to the following stochastic differential equation starting at point $x \in \mathbb{R}^m$:

$$\varphi_t(x) = x + \sum_{i=1}^d \int_0^t A_i(\varphi_s(x)) dW_s^i + \int_0^t B(\varphi_s(x)) ds. \quad (6.2)$$

In terms of the Stratonovich stochastic integral we can write

$$\varphi_t(x) = x + \sum_{i=1}^d \int_0^t A_i(\varphi_s(x)) \circ dW_s^i + \int_0^t A_0(\varphi_s(x)) ds.$$

We know (see, for instance, Kunita [54]) that there exists a version of $\varphi_t(x)$ such that $(t, x) \mapsto \varphi_t(x)$ is continuous, and we have

$$E(|\varphi_s(x) - \varphi_t(x')|^p) \leq C_{p,K}(|t - s|^{\frac{p}{2}} + |x - x'|^p).$$

for all $s, t \in [0, 1]$, $|x|, |x'| \leq K$, $p \geq 2$, $K > 0$. Then we can establish the following result

Theorem 6.1.1 *Assume that the coefficients A_i , $1 \leq i \leq d$ are of class C^3 , and A_i , $1 \leq i \leq d$, B have bounded partial derivatives of first order. Then for any random vector X_0 , the process $X = \{\varphi_t(X_0), t \in [0, 1]\}$ satisfies the anticipating stochastic differential equation (6.1).*

Proof: Under the assumptions of the theorem we know that for any $x \in \mathbb{R}^m$ Eq. (6.2) has a unique solution $\varphi_t(x)$. Set $X = \{\varphi_t(X_0), 0 \leq t \leq 1\}$. By a localization argument we can assume that the coefficients A_i , $1 \leq i \leq d$, and B have compact support. It suffices to show that for any $i = 1, \dots, d$, the process $u_i(t, x) = A_i(\varphi_t(x))$ satisfies the hypotheses of Theorem 5.3.3, suitably extended to a d -dimensional Brownian motion.

Condition (h1) is obvious. On the other hand, using Itô's formula one can easily check condition (h4). This completes the proof. $\square$

The uniqueness is a more difficult problem. We are going to show that there is a unique solution in the class of processes $\mathbb{L}_{2,\text{loc}}^{1,4}(\mathbb{R}^m)$, assuming some additional regularity conditions on the initial condition X_0 . We will denote by $L(\mathbb{R}^m)$ the space of m -dimensional processes (or random variables) whose components belong to the class L .

Lemma 6.1.1 *Suppose that $X_0 \in \mathbb{D}_{\text{loc}}^{1,p}(\mathbb{R}^m)$ for some $p > q \geq 2$, and assume that the coefficients A_i , $1 \leq i \leq d$, and B are of class C^2 with bounded partial derivatives of first order, then we claim that $\{\varphi_t(X_0), 0 \leq t \leq 1\}$ belongs to $\mathbb{L}_{2,\text{loc}}^{1,q}(\mathbb{R}^m)$.*

Proof: Let (Ω^n, X_0^n) be a localizing sequence for X_0 in $\mathbb{D}_{\text{loc}}^{1,p}(\mathbb{R}^m)$. Set

$$G^{n,k,M} = \Omega^n \cap \{|X_0^n| \leq k\} \cap \left\{ \sup_{|x| \leq k, t \in [0,1]} |\varphi_t(x)| \leq M \right\}.$$

On the set $G^{n,k,M}$ the process $\varphi_t(X_0)$ coincides with $\varphi_t^M(X_0^n \beta_k(X_0^n))$, where β_k is a smooth function such that $0 \leq \beta_k \leq 1$, $\beta_k(x) = 1$ if $|x| \leq k$, and $\beta_k(x) = 0$ if $|x| \geq k + 1$, and $\varphi_t^M(x)$ is the stochastic flow associated with the coefficients $\beta_M A_i$, $1 \leq i \leq d$, and $\beta_M B$. Hence, we can assume that the coefficients A_i , $1 \leq i \leq d$, and B are bounded and have bounded derivatives up to the second order, $X_0 \in \mathbb{D}^{1,p}(\mathbb{R}^m)$, and X_0 is bounded by k . Now we can apply Lemma 5.3.2. The following estimates hold:

$$E \left(\sup_{|x| \leq k, t \in [0,1]} [|\varphi_t(x)|^r + \|D[\varphi_t(x)]\|_H^r] \right) < \infty, \quad (6.3)$$

$$E \left(\sup_{|x| \leq k, t \in [0,1]} |\varphi'_t(x)|^r \right) < \infty, \quad (6.4)$$

for any $r \geq 2$. These estimate (6.3) and (6.4) follow from our assumptions on the coefficients A_i , $1 \leq i \leq d$, and B , taking into account that

$$\begin{aligned} D_s^j[\varphi_t(x)] &= A_j(\varphi_s(x)) + \int_s^t (\partial_k A_i)(\varphi_r(x)) D_s^j[\varphi_r^k(x)] dW_r^i \\ &\quad + \int_s^t (\partial_k B)(\varphi_r(x)) D_s^j[\varphi_r^k(x)] dr, \end{aligned}$$

for $1 \leq j \leq d$ and $0 \leq s \leq t \leq 1$. Hence, from Lemma 5.3.2 we obtain that $\{\varphi_t(X_0), 0 \leq t \leq 1\}$ belongs to $\mathbb{L}^{1,q}(\mathbb{R}^m)$ and that

$$D_s[\varphi_t(X_0)] = \varphi'_t(X_0) D_s X_0 + [D_s \varphi_t](X_0).$$

Finally, from the above expression for the derivative of $\varphi_t(X_0)$ it is not difficult to show that $\{\varphi_t(X_0), 0 \leq t \leq 1\}$ belongs to $\mathbb{L}_2^{1,q}(\mathbb{R}^m)$, and

$$(D[\varphi(X_0)])_s^{+,-} = \varphi'_s(X_0) D_s X_0 + (D\varphi)_s^{+,-}(X_0).$$

□

Theorem 6.1.2 *Assume that the coefficients A_i , $1 \leq i \leq d$ and B are of class C^4 , with bounded partial derivatives of first order. The initial condition X_0 belongs to $\mathbb{D}_{\text{loc}}^{1,p}(\mathbb{R}^m)$ for some $p > 4$. Then the process $X = \{\varphi_t(X_0), t \in [0, 1]\}$ is the unique solution to Eq. (6.1) in $\mathbb{L}_{2,\text{loc}}^{1,4}(\mathbb{R}^m)$ which is continuous.*

Proof: By Lemma 6.1.1 we already know that X belongs to $\mathbb{L}_{2,\text{loc}}^{1,4}(\mathbb{R}^m)$. Let Y be another continuous solution in this space. Suppose that (Ω^n, X_0^n) is a localizing sequence for X_0 in $\mathbb{D}_{\text{loc}}^{1,p}(\mathbb{R}^m)$, and $(\Omega^{n,1}, Y^n)$ is a localizing sequence for Y in $\mathbb{L}_2^{1,4}(\mathbb{R}^m)$. Set

$$H^{n,k,M} = \Omega^n \cap \Omega^{n,1} \cap \{|X_0^n| \leq k\} \cap \left\{ \sup_{|x| \leq k, t \in [0,1]} |\varphi_t(x)| \leq M \right\} \cap \left\{ \sup_{t \in [0,1]} |Y_t| \leq M \right\}.$$

The processes $Y_t^n \beta_M(Y_t^n)$, $(\beta_M A_i)(Y_t^n)$, and $(\beta_M A_0)(Y_t^n)$ satisfy hypothesis (A1). On the set $H^{n,k,M}$ we have the following equality

$$Y_t = X_0 + \sum_{i=1}^d \int_0^t A_i(Y_s) \circ dW_s^i + \int_0^t A_0(Y_s) ds, \quad t \in [0, 1].$$

Let us denote by $\varphi_t^{-1}(x)$ the inverse of the mapping $x \rightarrow \varphi_t(x)$. By the classical Itô's formula we can write

$$\varphi_t^{-1}(x) = x - \sum_{i=1}^d \int_0^t (\varphi'_s(\varphi_s^{-1}(x)))^{-1} A_i(x) \circ dW_s^i - \int_0^t (\varphi'_s(\varphi_s^{-1}(x)))^{-1} A_0(x) ds.$$

We claim that the processes $F_t(x) = \varphi_t^{-1}(x)$, $H_t^i(x) = (\varphi'_s(\varphi_s^{-1}(x)))^{-1} A_i(x)$, and $G_t(x) = (\varphi'_s(\varphi_s^{-1}(x)))^{-1} A_0(x)$ with values in the space of $m \times m$ matrices satisfy hypotheses (A2) to (A5). This follows from the properties of the stochastic flow associated with the coefficients A_j .

Consequently, we can apply Theorem 5.3.4 and we obtain, on the set $H^{n,k,M}$,

$$\begin{aligned}\varphi_t^{-1}(Y_t) &= X_0 + \int_0^t (\varphi_s^{-1})'(Y_s) A_i(Y_s) \circ dW_s^i + \int_0^t (\varphi_s^{-1})'(Y_s) A_0(Y_s) ds \\ &\quad - \int_0^t (\varphi_s'(\varphi_s^{-1}(Y_s)))^{-1} A_i(Y_s) \circ dW_s^i - \int_0^t (\varphi_s'(\varphi_s^{-1}(Y_s)))^{-1} A_0(Y_s) ds.\end{aligned}$$

Notice that

$$(\varphi_s^{-1})'(x) = -(\varphi_s'(\varphi_s^{-1}(x)))^{-1}.$$

Hence, we get $\varphi_t^{-1}(Y_t) = X_0$, that is, $Y_t = \varphi_t(X_0)$ which completes the proof of the uniqueness. $\square$

One-dimensional stochastic differential equations with random coefficients

Consider the one-dimensional equation

$$X_t = X_0 + \int_0^t \sigma(X_s) \circ dW_s + \int_0^t b(X_s) ds, \quad (6.5)$$

where the coefficients σ and b and the initial condition X_0 are random. If the coefficients and the initial condition are deterministic and we assume that the function σ is of class C^2 with bounded first and second derivatives, and that b is Lipschitz, then there is a unique solution X_t . This solution (see Doss [26]) can be written as $X_t = u(W_t, Y_t)$, where $u : \mathbb{R}^2 \rightarrow \mathbb{R}$ is the solution of the ordinary differential equation

$$\frac{\partial u}{\partial x} = \sigma(u(x)), \quad u(0, y) = y,$$

and Y_t is the solution to the ordinary differential equation with random parameters

$$Y_t = X_0 + \int_0^t f(W_s, Y_s) ds,$$

where $f(x, y) = b(u(x, y)) \exp(-\int_0^x \sigma'(u(z, y)) dz)$. Notice that

$$\frac{\partial u}{\partial y} = \exp(\int_0^x \sigma'(u(z, y)) dz).$$

Suppose that now the coefficients and the initial condition are random. We claim that in that case the process $X_t = u(W_t, Y_t)$ solves Eq. (6.5).

Consider a partition $\pi = \{0 = t_0 < t_1 < \dots < t_n = 1\}$ of the interval $[0, 1]$. We can introduce the polygonal approximation of the Brownian motion defined by

$$W_t^\pi = \int_0^t \sum_{i=0}^{n-1} \left(\frac{W(t_{i+1}) - W(t_i)}{t_{i+1} - t_i} \right) \mathbf{1}_{[t_i, t_{i+1})}(s) ds.$$

Set $X_t^\pi = u(W_t^\pi, Y_t)$. This process satisfies

$$X_t^\pi = X_0 + \int_0^t \sigma(X_s^\pi) \dot{W}_s^\pi ds + \int_0^t \exp\left(\int_{W_s}^{W_s^\pi} \sigma'(u(y, Y_s)) dy\right) b(X_s) ds = A_t^1 + A_t^2.$$

The term A_t^1 can be written as

$$A_t^1 = \int_0^t [\sigma(X_s^\pi) - \sigma(X_s)] \dot{W}_s^\pi ds + \int_0^t \sigma(X_s) \dot{W}_s^\pi ds = B_t^1 + B_t^2.$$

Note that B_t^2 is the Riemann sum that approximates the Stratonovich integral of $\sigma(X_s)$. We know that A_t^1 converges in probability to $X_t - X_0 - \int_0^t b(X_s)ds$ as $|\pi|$ tends to zero. As a consequence, if we show that B_t^1 converges in probability to zero as $|\pi|$ tends to zero this will imply $\sigma(X_t)$ is Stratonovich integrable on any interval $[0, t]$ and (6.5) holds.

We have

$$\sigma(X_s^\pi) - \sigma(X_s) = \sigma\sigma'(X_s)(W_s^\pi - W_s) + \frac{1}{2}[\sigma(\sigma')^2 + \sigma^2\sigma''](\xi)(W_s^\pi - W_s)^2,$$

where ξ is a random intermediate point between W_s^π and W_s . The last summand of the above expression does not contribute to the limit of B_t^1 . Finally the convergence to zero of B_t^1 follows from Lemma 6.1.2.

Lemma 6.1.2 *Let Φ_t be a continuous process. Then, the integral*

$$\int_0^1 \Phi_s \dot{W}_s^\pi (W_s^\pi - W_s) ds$$

converge in probability to zero as $|\pi|$ tends to zero, where $\pi = \{0 = t_0 < t_1 < \dots < t_n = 1\}$ runs over all partitions of the interval $[0, 1]$.

Proof: Fix a partition $\pi = \{0 = t_0 < t_1 < \dots < t_n = 1\}$. Consider another partition of the interval $[0, 1]$ of the form $\{s_j := \frac{j}{m}, 0 \leq j \leq m\}$. We can write

$$\begin{aligned} \left| \int_0^1 \Phi_s \dot{W}_s^\pi (W_s^\pi - W_s) ds \right| &\leq \left| \sum_{j=0}^{m-1} \Phi_{s_j} \int_{s_j}^{s_{j+1}} \dot{W}_s^\pi (W_s^\pi - W_s) ds \right| \\ &+ \left| \sum_{j=0}^{m-1} \int_{s_j}^{s_{j+1}} (\Phi_s - \Phi_{s_j}) \dot{W}_s^\pi (W_s^\pi - W_s) ds \right| = C_1 + C_2. \end{aligned}$$

The term C_1 converges to zero in probability as $|\pi|$ tends to zero, for any fixed value of m . In fact, if the points s_j and s_{j+1} belong to the partition π , we have

$$\begin{aligned} \int_{s_j}^{s_{j+1}} \dot{W}_s^\pi (W_s^\pi - W_s) ds &= \sum_{i: t_i \in [s_j, s_{j+1})} \frac{W(t_{i+1}) - W(t_i)}{t_{i+1} - t_i} \left(\int_{t_i}^{t_{i+1}} (W_s^\pi - W_s) ds \right) \\ &= -\frac{1}{2} \sum_{i: t_i \in [s_j, s_{j+1})} \frac{1}{t_{i+1} - t_i} \int_{t_i}^{t_{i+1}} [(W_{t_{i+1}} - W_s)^2 - (W_s - W_{t_i})^2] ds, \end{aligned}$$

and this converges to $\frac{1}{2}(s_{j+1} - s_j) - \frac{1}{2}(s_{j+1} - s_j) = 0$.

The term C_2 can be bounded as follows:

$$\begin{aligned} C_2 &\leq \sup_{|s-t| \leq \frac{1}{m}} |\Phi_s - \Phi_t| \sum_{j=0}^{m-1} \int_{s_j}^{s_{j+1}} |\dot{W}_s^\pi (W_s^\pi - W_s)| ds \\ &\leq \sup_{|s-t| \leq \frac{1}{m}} |\Phi_s - \Phi_t| \\ &\quad \times \sum_{j=0}^{m-1} \sum_{i: t_i \in [s_j, s_{j+1})} \frac{1}{2(t_{i+1} - t_i)} \int_{t_{i+1}}^{t_i} [(W_{t_{i+1}} - W_s)^2 + (W_s - W_{t_i})^2] ds. \end{aligned}$$

As n tends to infinity the right-hand side of the above inequality converges in probability to $\frac{1}{2}$. Hence, we obtain

$$\limsup_n C_2 \leq \frac{1}{2} \sup_{|s-t| \leq \frac{1}{m}} |\Phi_s - \Phi_t|,$$

and this expression converges to zero in probability as m tends to zero, due to the continuity of the process Φ_t . $\square$

The uniqueness requires the use of the Itô-Ventzell formula, and we have to impose regularity assumptions on the coefficients (cf. [51]).

6.2 Stochastic differential equations with boundary conditions

Consider the following Stratonovich differential equation on the time interval $[0, 1]$, where instead of giving the value X_0 , we impose a linear relationship between the values of X_0 and X_1 :

$$\begin{cases} dX_t = \sum_{i=1}^d A_i(X_t) \circ dW_t^i + A_0(X_t) dt \\ h(X_0, X_1) = 0 \end{cases} \quad (6.6)$$

We are interested in proving the existence and uniqueness of a solution for this type of equations. We will discuss two particular cases:

- (a) The case where the coefficients A_i , $0 \leq i \leq d$ and the function h are affine (see Ocone and Pardoux [86]).
- (b) The one-dimensional case (see Donati-Martin [25]).

Linear stochastic differential equations with boundary conditions

Consider the following stochastic boundary value problem:

$$\begin{cases} X_t = X_0 + \sum_{i=1}^d \int_0^t A_i X_s \circ dW_s^i + A_0 \int_0^t X_s ds, \\ H_0 X_0 + H_1 X_1 = h. \end{cases} \quad (6.7)$$

We assume that A_i , $i = 0, \dots, d$, H_0 , and H_1 are $m \times m$ deterministic matrices and $h \in \mathbb{R}^m$. We will also assume that the $m \times 2m$ matrix $H_0 : H_1$ has rank m .

Concerning the boundary condition, two particular cases are interesting:

Two-point boundary-value problem: Let $l \in \mathbb{N}$ be such that $0 < l < m$. Suppose that $H_0 = \begin{pmatrix} H'_0 \\ 0 \end{pmatrix}$, $H_1 = \begin{pmatrix} 0 \\ H''_1 \end{pmatrix}$, where H'_0 is an $l \times m$ matrix and H''_1 is an $(m-l) \times m$ matrix. Condition $\text{rank}(H_0 : H_1) = m$ implies that H'_0 has rank l and that H''_1 has rank $m-l$. If we write $h = \begin{pmatrix} h_0 \\ h_1 \end{pmatrix}$, where $h_0 \in \mathbb{R}^l$ and $h_1 \in \mathbb{R}^{m-l}$, then the boundary condition becomes

$$H'_0 X_0 = h_0, \quad H''_1 X_1 = h_1.$$

Periodic solutions to SDE's: Suppose $H_0 = -H_1 = I$ and $h = 0$. Then the boundary condition becomes

$$X_0 = X_1.$$

With (6.7) we can associate an $m \times m$ adapted and continuous matrix-valued process Φ solution of

$$\begin{cases} d\Phi_t = \sum_{i=1}^d A_i \Phi_t \circ dW_t^i + B \Phi_t dt, \\ \Phi_0 = I. \end{cases} \quad (6.8)$$

Using the properties of the extended Stratonovich integral, one can obtain an explicit solution to Eq. (6.7). By a solution we mean a continuous stochastic process X such that $A_i(X_s)$ is Stratonovich integrable with respect to W^i on any interval $[0, t]$ and such that (6.7) holds.

Theorem 6.2.1 *Suppose that the random matrix $H_0 + H_1 \Phi_1$ is a.s. invertible. Then there exist a solution to the stochastic differential equation (6.7), which is unique among those continuous processes whose components belong to the space $\mathbb{L}_{2,\text{loc}}^{1,4}$.*

Proof: Define

$$X_t = \Phi_t X_0, \quad (6.9)$$

where X_0 is given by

$$X_0 = [H_0 + H_1 \Phi_1]^{-1} h. \quad (6.10)$$

Then it follows from this expression that X^i belongs to $\mathbb{L}_{2,\text{loc}}^{1,4}$, for all $i = 1, \dots, m$, and this process satisfies Eq. (6.7).

In order to show the uniqueness, we proceed as follows. Let $Y \in \mathbb{L}_{2,\text{loc}}^{1,4}(\mathbb{R}^m)$ be a solution to (6.7). Then we have

$$\Phi_t^{-1} = I - \sum_{i=1}^d \int_0^t \Phi_s^{-1} A_i \circ dW_s^i - \int_0^t \Phi_s^{-1} A_0 ds.$$

By the change-of-variables formula for the Stratonovich integral (Theorem 5.2.2), we see that $\Phi_t^{-1} Y_t = Y_0$, namely, $Y_t = \Phi_t Y_0$. Therefore, Y satisfies (6.9). But (6.10) follows from (6.9) and the boundary condition $H_0 Y_0 + H_1 Y_1 = h$. Consequently, Y satisfies (6.9) and (6.10), and it must be equal to X . $\square$

Notice that, in general, the solution will not belong to $\mathbb{L}_2^{1,4}(\mathbb{R}^m)$. One can also treat more general linear equations. For instance, suppose that we have the equation:

$$\begin{cases} X_t = X_0 + \sum_{i=1}^d \int_0^t A_i X_s \circ dW_s^i + \int_0^t A_0 X_s ds + V_t, \\ H_0 X_0 + H_1 X_1 = h, \end{cases} \quad (6.11)$$

where V_t is a continuous semimartingale. In that case,

$$X_t = \Phi_t [H_0 + H_1 \Phi_1]^{-1} \left[h - H_1 \Phi_1 \int_0^1 \Phi_s^{-1} \circ dV_s \right] + \Phi_t \int_0^t \Phi_s^{-1} \circ dV_s,$$

is a solution to equation (6.11). The uniqueness in the class of processes $\mathbb{L}_{2,\text{loc}}^{1,4}(\mathbb{R}^m)$ can be established provided the process V_t belongs also to this space.

One-dimensional stochastic differential equations with boundary conditions

Consider the one-dimensional stochastic boundary value problem

$$\begin{cases} X_t = X_0 + \int_0^t \sigma(X_s) \circ dW_s + \int_0^t b(X_s) ds \\ a_0 X_0 + a_1 X_1 = a_2. \end{cases} \quad (6.12)$$

Applying the techniques of the anticipating stochastic calculus we can show the following result.

Theorem 6.2.2 *Suppose that the functions σ and $b_1 := b + \frac{1}{2}\sigma\sigma'$ are of class C^2 with bounded derivatives and $a_0 a_1 > 0$. Then there exists a solution to (6.12). Furthermore if the functions σ and b_1 are of class C^4 with bounded derivatives then the solution is unique in the class $\mathbb{D}_{2,\text{loc}}^{1,4}$.*

Proof: Let $\varphi_t(x)$ be the stochastic flow associated with the coefficients σ and b_1 . By Theorem 6.1.1 for any random variable X_0 the process $\varphi_t(X_0)$ satisfies

$$X_t = X_0 + \int_0^t \sigma(X_s) \circ dW_s + \int_0^t b(X_s) ds.$$

Hence, in order to show the existence of a solution it suffices to prove that there is a unique random variable X_0 such that

$$\varphi_1(X_0) = \frac{a_2 - a_0 X_0}{a_1}. \quad (6.13)$$

The mapping $g(x) = \varphi_1(x)$ is strictly increasing and this implies the existence of a unique solution to equation (6.13).

Let us now turn to the proof of the uniqueness. It suffices to show that the unique solution X_0 to equation (6.13) belongs to $\mathbb{D}_{\text{loc}}^{1,p}$ for some $p > 4$. Taking into account that the equation is one dimensional, one can represent the solution $\varphi_t(x)$ by means of the method of Doss. Hence, $\varphi_t(x)$ is Fréchet differentiable in ω . Using this fact, and the implicit function theorem it is not difficult to show that X_0 belongs to the space $\mathbb{D}_{\text{loc}}^{1,p}$. $\square$

6.3 Stochastic differential equations in the Skorohod sense

Let $W = \{W_t, t \in [0, 1]\}$ be a one-dimensional Brownian motion defined on the canonical probability space $(\Omega, \mathcal{F}, P)$. Consider the stochastic differential equation

$$X_t = X_0 + \int_0^t \sigma(s, X_s) dW_s + \int_0^t b(s, X_s) ds, \quad (6.14)$$

$0 \leq t \leq 1$, where X_0 is $\mathcal{F}_1$ -measurable and σ and b are deterministic functions. The stochastic integral $\int_0^t \sigma(s, X_s) dW_s$ is interpreted in the Skorohod sense. First notice that the usual Picard iteration procedure cannot be applied to show that this equation has a unique solution. Indeed, the Sokorohod integral is not a bounded operator in

L^p . In some sense Eq. (6.14) is an infinite dimensional hyperbolic partial differential equation. In fact, this equation can be formally written as

$$X_t = X_0 + \int_0^t \sigma(s, X_s) \circ dW_s + \frac{1}{2} \int_0^t \sigma'(s, X_s) [(D^+ X)_s + (D^- X)_s] ds + \int_0^t b(s, X_s) ds.$$

If the diffusion coefficient is linear one can show that there exist a unique global solution. In fact, let us consider the following particular case:

$$X_t = X_0 + \sigma \int_0^t X_s dW_s. \quad (6.15)$$

When X_0 is deterministic, the solution is given by the martingale

$$X_t = X_0 e^{\sigma W_t - \frac{1}{2} \sigma^2 t}.$$

We have seen in (5.4) that if $X_0 = \text{sign} W_1$, then, a solution to Eq. (6.15) is

$$X_t = \text{sign}(W_1 - \sigma t) e^{\sigma W_t - \frac{1}{2} \sigma^2 t}.$$

More generally, one can show (see Theorem 6.3.1 below) that if $X_0 \in L^p(\Omega)$ for some $p > 2$, then

$$X_t = X_0(A_t) e^{\sigma W_t - \frac{1}{2} \sigma^2 t}$$

is a solution to Eq. (6.15) where $A_t(\omega)_s = \omega_s - \sigma(t \wedge s)$. In terms of the Wick product (see [22]) one can write

$$X_t = X_0 \diamond e^{\sigma W_t - \frac{1}{2} \sigma^2 t} = X_0(A_t) e^{\sigma W_t - \frac{1}{2} \sigma^2 t}.$$

Consider the following equation:

$$X_t = X_0 + \int_0^t \sigma_s X_s dW_s + \int_0^t b(s, X_s) ds, \quad 0 \leq t \leq 1, \quad (6.16)$$

where $\sigma \in L^2([0, 1])$, X_0 is a random variable, and b is a random function satisfying the following condition:

(H.1) $b : [0, 1] \times \mathbb{R} \times \Omega \rightarrow \mathbb{R}$ is a measurable function such that there exist a nonnegative function γ_t on $[0, 1]$, $\gamma_t \geq 0$, a constant $L > 0$, and a set $N_1 \in \mathcal{F}$ of probability one, verifying

$$\begin{aligned} |b(t, x, \omega) - b(t, y, \omega)| &\leq \gamma_t |x - y|, \quad \int_0^1 \gamma_t dt \leq L, \\ |b(t, 0, \omega)| &\leq L, \end{aligned}$$

for all $x, y \in \mathbb{R}$, $t \in [0, 1]$ and $\omega \in N_1$.

Let us introduce some notation. Consider the family of transformations $T_t, A_t : \Omega \rightarrow \Omega$, $t \in [0, 1]$, given by

$$\begin{aligned} T_t(\omega)_s &= \omega_s + \int_0^{t \wedge s} \sigma_u du, \\ A_t(\omega)_s &= \omega_s - \int_0^{t \wedge s} \sigma_u du. \end{aligned}$$

Note that $T_t A_t = A_t T_t = \text{Identity}$. Define

$$\varepsilon_t = \exp\left(\int_0^t \sigma_s dW_s - \frac{1}{2} \int_0^t \sigma_s^2 ds\right).$$

Then, by Girsanov's theorem $E[F(A_t)\varepsilon_t] = E[F]$ for any random variable $F \in L^1(\Omega)$. For each $x \in \mathbb{R}$ and $\omega \in \Omega$ we denote by $Z_t(\omega, x)$ the solution of the integral equation

$$Z_t(\omega, x) = x + \int_0^t \varepsilon_s^{-1}(T_t(\omega)) b\left(s, \varepsilon_s(T_t(\omega)) Z_s(\omega, x), T_s(\omega)\right) ds. \quad (6.17)$$

Notice that for $s \leq t$ we have $\varepsilon_s(T_t) = \exp\left(\int_0^s \sigma_u dW_u + \frac{1}{2} \int_0^s \sigma_u^2 du\right) = \varepsilon_s(T_s)$. Henceforth we will omit the dependence on ω in order to simplify the notation.

Theorem 6.3.1 *Fix an initial condition $X_0 \in L^p(\Omega)$ for some $p > 2$, and define*

$$X_t = \varepsilon_t Z_t(A_t, X_0(A_t)). \quad (6.18)$$

Then the process $X = \{X_t, 0 \leq t \leq 1\}$ satisfies $\mathbf{1}_{[0,t]} \sigma X \in \text{Dom } \delta$ for all $t \in [0, 1]$, $X \in L^2([0, 1] \times \Omega)$, and X is the unique solution of Eq. (6.16) verifying these conditions.

Proof: We will only prove the existence. The uniqueness can be shown using similar arguments.

Let us prove first that the process X given by (6.18) satisfies the desired conditions. By Gronwall's lemma and using hypothesis (H.1), we have

$$|X_t| \leq \varepsilon_t e^{tL} (|X_0(A_t)| + L \int_0^t \varepsilon_s^{-1}(T_s) ds), \quad (6.19)$$

which implies $\sup_{t \in [0,1]} E(|X_t|^q) < \infty$, for all $2 \leq q < p$, as it follows easily from Girsanov's theorem and Hölder's inequality. Indeed, we have

$$\begin{aligned} E(|X_t|^q) &\leq c_q E\left(\varepsilon_t^q e^{qtL} \left(|X_0(A_t)|^q + L^q \int_0^t \varepsilon_s^{-q}(T_s) ds\right)\right) \\ &\leq c_q e^{qL} \left\{ E\left(\varepsilon_t^{q-1}(T_t) |X_0|^q + L^q \varepsilon_t^{q-1}(T_t) \int_0^t \varepsilon_s^{-q}(T_s^2) ds\right) \right\} \\ &\leq C \left\{ E(|X_0|^p)^{\frac{q}{p}} + 1 \right\}. \end{aligned}$$

Now fix $t \in [0, 1]$ and let us prove that $\mathbf{1}_{[0,t]} \sigma X \in \text{Dom } \delta$ and that (6.16) holds. Let $G \in \mathcal{S}$ be a smooth random variable. Using (6.18) and Girsanov's theorem, we obtain

$$\begin{aligned} E\left[\int_0^t \sigma_s X_s D_s G ds\right] &= E\left[\int_0^t \sigma_s \varepsilon_s Z_s(A_s, X_0(A_s)) D_s G ds\right] \\ &= E\left[\int_0^t \sigma_s Z_s(X_0) [D_s G](T_s) ds\right]. \end{aligned} \quad (6.20)$$

Notice that $\frac{d}{ds} G(T_s) = \sigma_s [D_s G](T_s)$. Therefore, integrating by parts in (6.20) and again applying Girsanov's theorem yield

$$\begin{aligned} E\left[\int_0^t Z_s(X_0) \frac{d}{ds} G(T_s) ds\right] &= E\left[Z_t(X_0) G(T_t) - Z_0(X_0) G\right. \\ &\quad \left. - \int_0^t \varepsilon_s^{-1}(T_t) b(s, \varepsilon_s(T_s) Z_s(X_0), T_s) G(T_s) ds\right] \\ &= E\left[\varepsilon_t Z_t(A_t, X_0(A_t)) G\right] - E\left[Z_0(X_0) G\right] \\ &\quad - \int_0^t E\left[b(s, \varepsilon_s Z_s(A_s, X_0(A_s))) G\right] ds \\ &= E[X_t G] - E[X_0 G] - \int_0^t E[b(s, X_s) G] ds. \end{aligned}$$

Because the random variable $X_t - X_0 - \int_0^t b(s, X_s)ds$ is square integrable, we deduce that $\mathbf{1}_{[0,t]}\sigma X$ belongs to the domain of δ and that (6.16) holds. $\square$

When the diffusion coefficient is not linear one can show that there exist a solution up to a random time. Existence and uniqueness results for one-dimensional Skorohod equations with random initial condition are established by Buckdahn in [20]. Let us state the main result of this paper. Its proof is based on Doss representation of the solution to a one-dimensional stochastic differential equation and on the change-of-variable formula for the Skorohod integral. We recall that i denotes the mapping from $H = L^2([0, 1])$ into $\Omega = C_0([0, 1])$, defined by $i(h)_t = \int_0^t h_s ds$.

Theorem 6.3.2 *Consider the stochastic differential equation in the Skorohod sense*

$$X_t = X_0 + \int_0^t \sigma(X_s) dW_s + \int_0^t b(X_s) ds. \quad (6.21)$$

Suppose that σ and b are functions of class C^2 with bounded derivatives of first and second order. Suppose that the initial condition X_0 is bounded and there exist a process $D_s X_0$ bounded by M such that

$$\begin{aligned} X_0(\omega + i(h)) &= X_0(\omega) + \int_0^1 D_s X_0 h_s ds + o(\|h\|_H), \\ |D_s X_0(\omega + i(h)) - D_s X_0(\omega)| &\leq M \|h\|_H. \end{aligned}$$

Then for any bounded ball $B_r(h)$ of radius $r > 0$ around $i(h)$ there exist a $t > 0$ such that Eq. (6.21) has a unique solution X on $[0, t] \times B_{r-3t\|\sigma'\|_\infty}(h)$ which belongs to $L^1([0, t] \times B_r(h))$ and it possesses and extension $\bar{X}$ in the space $\mathbb{D}_\infty^{1,2}$ (set of elements in $\mathbb{D}^{1,2}$ with a bounded derivative).

Bibliographical notes: There are several papers devoted to the study of the anticipating process $\varphi_t(X_0)$, composition of a stochastic flow with a random initial condition. The large deviation principle has been investigated in [68] and [69], the characterization of the support of the law of the solution in [67], the absolute continuity of the law was studied in [65], and the regularity of the density was treated in [23]. Stochastic differential equations in the Skorohod sense were analyzed by Buckdahn, in the linear case, in connection with the anticipating Girsanov theorem (see [19] and [21]). Other works on the Skorohod stochastic differential equations are [78] (multi-dimensional Brownian motion) and [22] (linear multidimensional case).